

MAGNETOM

Startup System

Start-up / Magnet System Avanto fit

Document Version

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The configuration of the IButton is only necessary for an Avanto^{fit} and Skyra^{fit} upgrade on-site.

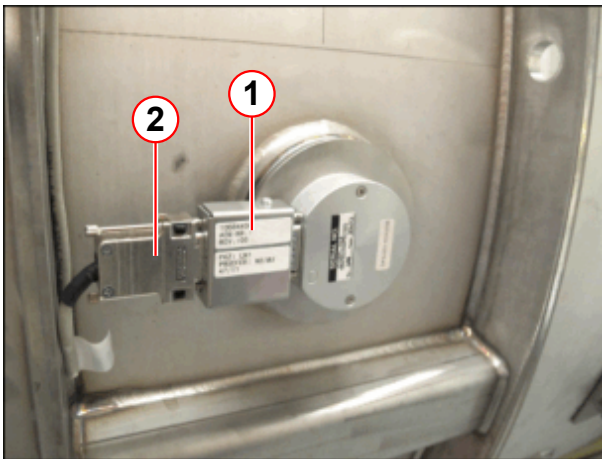
Avanto^{fit} with Magnet OR105.

Skyra^{fit} with Magnet OR97.

Prerequisite

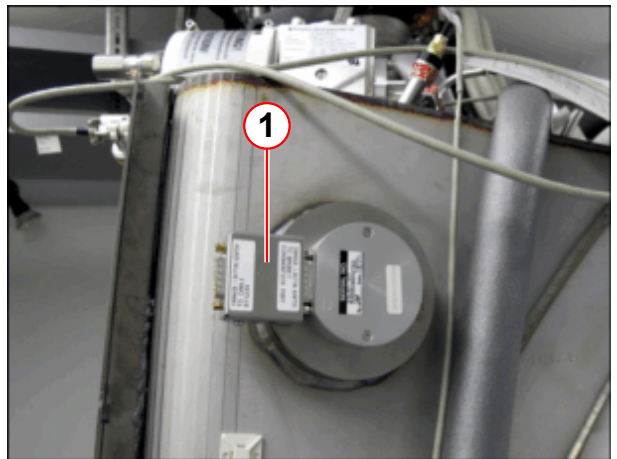
- The cabling is finished
- MR system is switched on
- MSUP has new LW from the system
- IButton is installed

Fig. 1: OR105- IButton(magnet right side)



- (1) IButton adapter
- (2) W11234

Fig. 2: OR97 IButton(side sock)



- (1) IButton adapter

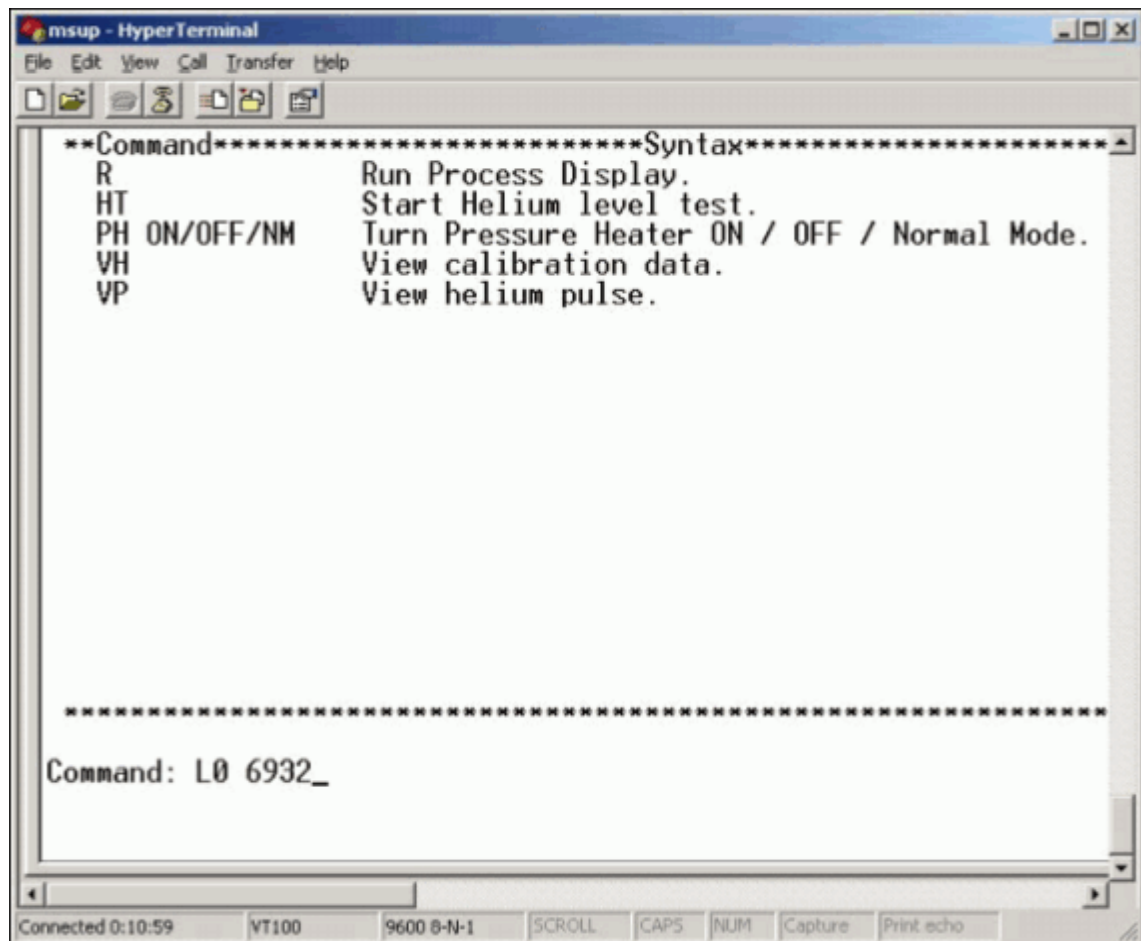
Tools

- Laptop
- RS232 cable

1.1 Procedure

- Turn on the laptop and open the Terminal emulation software (e.g. HyperTerminal).
 - » Set bits per second to 9600.
 - » Set data bits to 8.
 - » Set parity to 1.
 - » Set flow control to none.
 - » Set Emulation to VT100 (File: Properties: Settings).
- Connect the laptop to connector X6 on the MSUP via RS232 cable.
- Press **ESC**.
- Command: **L0 6932** <enter>

Fig. 3: IButton Configuration



- Command: **B** <enter>

Fig. 4: IButton Configuration

```

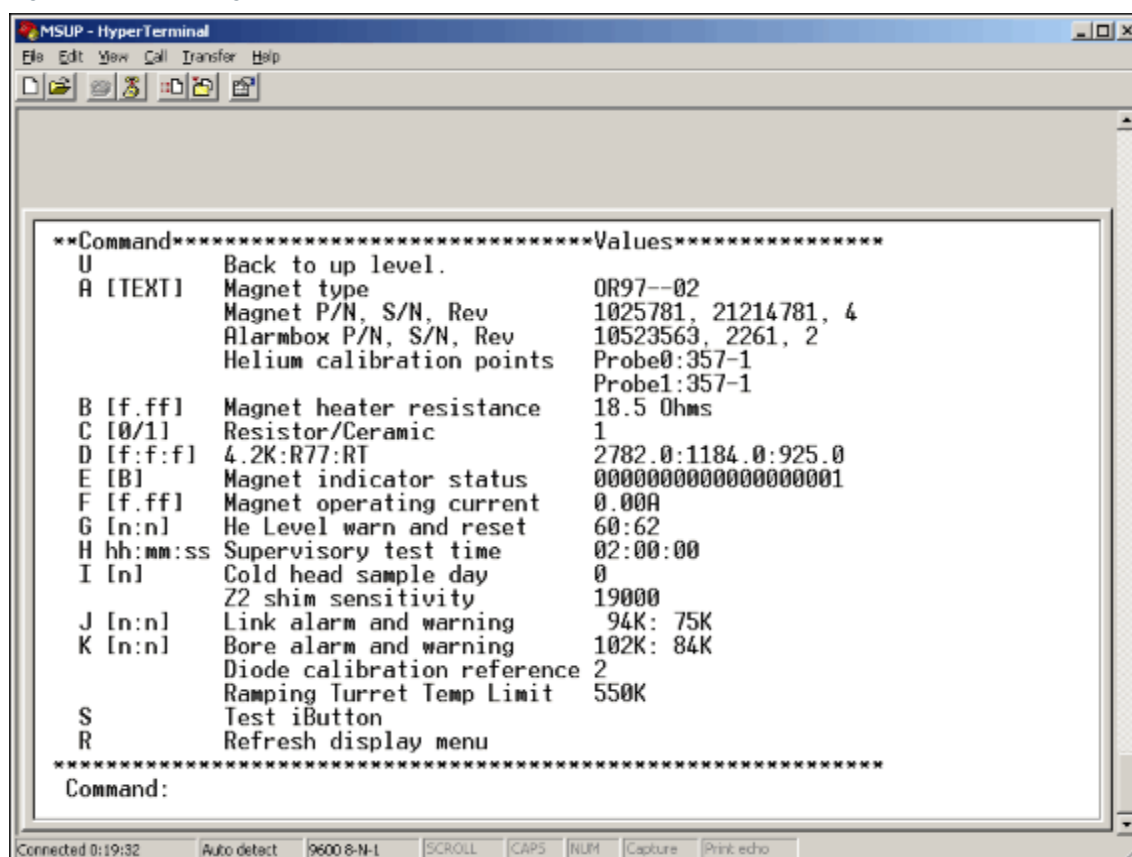
msup - HyperTerminal
File Edit View Call Transfer Help

--Command-----Syntax-----
R          Run Process Display.
B          Enter iButton menu.
T hh:mm:ss Set clock time.
D dd-mm-yy Set date.
xSN xxxxxxxx (A)larmbox, (M)agnet or (S)upervisory Serial Number.
xPN xxxxxxxx (A)larmbox, (M)agnet or (S)upervisory Part Number.
xRN xxx     (A)larmbox, (M)agnet or (S)upervisory Revision Number.
Hx nnn     Probe (1) or (2) ADC reading 100% He level.
Zx nnn     Probe (1) or (2) ADC reading 0% He level.
HT         Start Helium level test.
COMP ON/OFF Turn Compressor ON / OFF.
EIS ON/OFF  Turn EIS ON / OFF.
PH ON/OFF/NM Turn Pressure Heater ON / OFF / Normal Mode.
VA         View Log Files.
VQ         View Quench Log File.
VH         View Calibration data.
RD         Reset LVQD Triggered Channel.

-----
Command: B_
  
```

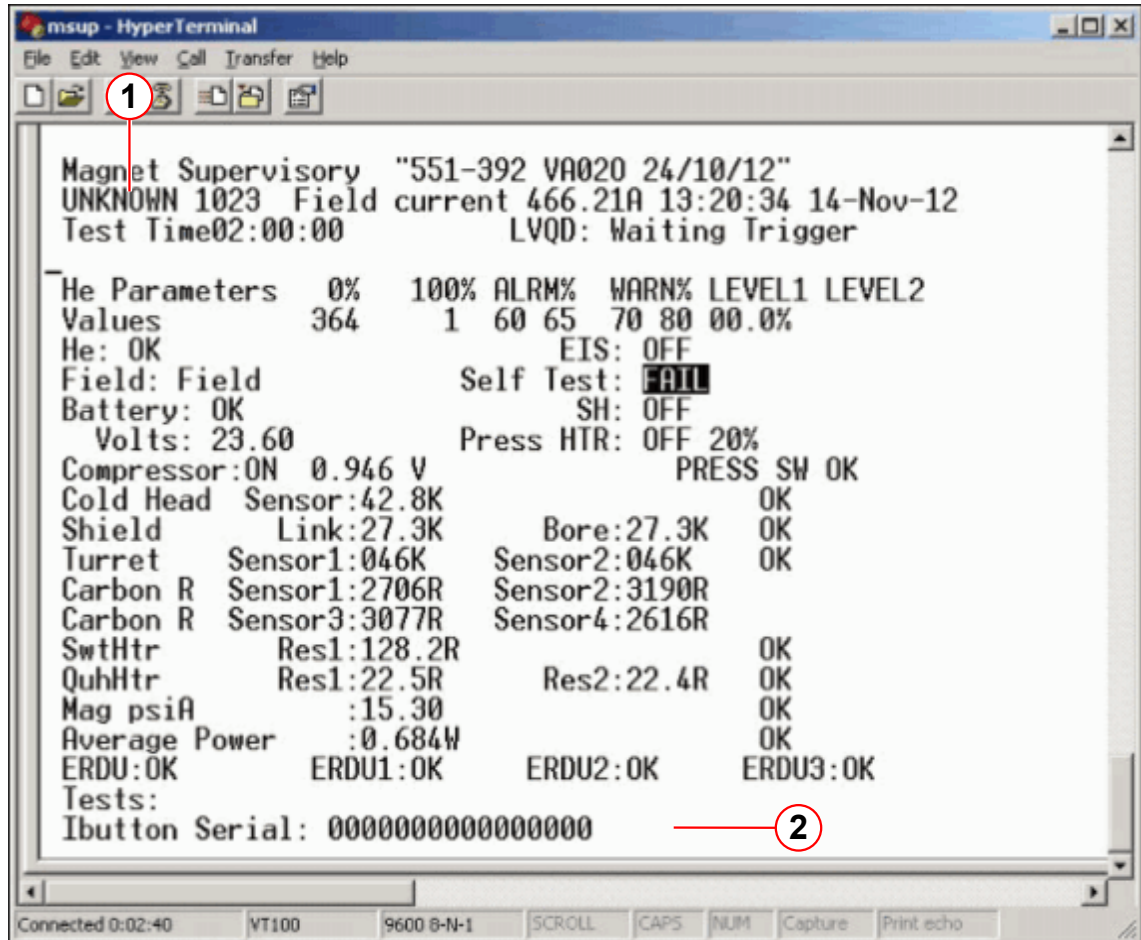
Connected 0:16:57 VT100 9600 B-N-1 SCROLL CAPS NUM Capture Print echo

Fig. 5: IButton Configuration



- Command: **R** <enter>

Fig. 6: IButton Configuration



- (1) Magnet Code
- (2) IButton Serial number

- Enter the correct magnet type.



Use always 8 characters for the correct format of the magnet code, e.g. OR105-00 (8 characters), OR97--01 (8 characters).

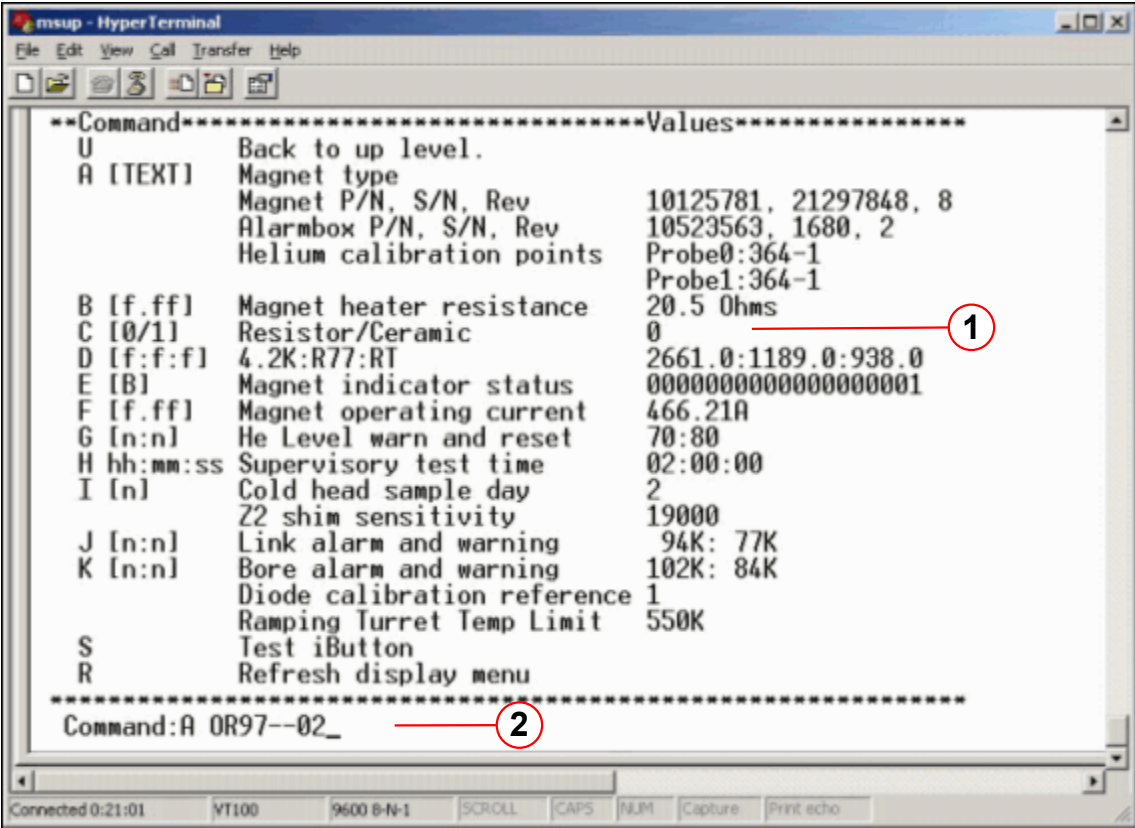
Tab. 1 Magnet code OR105 (1.5T)

Magnet code	Explanation	Command
OR105-00	OR105 with standard carbons, ES <42, Serial no. < 21241848. Cold value about 1000 Ohms.	A OR105-00 (8 characters for magnet code)
OR105-01	OR105 with ceramical carbons, ES >41, Serial no. > 21241847. Cold value about 3000 Ohms.	A OR105-01 (8 characters for magnet code)

Tab. 2 Magnet code OR97 (3T)

Magnet code	Explanation	Command
OR97--01	OR97 with standard carbons, ES <2, Serial no. < 21199820. Cold value about 1000 Ohms.	A OR97--01 (8 characters for magnet code)
OR97--02	OR97 with ceramical carbons, ES >1, Serial no. > 21199819. Cold value about 3000 Ohms.	A OR97--02 (8 characters for magnet code)

Fig. 7: IButton Configuration



- (1) Resistor type
- (2) Magnet Code
- Reset the MSUP
 - » Power off/on the MSUP with F40

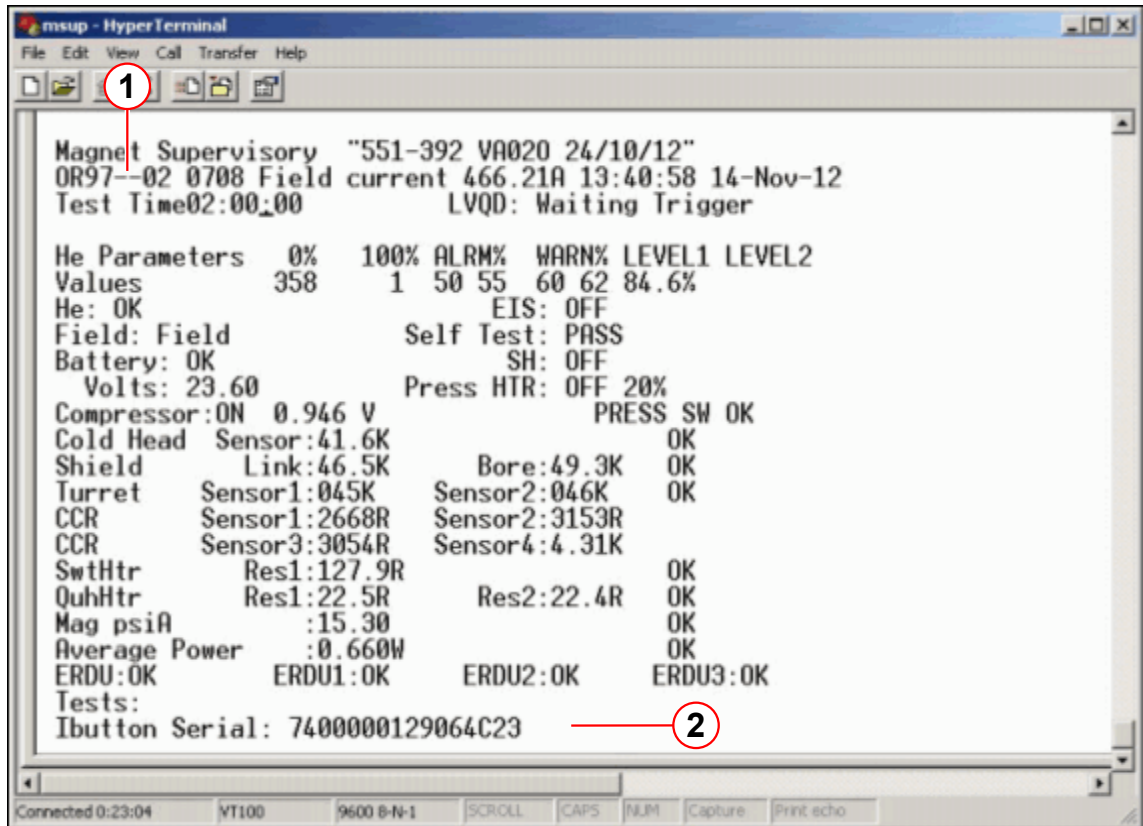
i Power off/on only required if MSUP-LW < VA02F!

- Check the IButton serial number (→ 2/Fig. 8 Page 11)



Only "0" as IButton serial number indicates an access problem to the IButton.
(→ 2/Fig. 6 Page 9)

Fig. 8: IButton Configuration

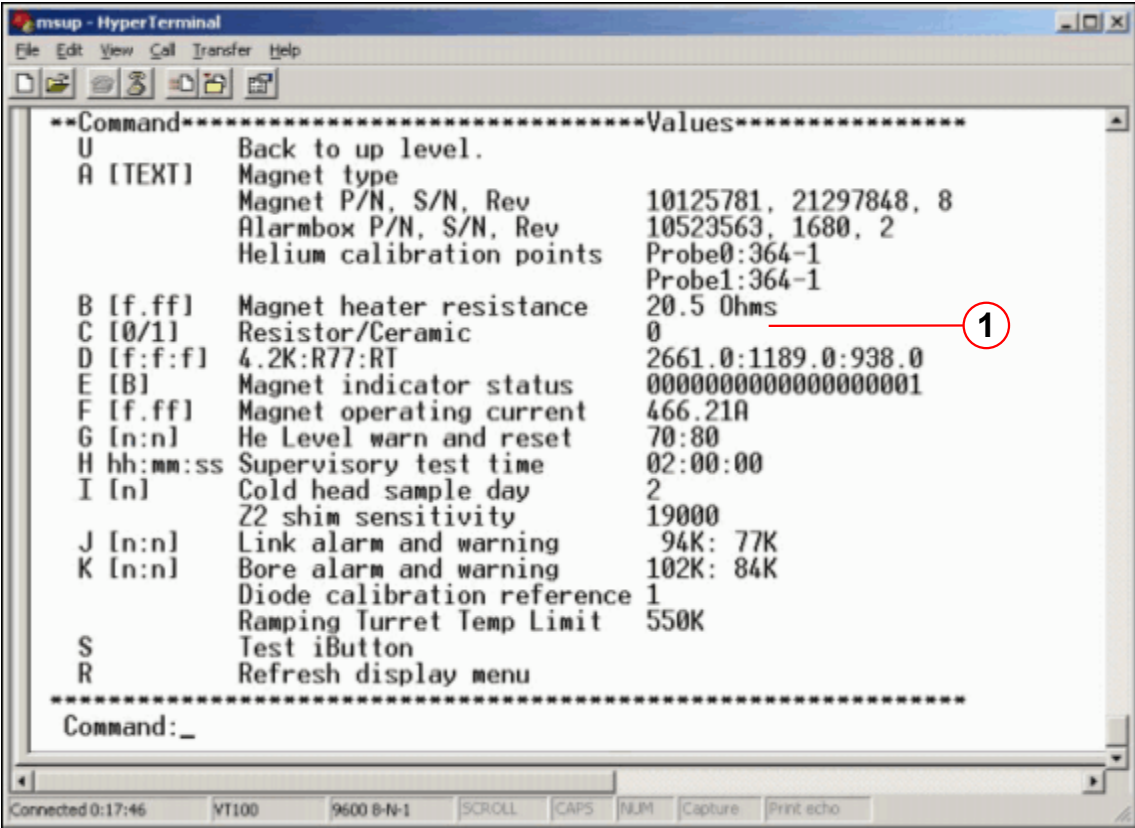


- (1) Magnet Code
- (2) IButton Serial number

- Check the Magnet type. (→ 1/Fig. 8 Page 11)

- Checking the correct resistor type setting. (→ 1/ Fig. 10 Page 13)(→ 1/ Fig. 7 Page 10)

Fig. 9: IButton Configuration



- (1) Resistor type
- Type the correct resistor type.

Tab. 3

Resistor type	Magnet type	command	CCR calibration values
Standard carbons	OR105-00 OR97--01	c 0	Default values in the MSUP LW
Ceramical carbons	OR105-01 OR97--02	c 1	Refer to (→ 1/ Fig. 12 Page 16)

- Check the correct setting. (→ 1/ Fig. 10 Page 13)

Fig. 10: IButton Configuration

```

msup - HyperTerminal
File Edit View Call Transfer Help

*****Command*****Values*****
U      Back to up level.
A [TEXT] Magnet type          OR97--02
        Magnet P/N, S/N, Rev  10125781, 21297848, 8
        Alarmbox P/N, S/N, Rev 10523563, 1680, 2
        Helium calibration points Probe0:358-1
                                   Probe1:358-1
B [f.ff] Magnet heater resistance 18.5 Ohms
C [0/1]  Resistor/Ceramic        1
D [f:f:f] 4.2K:R77:RT           2661.0:1189.0:938.0
E [B]      Magnet indicator status 00000000000000000001
F [f.ff]   Magnet operating current 466.21A
G [n:n]    He Level warn and reset 60:62
H hh:mm:ss Supervisory test time  02:00:00
I [n]      Cold head sample day    2
        Z2 shim sensitivity      19000
J [n:n]    Link alarm and warning  94K: 75K
K [n:n]    Bore alarm and warning  102K: 84K
        Diode calibration reference 2
        Ramping Turret Temp Limit 550K
S          Test iButton
R          Refresh display menu

*****
Command:
  
```

(1) Resistor Type

Check the Magnet parameters

- Open Seso / Magnet & Cooling / Magnet Setup

Fig. 11: Magnet Setup

Magnet & Cooling

Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

Actions

- Cooling Status
- Magnet Status
- Compressor Status
- Cooling Setup
- Magnet Setup

Magnet & Cooling - Magnet Setup

Magnet

Helium Level Probe 1

Min. (0%)

85.5

Ohm

Max. (100%)

0.2

Ohm

Helium Level Probe 2

Min. (0%)

85.5

Ohm

Max. (100%)

0.2

Ohm

Helium Level Probe Selection

☒ Probe 1

☐ Probe 2

☐ Both Probes

Helium Level Warning

Min.

55

%

Shield Temperature Warning 50K Link

Max.

75.0

K

Shield Temperature Warning 50K Bore

Max.

84.0

K

CCR Calibration

Room Temperature

938.0

Ohm

Nitrogen Temperature

1189.0

Ohm

Helium Temperature

2661.0

Ohm

Pressure Heater Resistance

18.5

Ohm

Time Setting

Daily at

02:00

O'Clock

MSUP

Serial Number

1627

Revision Number

7

Part Number

10523587

Alarmbox

Serial Number

1680

Revision Number

2

Part Number

10523563

Magnet

Magnet Type

OR97--02

Serial Number

21297848

Revision Number

8

Part Number

10125781

Magnet Current

466.21 A

- (1) CCR Calibration
- (2) SMT Reference Number

- Check the parameters with the Magnet Acceptance Documentation data sheet.



Magnets without standard carbon resistors do not have **TDC Calibration Values** in the magnet acceptance documentation. The **SeSo / Magnet&Cooling / Magnet Setup** shows default values for the **CCR Calibration**.

- Helium Level Probes (Ohms)
- CCR Calibration values (→ 1/ Fig. 12 Page 16)
- Pressure Heater Resistance

Tab. 4 Pressure Heater Resistance

Magnet type	Resistance (Ohm)
OR97	18.5
OR105	18.3

Fig. 12: Magnet Acceptance Documentation

SIEMENS Magnet Acceptance Documentation
Installation Data
MR Magnet Technology Quality Regulations

Cryostat Number 21296456 **SMT Reference** 21297848 06.11

2

System Description
2.894T Actively Shielded Superconducting Twin Turret Magnet System **OR97**

Measured Magnet Centre (Offset) mm
Gradient Coil Position mm
Overall System Length mm

This system requires rings and iron to shim. This is a theoretical calculation that may change due to site conditions.

Operating Parameters

Description	Specification value	Measured value
Operating Current @ 2.894T	< 500 AMPS	466.09 AMPS
Field/Current Ratio	N/A	26.44 kHz/0.1 Amp

Carbon Resistor Values

POSITION	Room Temperature	Nitrogen	Helium Temperature
CCR4 TDC (D-E)	980		2578
CCR3 Top (C-E)	1059		3053
CCR2 Middle (B-E)	1066		3182
CCR1 Bottom (A-E)	959		2689
TDC Calibration Values:	938	1189	2621

1

Helium Level Probe Resistance (When using a 250 mA supply)

	Probe 1	Probe 2
0% Resistance	85.35	85.35 Ohms
0 % Calibration	774	774
100 % Resistance	0.22	0.22 Ohms
100 % Calibration	2	2
Diode Table No	-2	
Pressure Heater Res	20.5 Ohms	

Final Temperature Sensor Recon Values

50Kbore	50K Link	Coldhead	Recon Margin
67	56.7	45.8 K	651 Milliwatts

Checked by Signature

Date

Page 1 of 1 25-AD EPS 000 01

- (1) CCR Calibration Values
(2) Magnet Serial Number

- Check the correct serial numbers
 - Magnet (→ 2/Fig. 12 Page 16)
 - Avanto^{fit} OR105, Siemens part number: 7758167
 - Skyra^{fit} OR97, Siemens part number: 10125781
 - MSUP
 - Alarmbox
- Enter the Alarmbox P/N, SN, RN

Alarmbox	Command
Alarmbox Part number	APN xxxxxxxx
Alarmbox Serial number	ASN xxxxxxxxxx
Alarmbox Revision number	ARN xxx

Verifying the electronic low pressure switch (PressContrInterlCheck) is part of the filling procedure.

If the magnet does not require a helium fill, the test must be performed as a stand-alone procedure.

This procedure may be carried out in conjunction with any other process involving depressurization of the helium vessel, i.e fitting or removing the service syphon, cold head replacement.



If the magnet does not require a helium fill at start-up, the Low Pressure Switch must be tested.

Refer to (→ Verifying the electronic low pressure switch / Page 55) in the Filling Procedure

3.1 Quench tube check

- Perform a visual inspection of the quench tube.
 - » Refer to (→ Quench tube / M6-024.831)
- Check if the Installation protocol is signed by the Quench line manufacturer.



Ensure that the quench line installation is complete before performing the first ramping procedure.

The quench line acceptance protocol must be checked before the first ramp.

Check that the Quench Line Acceptance Protocol is signed off by the Quench Line Manufacturer.

The acceptance protocol can be found under:

Product Information Home > Planning > Planning Guides > MR Systems > Quench Line Design and Calculation Tool > Quench Line Calculation Tools

3.2 Filling the static system (OR105)

3.2.1 General

The magnet design does not include a permanently fitted syphon. A short magnet service syphon must be fitted when filling with liquid helium. The service syphon is **tooling** only and must be **removed immediately** after filling has been completed.

This procedure can be performed on magnets **at field** and magnets **at zero field**.

Verifying the low pressure switch



If the magnet does not require a helium fill at start-up, the Low Pressure Switch must be tested.

Refer to (→ Verifying the electronic low pressure switch / Page 55) in the Filling Procedure

Bottom filling for zero or low helium levels



To recover a potentially warm magnet, refer to magnet temperatures in the Bottom filling procedure (→ Bottom filling the magnet / Page 48). Contact Magnet HSC for advice.

3.2.2 Safety



Check that the Quench line installation is complete and the magnet connected before helium filling.



WARNING

Strong magnetic field.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Only non-magnetic equipment may be taken into the magnet room.

**WARNING**

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogen.

- » Oxygen monitors must be activated at all times.
- » Do not vent helium gas continuously into the magnet room during helium transfer.
- » Check room ventilation I turned on.

**CAUTION**

Cryogenic liquids.

Failure to observe the following safety warning may result in cold burns.

- » Protective clothing, gloves and eye protection must be worn.
- » Filling must be performed only by trained personnel.

**CAUTION**

Pressurized helium vessel.

Failure to observe the following safety information may result in physical injury.

- » Do not disconnect any of the turret venting or fittings without first depressurizing the helium vessel.

**CAUTION**

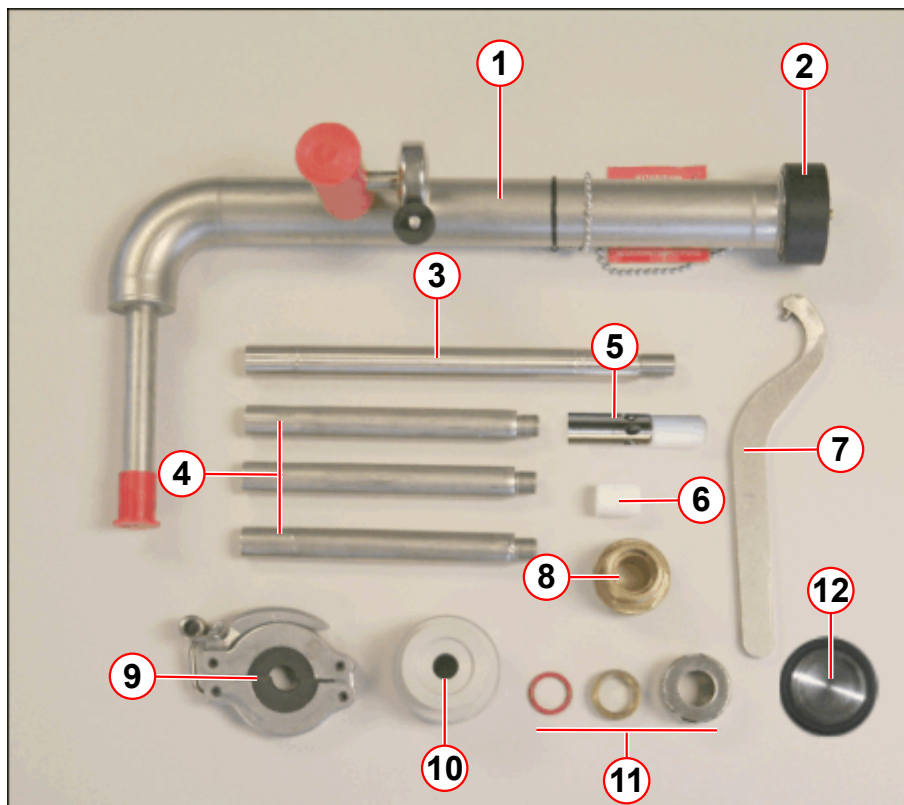
Risk of quench during helium filling

If the magnet quenches:

- » Exit room and wait 15 to 20 minutes for magnet pressure to fall.
- » Stop the helium transfer as soon as it is safe.
- » Follow Post Quench procedures
- » Report quench and fill in quench database

3.2.3 Equipment required

Fig. 13: Magnet service syphon kit



- (1) Service syphon assembly
- (2) Syphon end cap
- (3) 200 mm extension leg (1 off)
- (4) 140 mm extension legs (3 off)
- (5) 6 hole helium fill diffuser tip (top fill)
- (6) 22mm PTFE seal (bottom fill)
- (7) Syphon C-spanner/wrench
- (8) M28 brass syphon adaptor bush (OR105, OR122)
- (9) Syphon leg support clamp (low ceiling)
- (10) Nitrogen fill adaptor and O-ring seal
- (11) Syphon O-ring seal, brass syphon flange, syphon ring nut
- (12) NW25 flow restricter

- Magnet service syphon kit, (part no **10126651**).
- Dewar syphon.
- Dewar neck adaptor, clamps, seals, and hose.
- Heat gun.
- Paper towelling.
- Helium gas supply (at least 99.9% pure).
- Pressure regulator, (capable of 12 psi (G)).
- Protective clothing: cryogenic gloves, safety glasses, face shield, lab coat, safety shoes.
- Replacement syphon seal, (part number **03108094**).

- Service platform.
- Gas bottle wrench.
- Syphon 'C' spanner, (part number **03108102**)
- Personal Oxygen monitor.

3.2.4 Preparation

Site preparation

- Ensure warning signs are correctly positioned.
- Ensure there is a clear path to exit the imaging suite.

System preparation

- **Remove** the appropriate system **covers** to access the magnet turret and MSUP.
- **Position** the **Service platform** next to the magnet.
 - Secure the platform to the magnet, using the 2 x long straps through the magnet bore.
- Check that there is **open and clear access** to the magnet for **helium dewar(s)**.
- Check that there is **open and clear access** to the **syphon**.
- Take a helium sample.

3.2.5 Helium level check

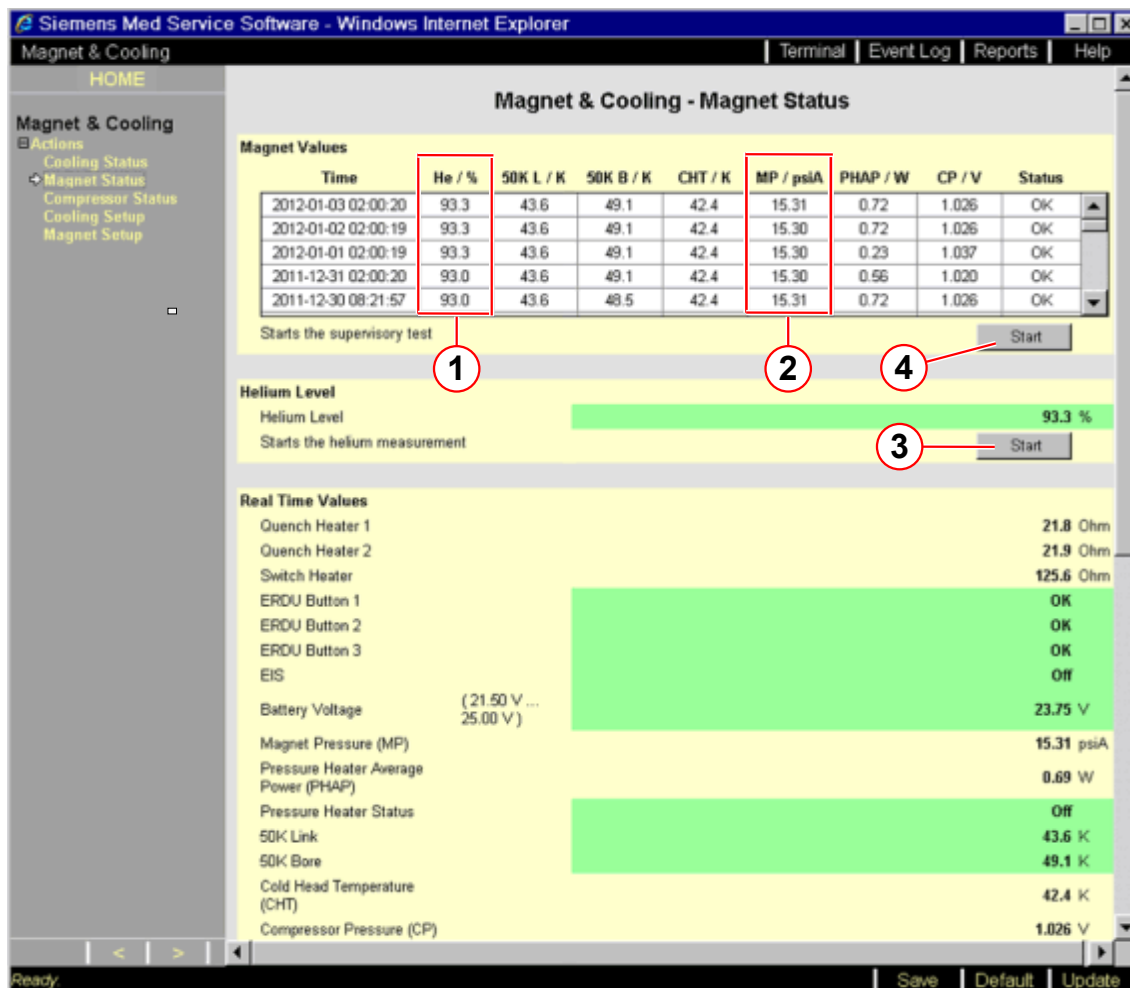
- During installation, if the **MRC (MRAWP) Host** is not available, take a helium sample from the **MSUP**.
 - If there is no site power available a 36 volt external power supply must be fitted to power the MSUP.
 - A laptop PC can also be attached to the MSUP RS232 using simulation software, (for example 'hyperterminal').
 - Refer to Troubleshooting guide.

3.2.5.1 Helium level check at the MRC Host

- Access the **Service Software** at the MRC (MRAWP) Host.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- Press the **Start** button to initiate the helium measurement.
 - This will perform a new helium sample. The MSUP LEDs will be refreshed.

- Press the **Start** button to update the magnet values.
 - This will refresh the screen values.

Fig. 14: Checking liquid helium and magnet pressure levels



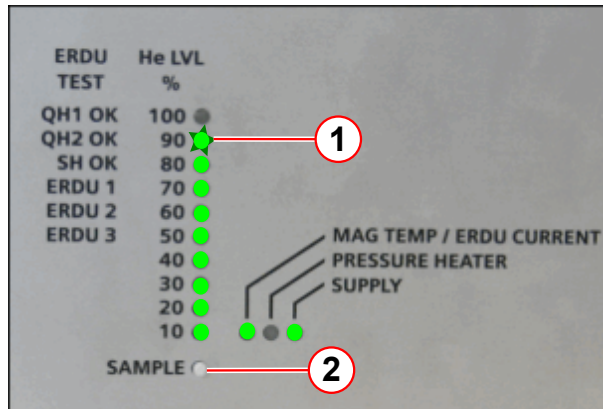
- (1) Liquid helium level
- (2) Magnet pressure
- (3) Start button - helium measurement
- (4) Start button - magnet value refresh



Data is only accurate from the last refresh of the page, or from when the page was opened last.

3.2.5.2 Helium level check at the MSUP

Fig. 15: MSUP - Helium level sample



- (1) Helium level
(2) Sample switch

- Locate the **SAMPLE** switch on the MSUP lid.
- Depress the sample switch for 2 s, and release.
- The green LEDs illuminate and display the actual liquid **helium level percentage**.

3.2.6 Magnet depressurization



Before opening the magnet bypass valve take note of the magnet **gauge pressure**.

Depressurizing the magnet will take:

- 20 to 30 minutes with a magnet site at sea level.
- Up to 2 hours depending on the altitude of the magnet site above sea level.



Depressurization time can be utilized to prepare the helium dewar and Service syphons.

Verifying the electronic low pressure switch

- The functionality of the low pressure switch can be checked during magnet depressurization for the helium fill:-
- Check when pressure in the helium vessel drops to 0.2 psi (G) above atmospheric pressure, the electrical pressure switch opens and turns off the compressor.
- Check when the helium vessel pressure rises to 0.2 psi (G), the switch closes and the compressor restarts.
- At normal operating pressure check at the MRC Host that the pressure has returned to 15.3 psi (A).

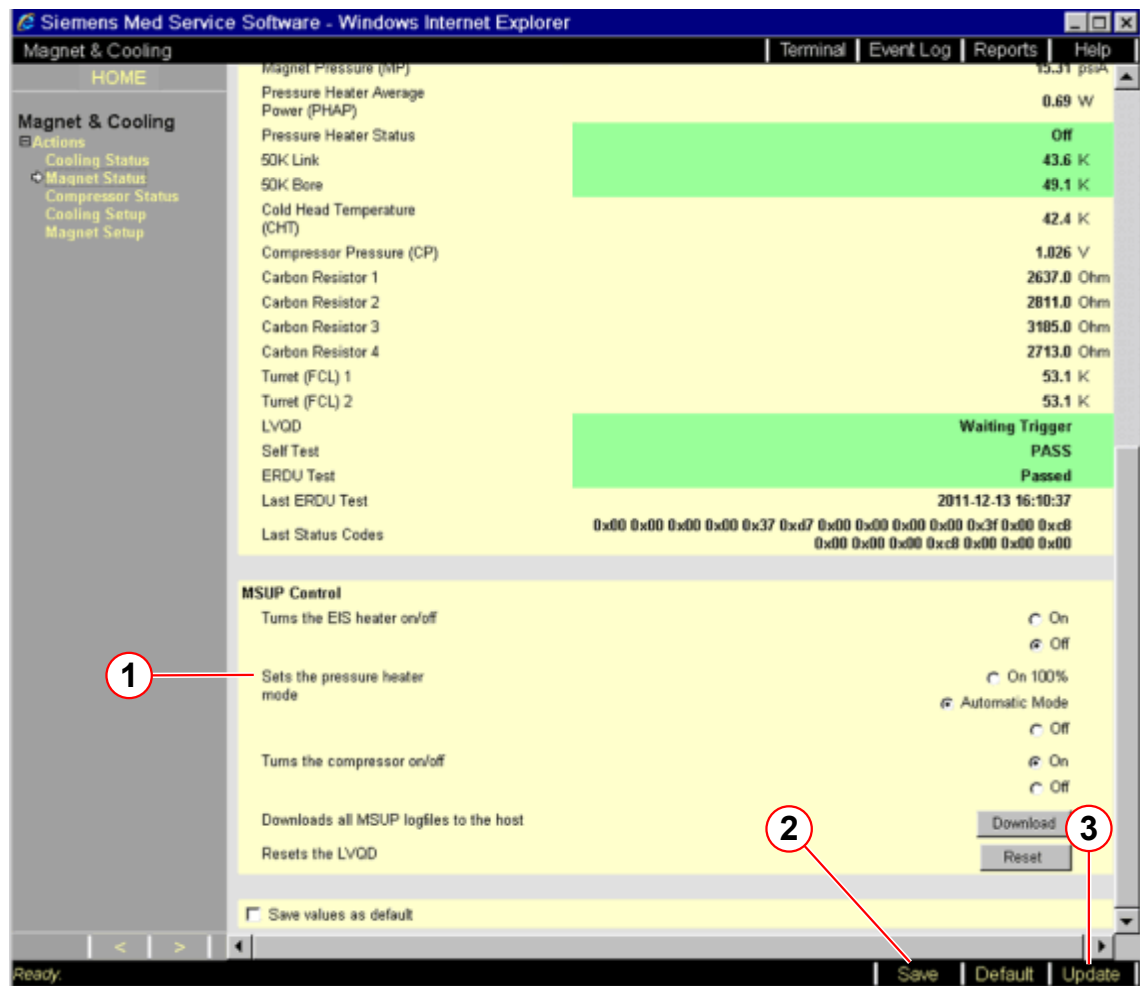
Pressure heater de-activation

To de-activate the pressure heater for filling procedure:-

- Access the service software at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.

- At **MSUP Control** - Sets the pressure heater mode select the **OFF** button.
- Select **SAVE**.

Fig. 16: Service screen Mag&Cooling - Magnet Status - MSUP Control



- (1) Sets the pressure heater mode
- (2) Save
- (3) Update



The pressure heater will automatically activate after 60 minutes.

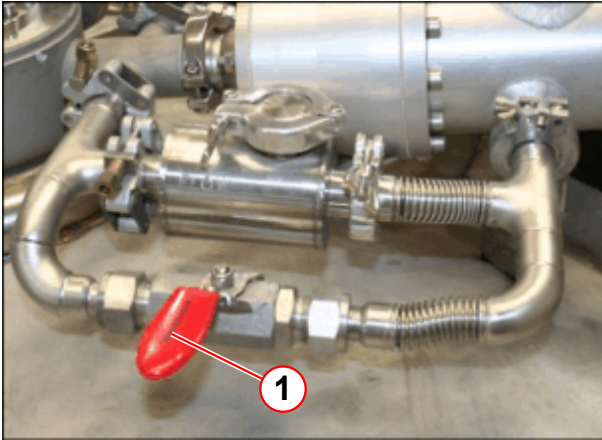
To depressurize the magnet:-

- **Fully open** the magnet bypass valve.
- Check at 0.2 psi (G) above atmospheric pressure the compressor stops.
- Wait until zero psi is displayed on the pressure gauge.
- Warm up the vent assembly using a heat gun.

- Close the magnet bypass valve, (Pos. 1).

Mk1 Vent valve assembly

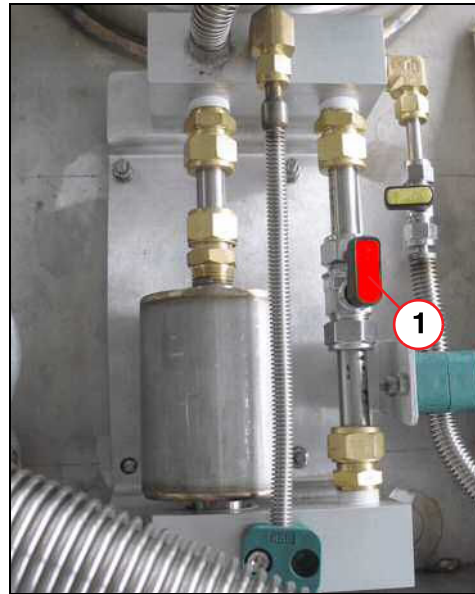
Fig. 17: Mk1 Vent valve assembly



(1) Bypass valve

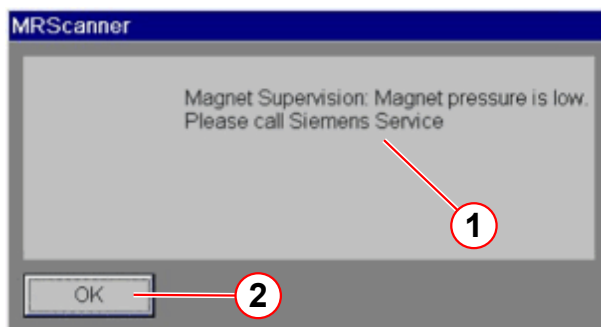
Mk2 Vent valve assembly

Fig. 18: Mk2 Venting bypass valve



(1) Bypass valve (show open)

Fig. 19: MSUP - Magnet pressure is low



(1) Pop-up message
(2) Click OK to clear

At the MR Host:

- A pop-up **alarm message** is displayed.
 - » **Magnet pressure is low.**
 - » Click **OK** to clear the alarm.
- The **alarm box** sounds when **pressure falls** to alarm limits.
 - » The yellow **WARNING LED** lights on the **alarm box**.
 - » The alarm **buzzer** is **activated**
 - » Press **AUDIO ALARM OFF** to silence the alarm.



CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmosphere.

- » Always ensure that the cold head bypass valve is **CLOSED** during helium filling.

3.2.7 Dewar preparation

This procedure describes the use of **top fill** helium dewars used in conjunction with the Siemens Magnet Technology dewar syphon.

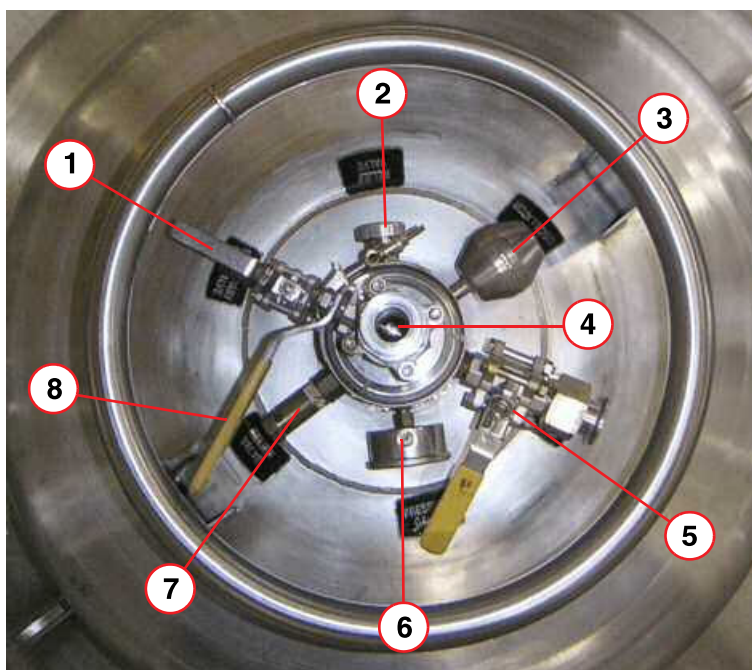
The procedure covers all dewars that are externally pressurized with helium gas.



Refer to the manufacturer's documentation when using side fill dewars and dewars fitted with electric pressure builders.

Refer to the dewar manufacturer's safety guidelines and check the dewar valve functions before commencing the LHe fill.

Fig. 20: Dewar Ports (typical)



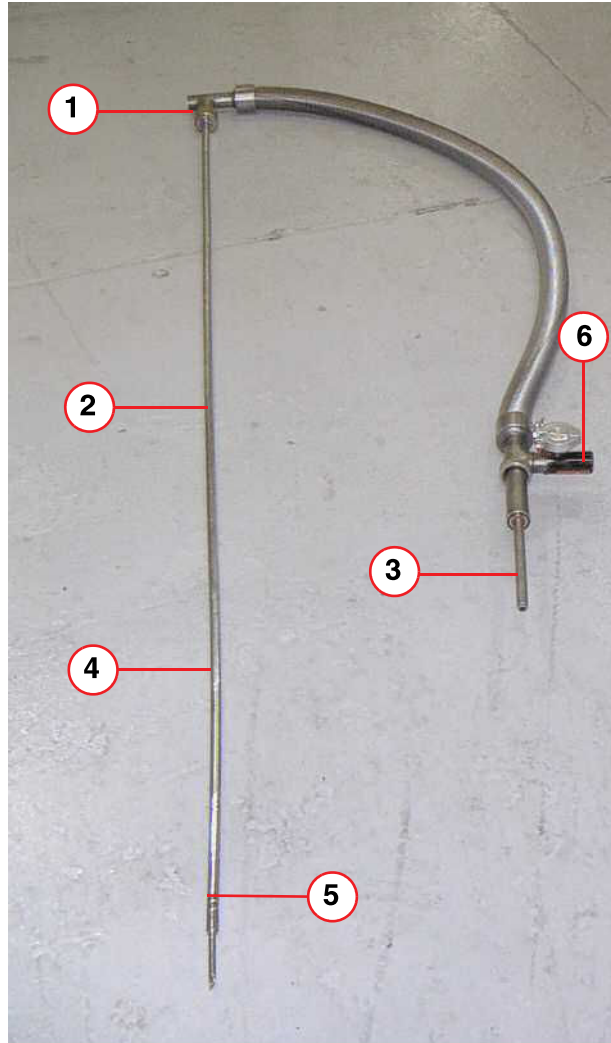
- (1) Low pressure relief valve (shown open)
- (2) Relief valve
- (3) Anti-oscillator
- (4) Syphon entry port (shown closed)
- (5) Primary ball valve (shown closed)
- (6) Pressure gauge
- (7) Relief valve
- (8) Syphon entry valve

- If the dewar is **highly pressurized**, (greater than 0.3 bar (4.4 psi)):
 - **Slowly open** the primary ball valve release the dewar pressure.
 - Depressurization must be done in a **well ventilated area - NOT in the imaging suite.**
 - Check for the cause of the high pressure. Inform the supplier if there are faulty dewar valves.
- Fit the **dewar syphon adaptor** to the top of the dewar.

- Close the primary ball valve and low pressure relief valve.

3.2.7.1 Dewar syphon preparation

Fig. 21: Dewar syphon



- Fit the extension leg (s) and sludge inhibitor, to the dewar **syphon leg**.

- (1) Syphon flow valve
- (2) Syphon leg
- (3) Bayonet connection (male)
- (4) Extension leg
- (5) Sludge inhibitor
- (6) Syphon vacuum valve

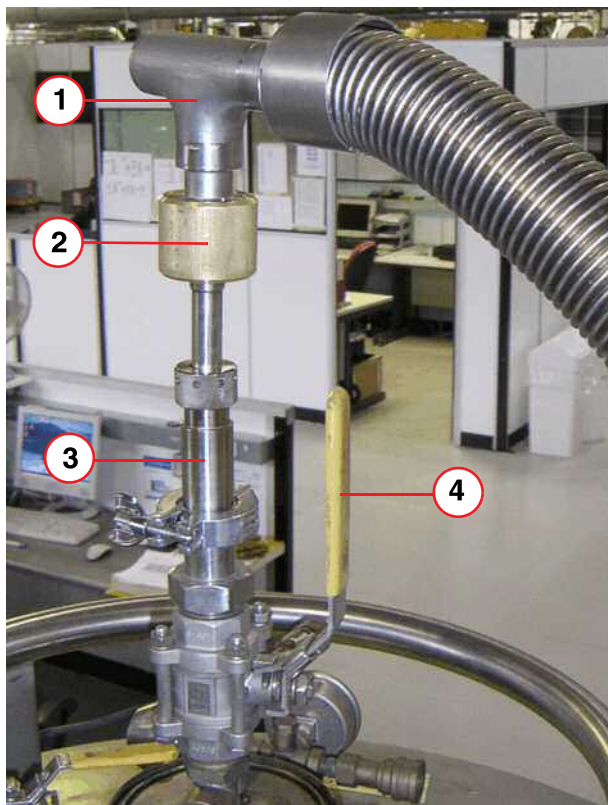


Do not open the syphon vacuum valve (6).

- The helium dewar must be positioned close to the magnet with the dewar syphon fitted, ready for use.
- There must be no excessive bending of the syphons.

3.2.7.2 Precool the dewar syphon

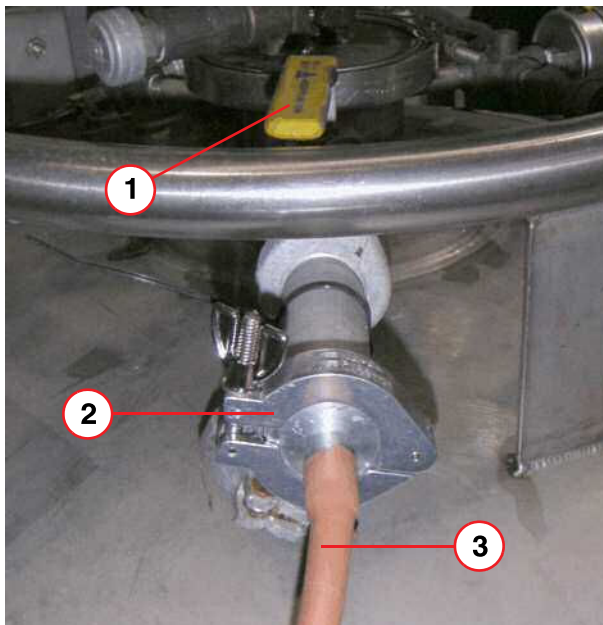
Fig. 22: Dewar syphon fitment



- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

- **Open the syphon flow valve** (pos 2), two full turns.
- Insert the syphon leg into the **dewar syphon adaptor** (pos 3), and tighten the ring nut and seal in order to create a gas seal that the syphon can slide through.
(Do not tighten the ring nut at this stage).
- **Open the syphon entry valve** (pos 4), and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.
- When the sludge inhibitor touches the bottom, **fully** tighten the ring nut and seal.
- The syphon will now be purged. The syphon flow valve (pos.2) can be **closed**.
- Open the dewar relief valve if required to keep pressure at 4–5 psi (G).

Fig. 23: Dewar gas connection

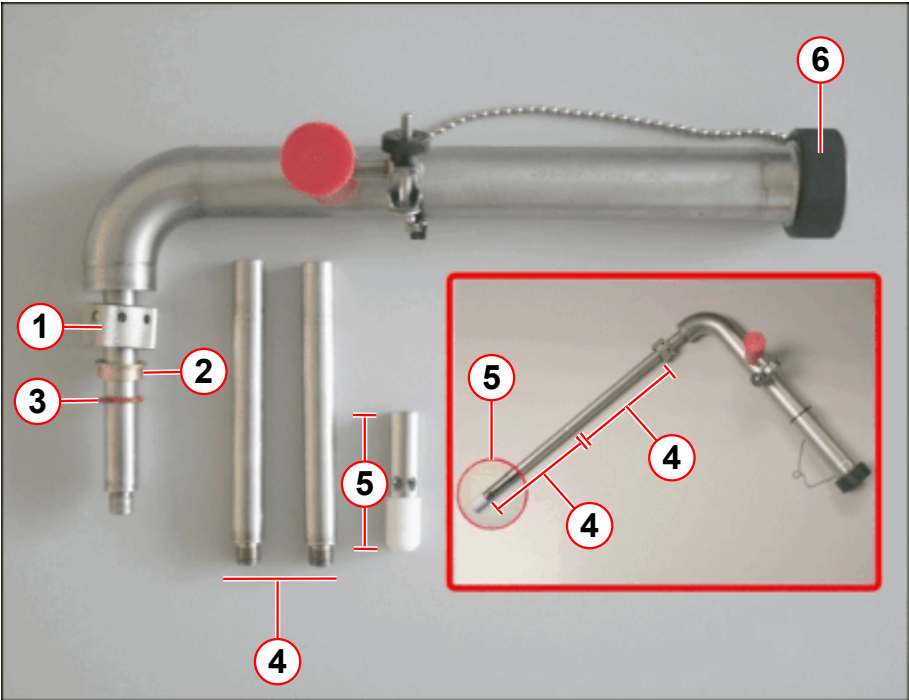


- (1) Primary ball valve
- (2) NW25 clamp and seal
- (3) Gas supply line

- Position the helium gas cylinder in a safe position outside the magnet room.
- Fit the helium regulator to the gas bottle.
- Ensure that the **gas supply line** (pos 3) is long enough to allow the dewar to be moved close to the magnet if the magnet is at field.
- **Purge** the helium **gas supply line** of air.
- Connect it to the **primary ball valve** (1) on the dewar, using an **NW25** clamp and seal (pos 2).
- **Open** the primary ball valve and pressurize the dewar to **0.27 bar (4 psi (G))**.

3.2.8 Service syphon assembly and fitment

Fig. 24: Magnet service syphon assembly - top filling



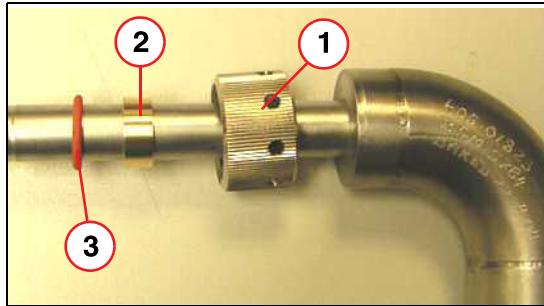
- (1) Ring nut
- (2) Brass flange
- (3) O-ring seal
- (4) 140 mm syphon legs (2 off)
- (5) De-ice and helium fill diffuser and PTFE tip
- (6) Syphon end cap

Tab. 5 OR105 TOP filling syphon extension legs

Magnet type	140 mm leg	60 mm De-ice & helium fill diffuser & PTFE tip
Avanto (OR105)	2	1
The syphon diffuser is optimally at 50-80 mm above the syphon cone for top filling . If damaged, the PTFE diffuser tip must be replaced.		

- Check the **ceiling height**. (→ Low ceiling height / Page 47)
 - Assemble the **syphon** legs.

Fig. 25: Assembling the Ring Nut Seal



- (1) Ring nut
- (2) Brass spacer
- (3) Silicon O-ring

- Fit the ring nut, brass spacer flange, and the O-ring to the **service syphon leg**.

Connect the extension legs and diffuser to the syphon:

- Screw the (2 off) **140 mm legs** together.
- Fit the 60 mm **diffuser** and PTFE tip.

Fig. 26: Syphon end cap



- **Remove the syphon end cap.**



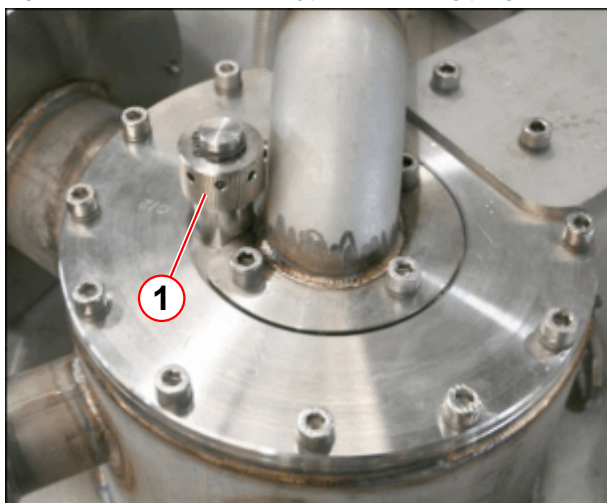
CAUTION

Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

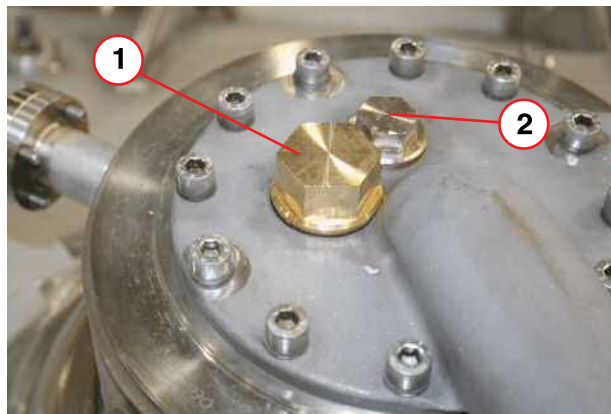
Fig. 27: Current and Mk1 Syphon blanking plug



(1) Syphon blanking plug

- Warm and dry the **syphon blanking plug** using a heat gun
- Remove the **syphon blanking plug** using the syphon 'C' spanner.

Fig. 28: Mk2 Syphon port entry



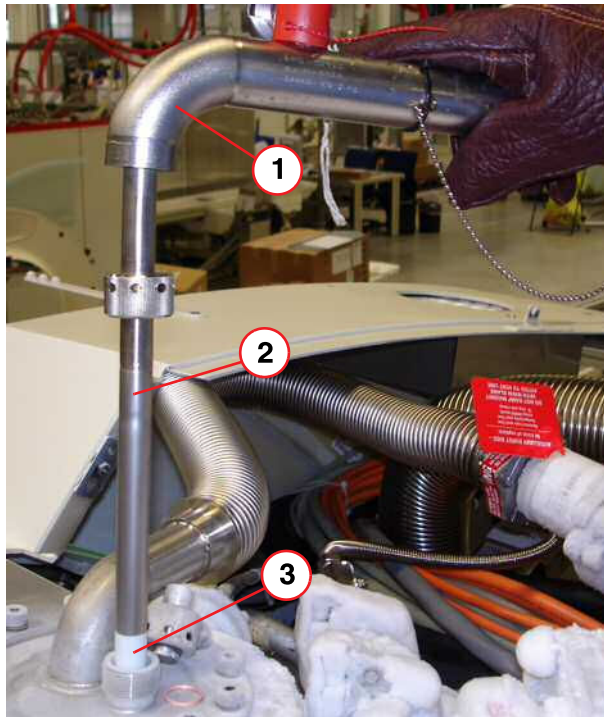
(1) Syphon blanking plug
(2) Sight glass

Fig. 29: Mk2 Syphon port port



(1) Syphon blanking plug

Fig. 30: Helium syphon



- (1) Service Syphon
- (2) Two Extension Legs
- (3) PTFE Syphon Diffuser

- **Insert** the service syphon into the port without delay.
- Slowly lower the service syphon fully into the magnet.
- Tighten the **syphon port ring nut** by hand, then apply a half turn using the syphon C-spanner to form a **gas tight seal**.
 - Do not overtighten. The O ring seal will still allow the syphon to move.
- Re-fit the syphon end cap.

**CAUTION****Risk of quench**

Warm helium gas can cause a quench.

- » If the magnet is at field both syphon's must be pre-cooled before joining together.
- » Frost or condensation on the syphon body is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

3.2.9 Helium transfer - top fill

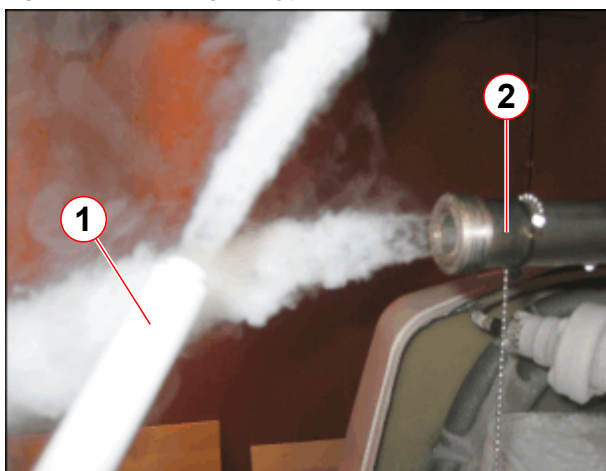
The cool down of the dewar and service syphon must be performed at the same time. With less pressure in the magnet the service syphon will take longer to cool down.

Dewar syphon cool down

- Check the **dewar gas supply pressure** is set at **0.27 bar (4psi (G))**.
 - Open the **dewar syphon flow valve** half way (a full 2 turns).
 - **Precool the syphon** until a dense white "**cascading plume**" appears.
 - **Close** the flow valve. Wait for the service syphon to complete its cool down.

Service syphon cool down:-

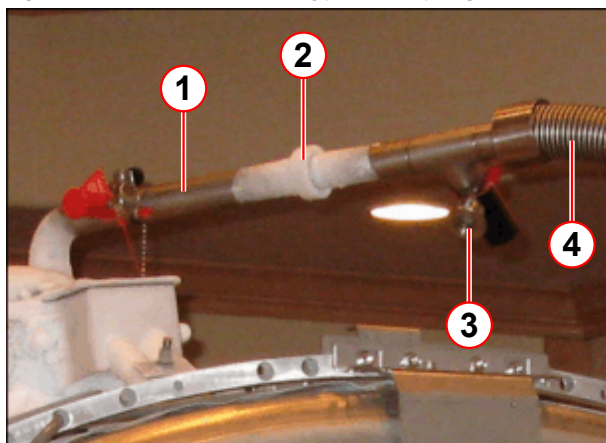
- Check the end cap is fitted to the service syphon.
- Check the magnet bypass valve is **closed**.
- If required activate the **pressure heater** to achieve a positive pressure of approximately **0.3 psi (G)** in the magnet.
- **Deactivate** the pressure heater, (if used to build pressure).

Fig. 31: Pre-cooling the syphons

(1) Dewar syphon
(2) Service syphon

When there is sufficient magnet pressure to cool the service syphon:-

- **Remove the syphon end cap.**
- Allow the syphon to vent until **fully cold**.
 - » a white “**cascading plume**” of helium gas appears.
- As the service syphon nears the point of being fully cold:-
 - » **Open** the dewar syphon flow valve half way.

Fig. 32: Dewar and service syphon coupling

(1) Service syphon
(2) Lock nut
(3) Dewar syphon vacuum valve
(4) Dewar syphon

- When **both** syphons are giving a dense white “**cascading plume**” of helium gas, connect them together and secure the lock nut.
- **OPEN** the magnet bypass valve. This will:-
 - » Improve efficiency.
 - » Improve the filling speed.

Fig. 33: Helium filling



- **Fully open** the dewar syphon flow valve until the guide pins are flush with the top of the stainless steel body.-
- Check and maintain the dewar pressure at 4 psi (G).

Dewar transfer pressure

The optimum pressure differential between magnet and dewar for a helium transfer is 4 psi (G). To calculate the required filling pressure on the helium dewars:-

- Take the magnet operating gauge pressure **before depressurizing** the helium vessel.
- Add 4 psi for the required transfer pressure from the dewar.



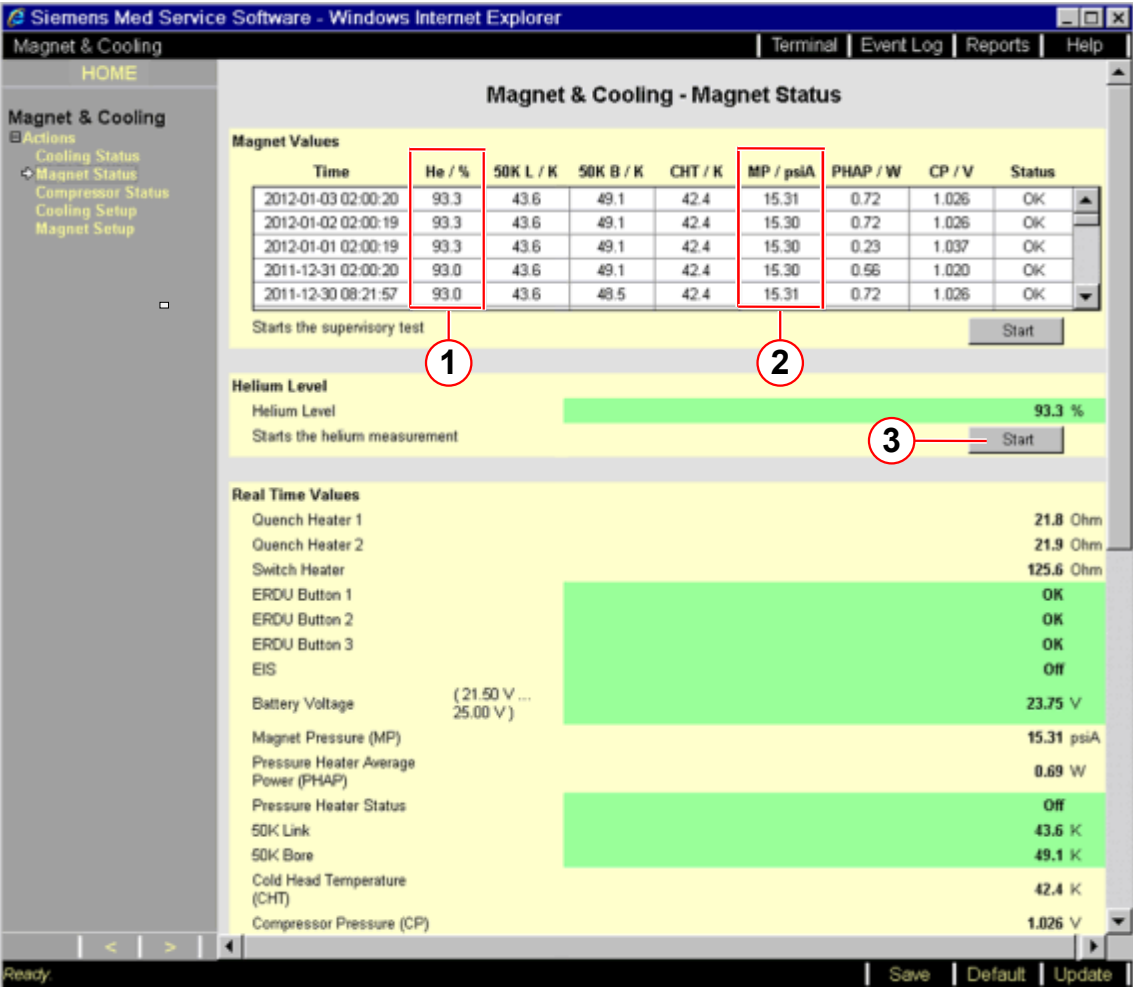
Check manufactures specifications for maximum dewar pressure.

3.2.10 Checking the helium fill level

- Check the **liquid helium level** by using **Service Software** on **MRC (MRAWP) Host.** or from the **MSUP**.
 - Access the **Service Software** at the MRC.
 - Select the menu item **Magnet & Cooling - Magnet Status**.

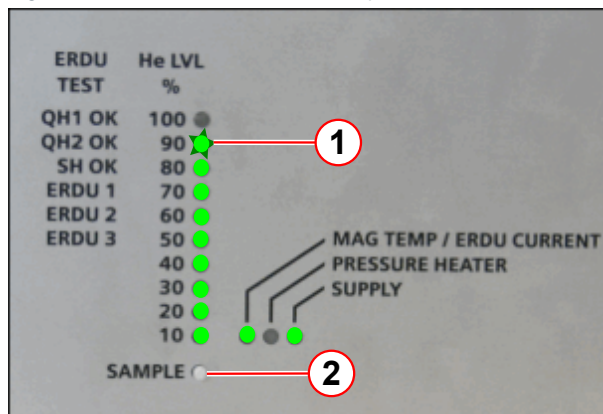
- Press the **Start** button to initiate the helium measurement.

Fig. 34: Checking liquid helium and magnet pressure levels



- (1) Liquid helium level
- (2) Magnet pressure
- (3) Start button

Fig. 35: MSUP - Helium level sample



- (1) Sample switch
(2) Helium level

- Locate the **SAMPLE** switch on the MSUP lid.
- Depress the sample switch for 2 s, and release.
- The green LEDs illuminate and display the actual liquid **helium level percentage**.
- If the 90% LED is flashing, **STOP** the transfer.

Tab. 6 MSUP Helium display

MSUP LEDs	Helium level	Comments or actions
0 - 80% ON	As displayed	Continue filling.
90% ON	As displayed	Stop transfer.
*90% FLASHING 10 - 80% ON	95% - 97%	STOP the transfer IMMEDIATELY!
*100% FLASHING 10% - 90% ON	97% and over	Magnet is OVERFILLED .
* The flashing LEDs return to ON after approximately 30 seconds.		

- Maintain a pressure of 0.27 bar 4 psi (G) on the helium dewar.
 - » Too much or too little pressure will result in inefficiency.
- Repeat the helium measurement **every 5 minutes**.
 - » An increase in level should be seen within 20 minutes.
- If not, **stop the transfer** and investigate the reason for poor transfer efficiency.

Poor fill efficiency can be caused by:

- Internal ice in the turret syphon port.
- Condensation or frost along the length of the syphon(s) indicating a poor vacuum. A replacement syphon will be required.
- Partially obstructed or blocked quench pipe.
- Partially obstructed or blocked syphon due to no diffuser fitted to the dewar syphon.

At 4 psi (G), the transfer rate is approximately 10 liters per minute.

- » **Do not leave the transfer unattended.** A 250 liter dewar will empty in 25 - 30 minutes.
- » Continue the helium fill until the desired helium level has been reached.

3.2.11 Stopping the helium transfer

- As soon as the helium level reads 95%, or the dewar is empty, **stop the transfer** by closing the flow valve at the dewar syphon.
- Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
- The dewar syphon flow valve must be **closed** immediately the dewar empties - if left open there is a risk of quenching.
- Set the dewar gas pressure regulator to zero.

NOTICE

Risk of quench.

Warm helium gas can cause a quench.

- » **If the magnet is still at field, stop the transfer immediately the syphon whistles (a high pitched sound), or the dewar pressure drops quickly.**

- **Close** the dewar primary ball valve.
- Remove the gas line from the dewar.
- If the magnet is not full, refer to (→ Changing supply dewars during transfer / Page 42)

3.2.12 Removing the service syphon

- Allow the helium vessel to depressurize before removing the service syphon..

Fig. 36: Warm up the syphon joint with heat gun.



- **Warm** the connecting **syphon joint** to room temperature using a **heat gun**.

- Warm the syphon **entry valve** to room temperature and dry.
- Warm the syphon **end cap** ready to fit onto the syphon.

CAUTION

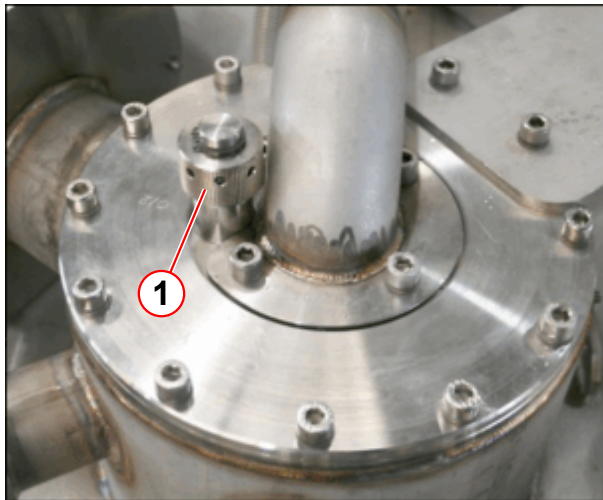
Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

- Close the magnet bypass valve.
- Disconnect the two syphons.
- Close off the end of the service syphon with the **syphon end cap**.
- Undo the turret syphon ring nut.
- Remove the service syphon.

Fig. 37: Current and Mk1 Syphon blanking plug



(1) Syphon blanking plug

- **Immediately fit** the syphon blanking plug using a **new copper seal**.
- To ensure a **leak tight seal**, warm up the syphon ring nut and fully tighten with the 'C' spanner.

Pressure heater activation

- Pressurize the helium vessel to 15.3 psi (A).
- Access the Service Software at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- At MSUP Control - Sets the pressure heater mode select the On 100% button.
- Select **SAVE**.
- Select Update to refresh data.
- At MSUP Control - Sets the pressure heater mode select the **Automatic mode** button.
- Select **SAVE**.

3.2.13 Changing supply dewars during transfer

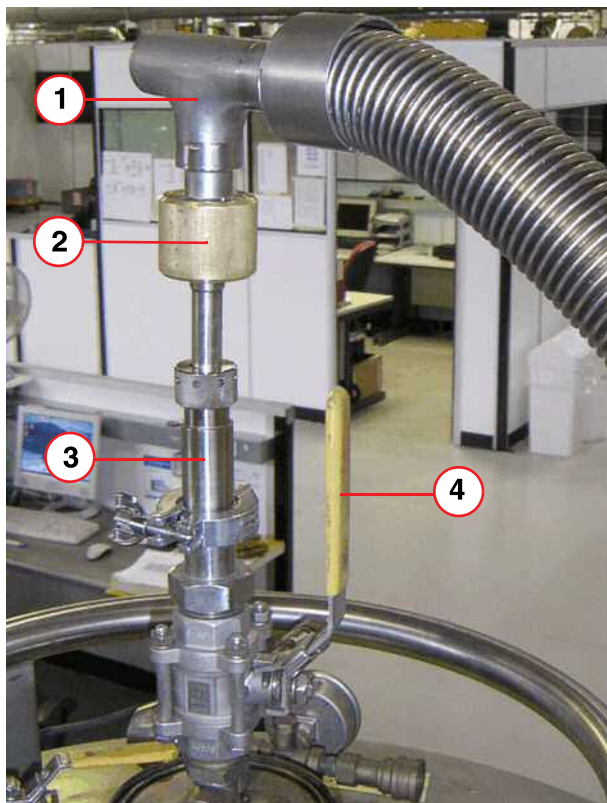
**CAUTION**

Risk of quench.

Failure to close the syphon flow valve immediately, when the magnet is at field, will result in a quench.

- » Close the syphon flow valve immediately.

Fig. 38: Dewar syphon fitment



- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

If the supply dewar empties before the magnet is filled, replace the supply dewar with a full one.

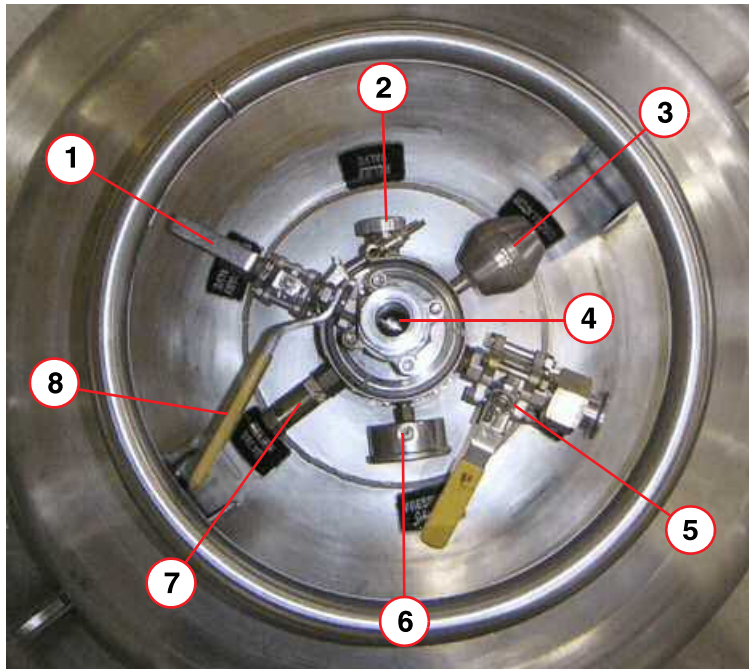
- Ensure that a full dewar is available and depressurized, ready to receive the syphon.
- When the dewar empties, stop the transfer **immediately** by closing the dewar syphon flow valve (pos 2).

Turn off the dewar gas supply:

- Close the primary ball valve and remove the **gas connection**.

- Vent the dewar down to atmospheric pressure in a well ventilated area.

Fig. 39: Dewar Ports (typical)



- (1) Low pressure relief valve (shown open)
- (2) Relief valve
- (3) Anti-oscillator
- (4) Syphon entry port (shown closed)
- (5) Primary ball valve (shown closed)
- (6) Pressure gauge
- (7) Relief valve
- (8) Syphon entry valve

- Using a heat gun, **warm** the **syphon connecting joint** to room temperature.
- **Disconnect** the syphons.



CAUTION

Risk of cold burns.

A pressurized flow of cold gas will exhaust from the syphons when they are disconnected.

- » Stand clear of the syphon joint when disconnecting.
- » Wear protective gloves and safety glasses.

- Remove any moisture from the end of the **service syphon** with paper towelling.
- Fit the **service syphon end cap**. (→ Fig. 26 Page 33)

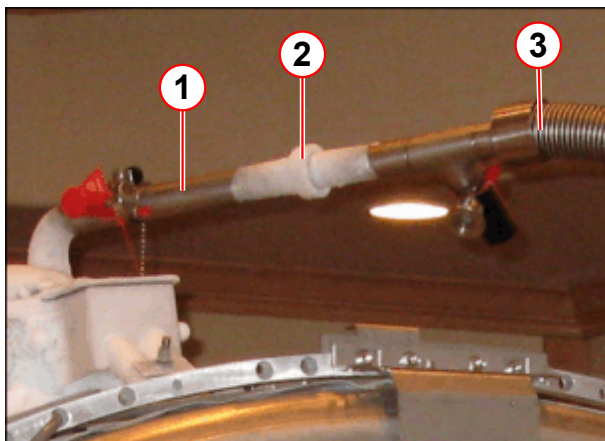


Syphon removal and fitment on top fill dewars where there is restricted ceiling height will need to be performed in a ventilated area outside the magnet room.

- Move the full dewar next to the empty dewar.

- **Open** the low pressure relief valve on the empty dewar to fully depressurize, .
(→ 1/ Fig. 39 Page 43)
- **Remove** the dewar syphon.
- **Close** the syphon entry valve on the empty dewar and **open** the low pressure relief valve.
- Remove the syphon adaptor from the empty dewar and fit onto the full dewar.
- **Close** the low pressure relief valve of the full dewar.
- Open the syphon flow valve two turns.
- Open the syphon entry valve and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.
- **Purge** the gas supply line and connect to the full dewar.
- **Open** the dewar primary ball valve.
- Set the gas pressure to 0.27 bar – 4 psi (G).
- Move the dewar next to the magnet.
- Remove the syphon end cap and precool the service syphon.

Fig. 40: Dewar and service syphon coupling



- (1) Service syphon
- (2) Lock nut
- (3) Dewar syphon

- When **both** syphons are giving a dense white "cascading plume" of helium gas, **connect** them together and **secure** the lock nut.
- Continue with the transfer.

3.2.14 Final work steps

Helium dewar :

- **Move the dewar** out of the imaging suite into a **well ventilated area**.
- Complete **depressurizing** the dewar by opening the **primary ball valve**.
(→ 5/ Fig. 39 Page 43)
- When at the gas pressure is at **zero**, **close** the **primary ball valve** on the dewar.
- **Remove** the dewar syphon.
- **Immediately** close the **syphon entry valve** on top of the dewar.
- **Open** the **low-pressure dewar relief valve**.

- The **dewar valves** are correctly **closed**, with their low pressure relief valves **open**.

After transfer check:-

- The **magnet bypass valve** is **closed**.
- Press the **LVQD reset button** to ensure the LVQD shows **waiting trigger**.

Fig. 41: LVQD reset button

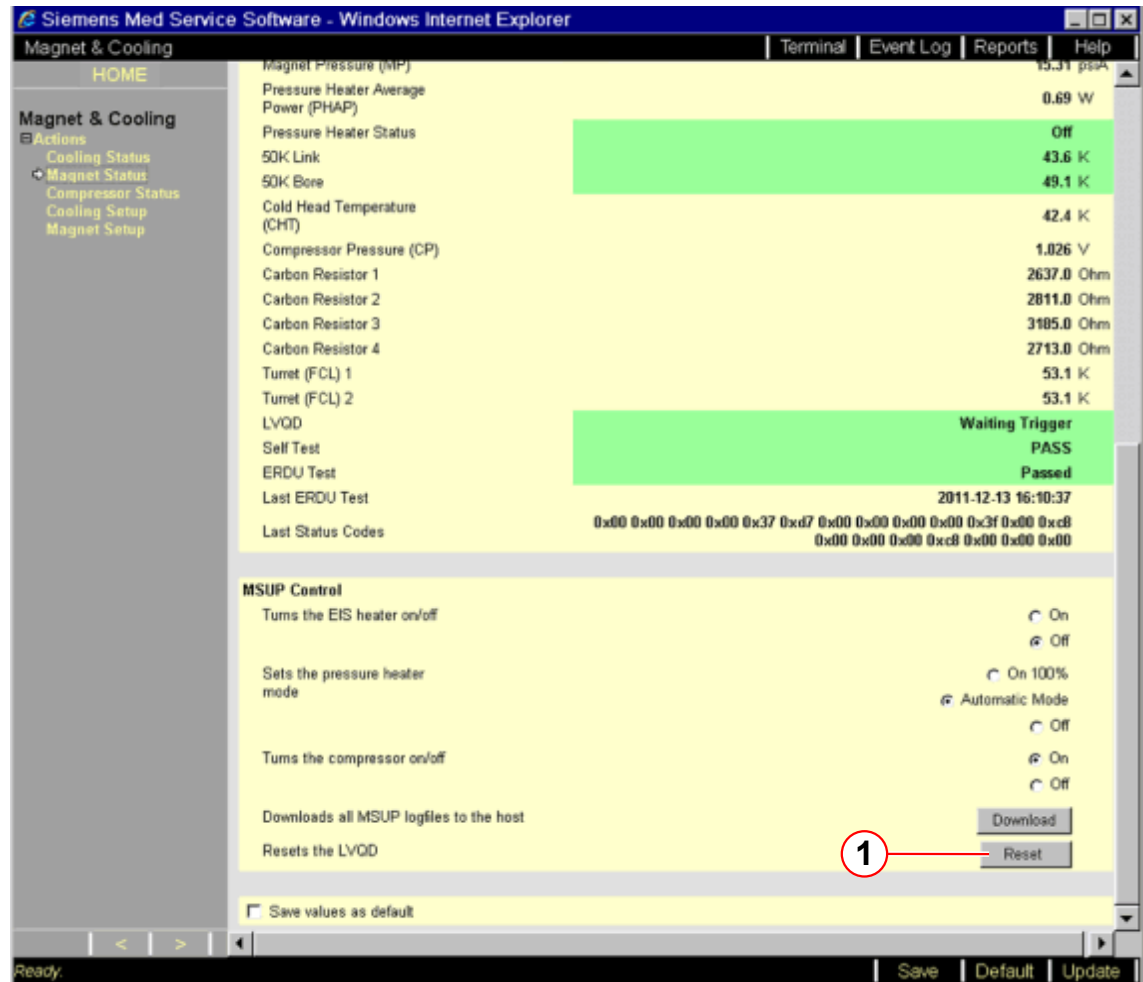
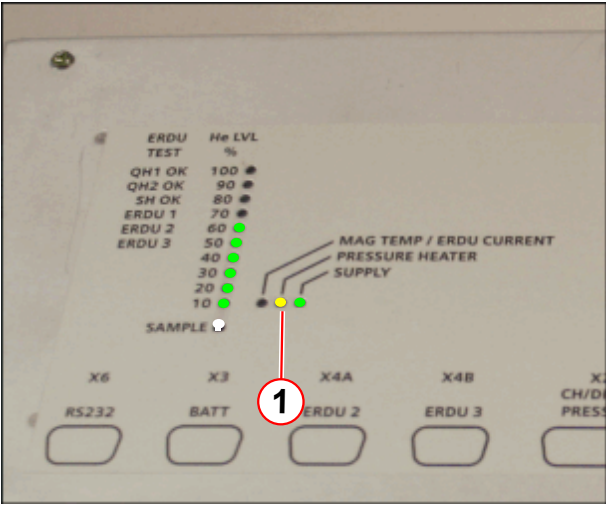


Fig. 42: MSUP - Pressure heater LED



(1) Pressure heater LED - yellow

- Check the pressure heater is operating.
- The pressure heater **LED** on the **MSUP** is illuminated, or switching on and off.

- **No alarm** signals are evident at the **MSUP** or the **alarm box**.
- The **compressor** is switched **on** and **running**. (Note: 20 minute time delay may be active).
- The **magnet pressure** is stable at **15.3 psi (A)**.
- The **turret** and **venting** system are **warm** and **dry**.
- The filling **port** and **venting** system have **no leaks**. (Refer to **Venting Leak Check procedure**).
- **Warm** both **syphons** with the heat gun **until dry**, and store safely.

3.2.15 Helium levels and volumes

The table below provides a guide for the average helium quantities required for filling the helium vessel.

Tab. 7 OR105 Helium level and volumes

Helium probe %	Helium volume liters	Helium required
0	6	1500 ltrs
10	99	1400 ltrs
20	261	
30	452	1025litrs
40	606	
50	732	725 ltrs
60	868	
70	1001	432ltrs
80	1140	

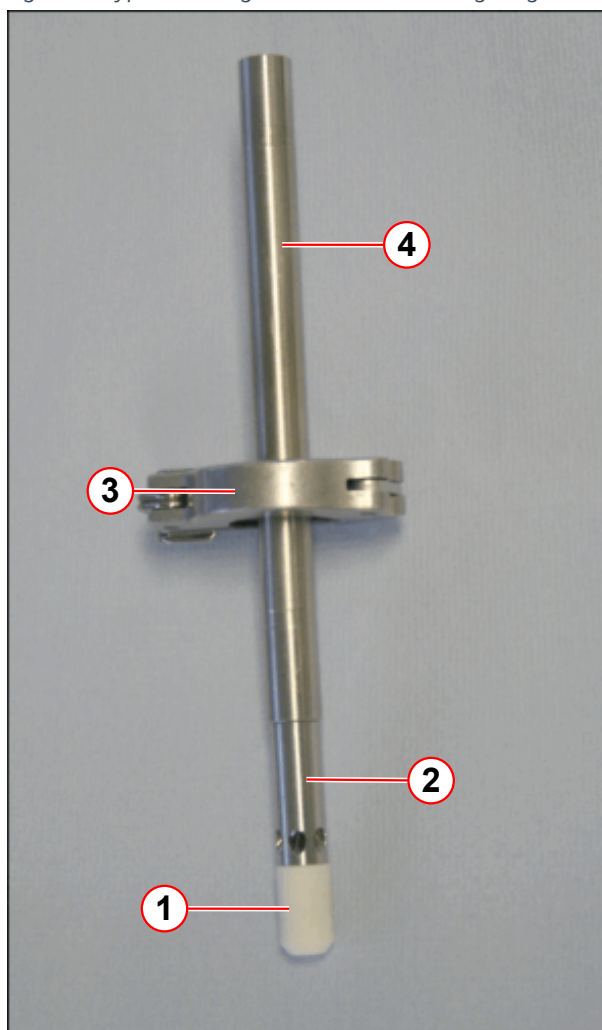
Helium probe %	Helium volume liters	Helium required
90	1274	
100	1400	
Note: 14 Liters corresponds to 1% increase		

3.2.16 Low ceiling height

Where magnets have been installed into rooms with a **low ceiling height**, it can be difficult to fit the assembled service syphon. To avoid dropping the extension leg(s) down into the turret through the syphon port, the **syphon leg support clamp** must be used (→ 3/ Fig. 43 Page 48).

- Check the **ceiling height**:
 - Assemble the diffuser, bottom extension and support clamp

Fig. 43: Syphon configuration for low ceiling height



- (1) PTFE extension tip (top fill)
- (2) Helium fill diffuser tip (top fill)
- (3) Syphon leg support clamp
- (4) Extension leg

Refer to Magnet Service syphon kit

- Remove the **syphon end cap**.
- Ensure the **ring nut, brass spacer and O-ring** are fitted correctly.
- Fit the **diffuser** to the extension leg.
- Fit **support clamp** to the extension leg.
- **Warm** the syphon **blanking plug** to ease removal.
- **Remove** the syphon **blanking plug** using the C-spanner.
- **Lower** the syphon leg(s) into the neck.
- **Screw** on the second **extension leg**.
- **Move** the syphon **clamp** up the extension leg and lower extensions into the entry port, to allow room for the syphon body.
- **Screw** the **diffuser leg** to the syphon body
- **Remove** the syphon **clamp** and slowly **lower the syphon fully** into the syphon entry.
- **Screw** the **syphon port ring** nut into place and **tighten** to make a gas-tight seal.
- **Re-fit** the syphon **end cap**.



Use the support clamp for removing the syphon.

3.2.17 Bottom filling the magnet



The bottom fill procedure must **ONLY** be performed when the magnet is at **ZERO FIELD**.

Bottom filling of the magnet is required if:-

- The liquid helium level displays **zero** percent.
- Check the magnet carbon resistor(CCR) temperatures from the MRC Host or MSUP down load:-
 - The temperatures measured must be **no warmer than** 110 K.
 - Reference measured magnet temperatures to those on the original **Magnet Acceptance Document**.

The magnet must be filled through the **magnet syphon cone** to allow filling from the bottom of the helium vessel.



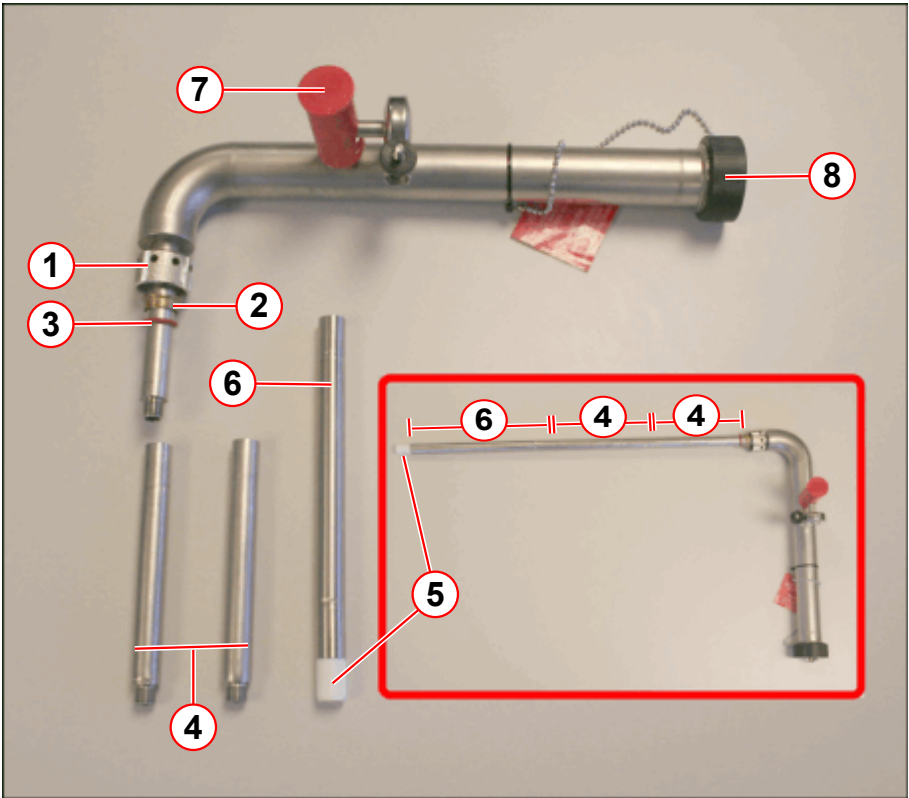
Inform Magnet HSC. Magnet temperatures must be confirmed before continuing with the helium fill.

3.2.17.1 Preparation

- Prepare the **dewar**, (→ Dewar preparation / Page 28) and **dewar syphon**, (→ Dewar syphon preparation / Page 29)
- Check the **ceiling height**. (→ Low ceiling height / Page 47)
 - Assemble the **syphon**.

3.2.17.2 Assemble the service syphon

Fig. 44: Magnet service syphon assembly - bottom filling

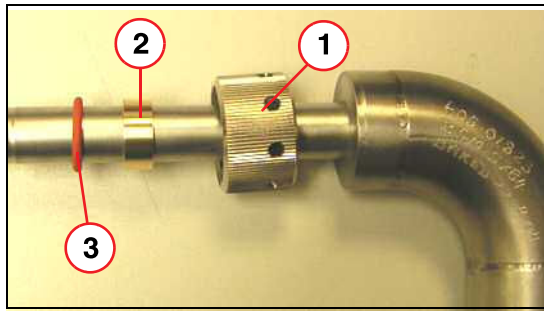


- (1) Ring nut
- (2) Brass spacer
- (3) O-ring seal
- (4) 140 mm syphon legs (2 off)
- (5) 22 mm PTFE seal (bottom fill)
- (6) 200 mm syphon leg (1 off)
- (7) Vacuum valve
- (8) Syphon end cap

Tab. 8 OR105 BOTTOM filling syphon extension legs

Magnet type	140 mm leg	200 mm leg	22 mm PTFE seal
OR105	2	1	1

Fig. 45: Assembling the Ring Nut Seal



- (1) Ring nut
- (2) Brass spacer
- (3) Silicon O-ring

- Fit the ring nut, brass spacer and the O-ring to the **service syphon leg**.
- Connect the **extension legs** and the 22 mm **PTFE seal**:
 - Screw the 2 x **140 mm legs** together, and connect to the **syphon**.
 - Fit the 1 x **200 mm leg**.
 - Fit the **22 mm PTFE seal** to the 200 mm leg.



The minimum length required to reach the syphon cone is:
OR105 - 485 mm (+/- 5 mm)

3.2.17.3 Fit the service syphon

- **De-activate** the **pressure heater**.
- **Depressurize** the helium vessel .
- Note the **pressure** (using the **MRC**, or **Laptop**) when the **compressor** switches **OFF**.
- **Check** the helium vessel has a **small positive pressure**.
- **Warm** and dry the **syphon blanking plug** using a heat gun to ease removal.
- **Dry** the syphon **port**.
- **Remove** the syphon **end cap**.
- **Remove** the syphon **blanking plug**.



CAUTION

Risk of cold burns.

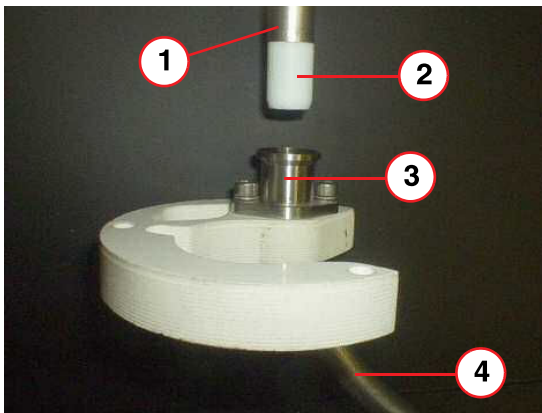
Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

- **Slowly** lower the **service syphon** fully into the neck.

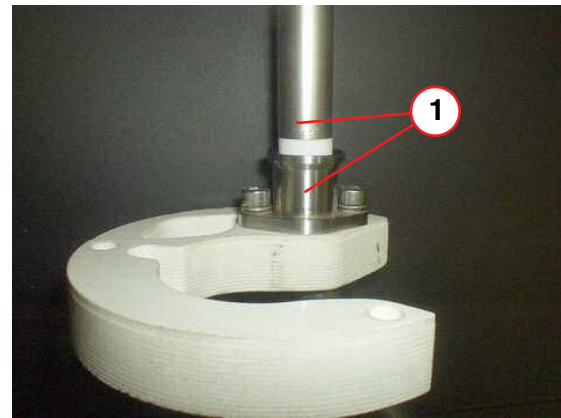
- The **syphon leg** must locate fully into the **syphon cone** - make sure it is **fully engaged**.

Fig. 46: Bottom fill syphon cone (pictorial view)



- (1) Syphon leg
- (2) PTFE Seal (Bottom Fill)
- (3) Syphon cone
- (4) Syphon tube

Fig. 47: Bottom fill - syphon leg fully engaged (pictorial view)



- (1) Syphon leg - fully engaged



Helium efficiency will be reduced if the syphon leg is not fully engaged in the syphon cone.

- Tighten the **syphon port ring nut** by hand, then apply a half turn, (using the appropriate spanner), to form an air-tight seal.
- Re-fit the **syphon end cap**.

3.2.17.4 Bottom fill helium transfer

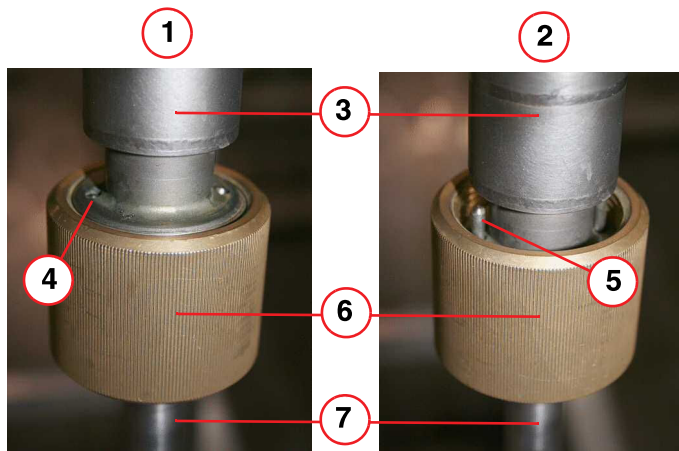


For bottom filling a magnet at **zero field**, there is no need to pre-cool the syphons.

- Position the helium dewar close to the magnet with the dewar syphon fitted.
- **Check** the **dewar gas supply pressure** is set at **0.27 bar (4 psi (G))**.

- **Open** the **dewar syphon flow valve** half way (2 full turns) to **purge** the syphon of air.

Fig. 48: Dewar syphon flow valves - open and closed positions



- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg

Fig. 49: Helium filling



- **Close** the dewar syphon **flow valve**.
- **Remove** the service syphon end cap and **couple the two syphons together** and secure the **locknut**.
- **Open** the dewar syphon **flow valve**.
- Maximize the transfer by fully opening the **dewar syphon flow valve**.
 - » The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body. (→ 4/ Fig. 48 Page 53)
 - » From **fully closed** to **fully open**, is **4 full turns**.
- Ensure the transfer dewar pressure is **4 psi (G)**.

Monitor CCR temperatures

- Monitor the **CCRs** every **15 minutes**, to measure the **temperature drop**.
 - » Refer to the **Acceptance documentation** (Installation Protocol) for required temperature values.
- When the required helium temperature is reached, **monitor** the **helium level** at the **Host** or **MSUP**.

3.2.17.5 Bottom fill completion

When the helium level is visibly rising on the helium level meter complete the fill following the normal '**top**' **fill procedures**.

- Monitor the helium fill level, (→ Helium level check / Page 23)
- Stopping the helium transfer, (→ Stopping the helium transfer / Page 40)
- Removing the service syphon, (→ Removing the service syphon / Page 40)

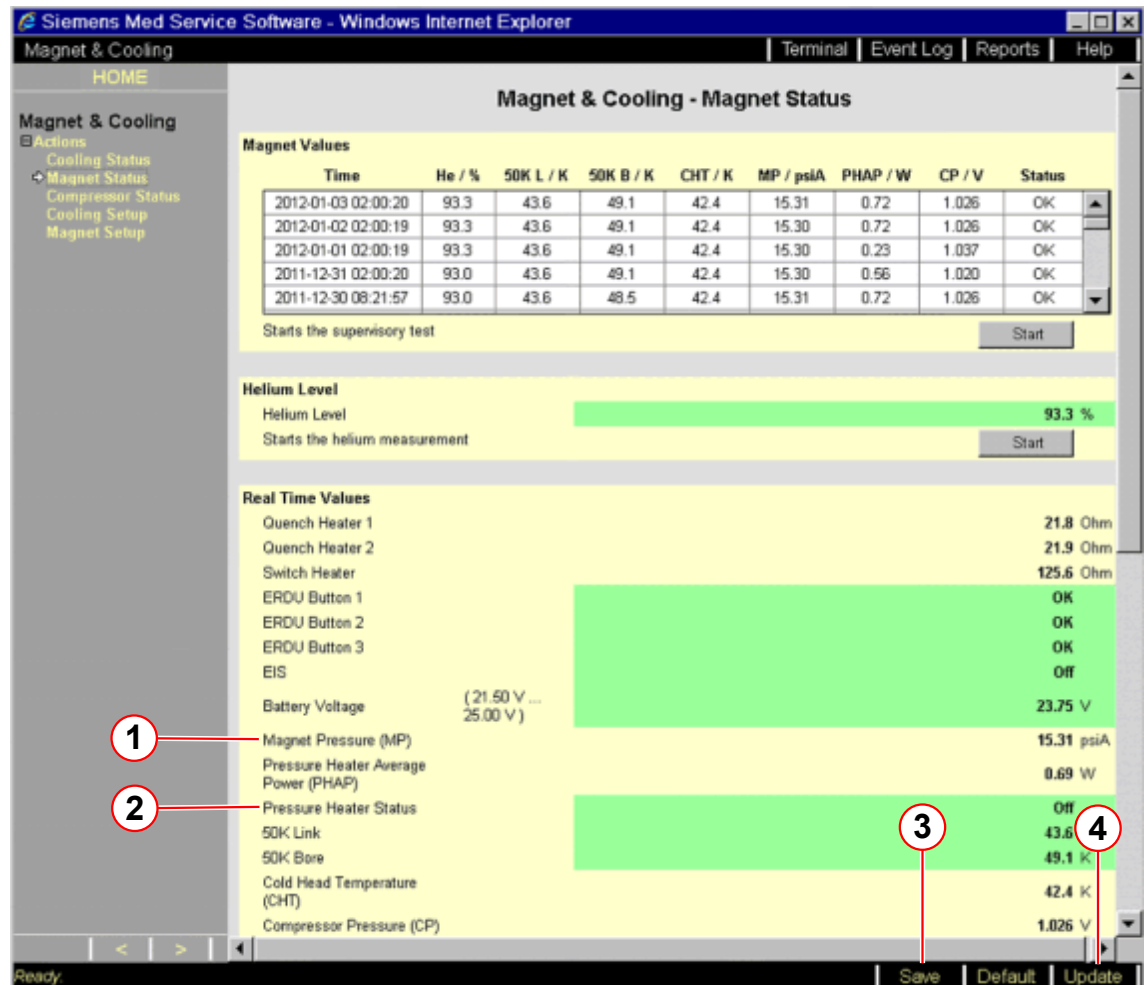
- Activate the pressure heater, (→ Activate the pressure heater / Page 54)
- Changing supply dewars during transfer, (→ Changing supply dewars during transfer / Page 42)
- Final work steps, (→ Final work steps / Page 44)

3.2.18 Activate the pressure heater

- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- At **MSUP Control - Sets the pressure heater mode** select the **On 100%** button.
- Select **SAVE**.
- **At Real-Time Values** - Monitor the **magnet pressure**.
 - Select **Update** to refresh data until pressure reaches ≥ 15.3 psi (A).
- At **MSUP Control - Sets the pressure heater mode** select the **Automatic mode** button.

- Select **SAVE**.

Fig. 50: Check magnet pressure level



- (1) Magnet Pressure (MP)
- (2) Pressure Heater Status
- (3) Save
- (4) Update

3.2.19 Verifying the electronic low pressure switch

Pressure control of the helium vessel is managed by the **MSUP**.

The low pressure switch must be checked with any process involving depressurization of the helium vessel, i.e. fitting the service syphon or cold head replacement.

- When pressure in the helium vessel drops to **0.2 psi (G)** above atmospheric room pressure, the **electrical pressure switch** opens and turns **off** the compressor.
- When the helium vessel **pressure** rises to above **0.2 psi (G)** the switch closes and the **compressor** will **re-start**.

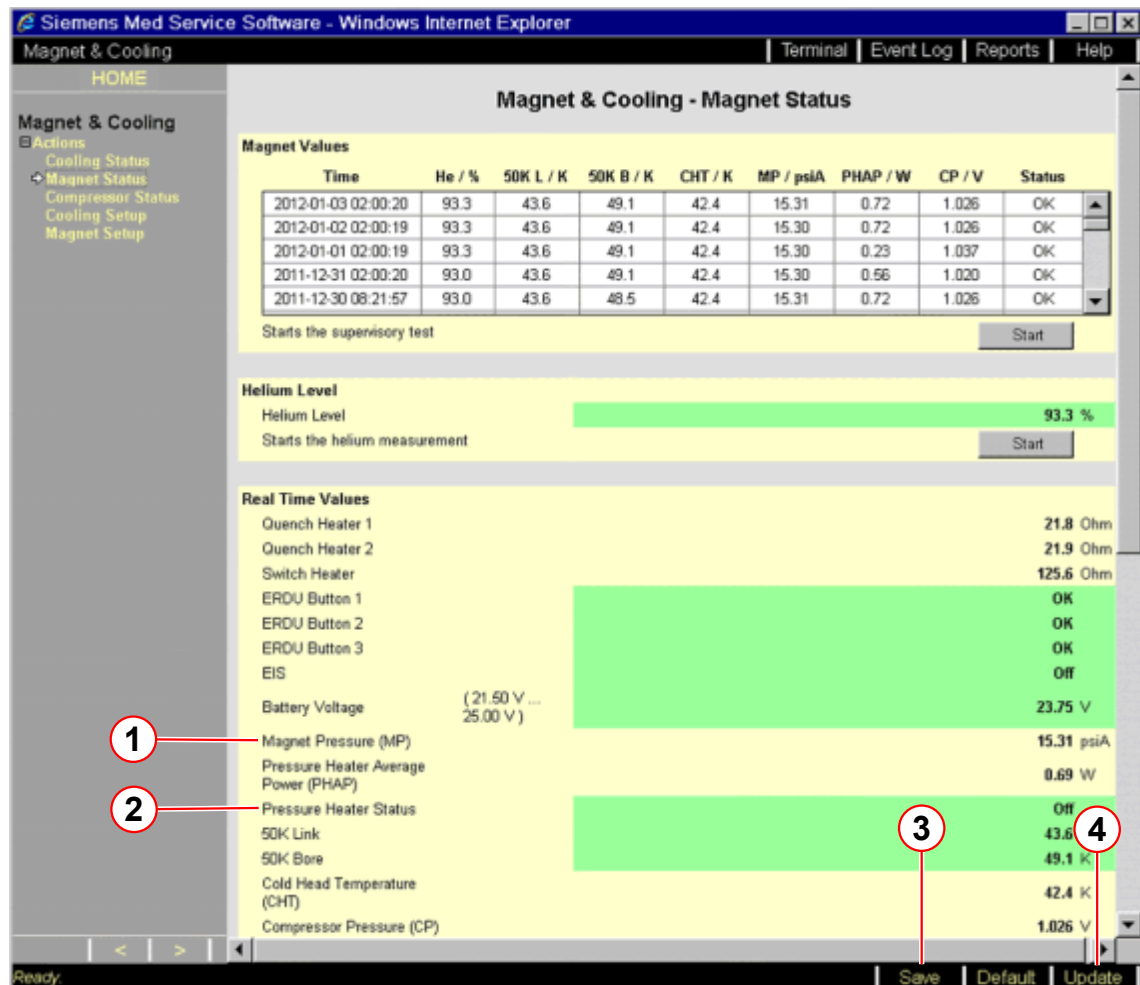
- Access the **Service Software** at the **MRC (MRAWP) Host**.



There is a delay between on and off signals to the compressor, this delay can lead to the compressor not starting instantly when 0.2 psi (G) but once the time cycle of approximately 20 minutes has lapsed.

- Select the menu item **Magnet & Cooling - Magnet Status**.
- Check **magnet pressure** is normal (**15.3 psi (A)**).

Fig. 51: Check magnet pressure



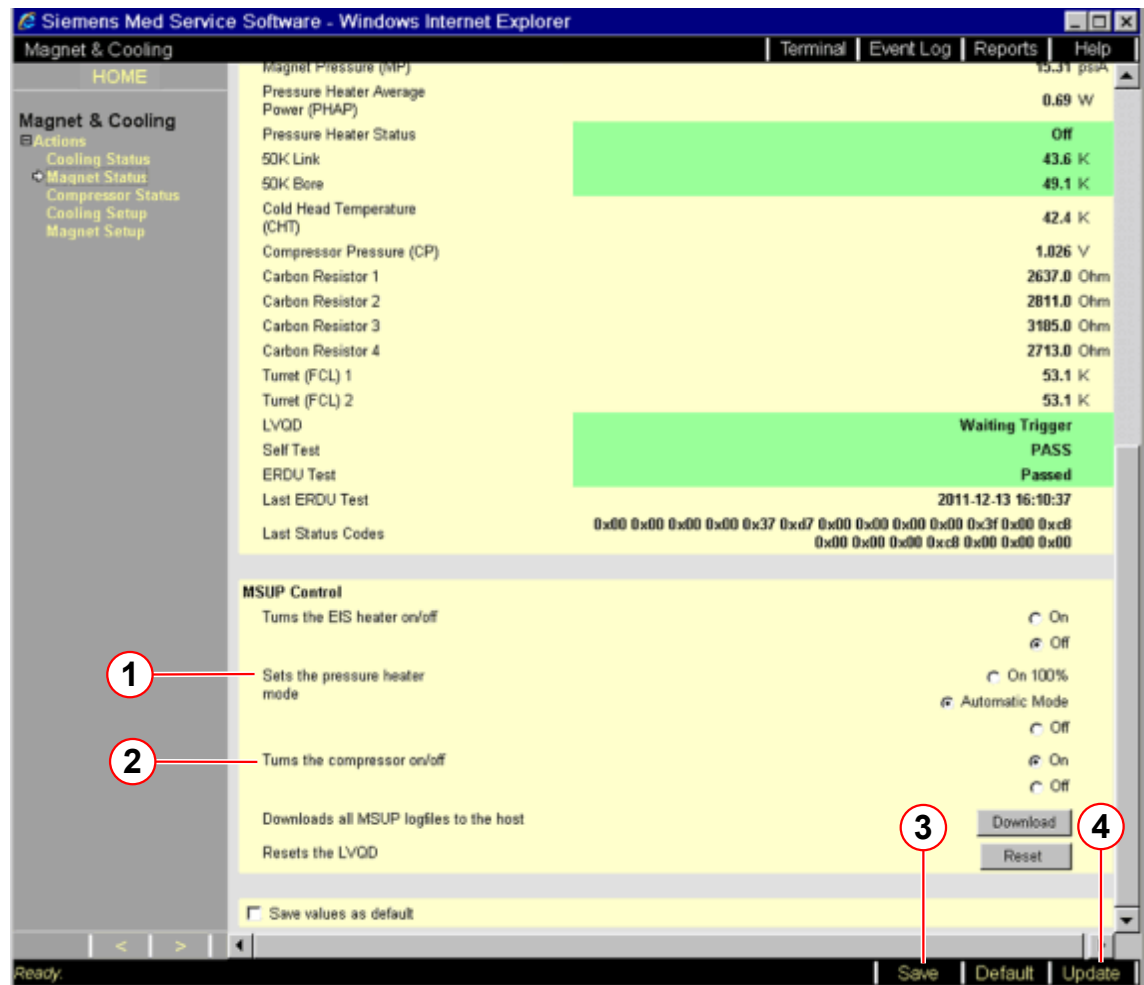
- (1) Magnet Pressure (MP)
- (2) Pressure Heater Status
- (3) Save
- (4) Update

- Check the **compressor** is switched **on** and **running**.

Deactivate the pressure heater

- At MSUP Control - Set the pressure heater mode select the **OFF** button.

Fig. 52: Service screen Mag&Cooling - Magnet Status - MSUP Control

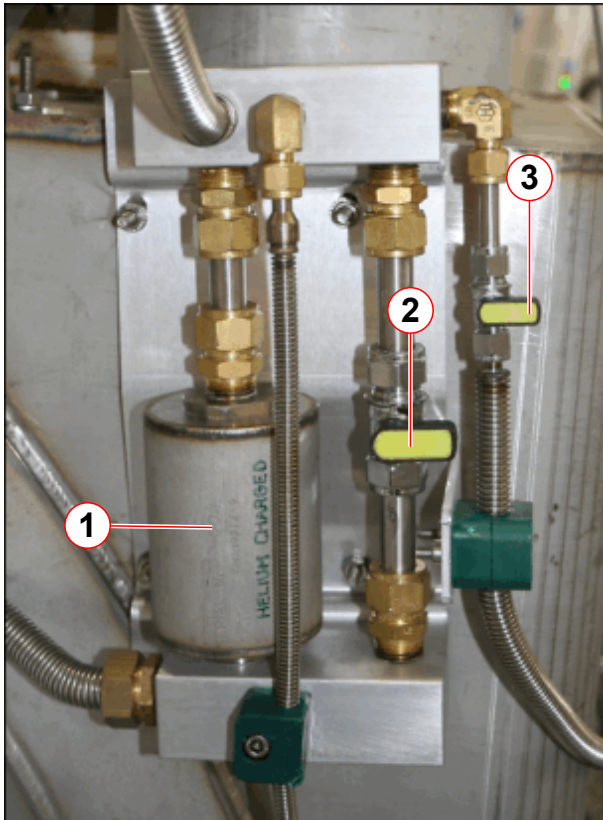


- (1) Sets the pressure heater mode
- (2) Sets the compressor on/off
- (3) Save
- (4) Update

Depressurization of the helium vessel

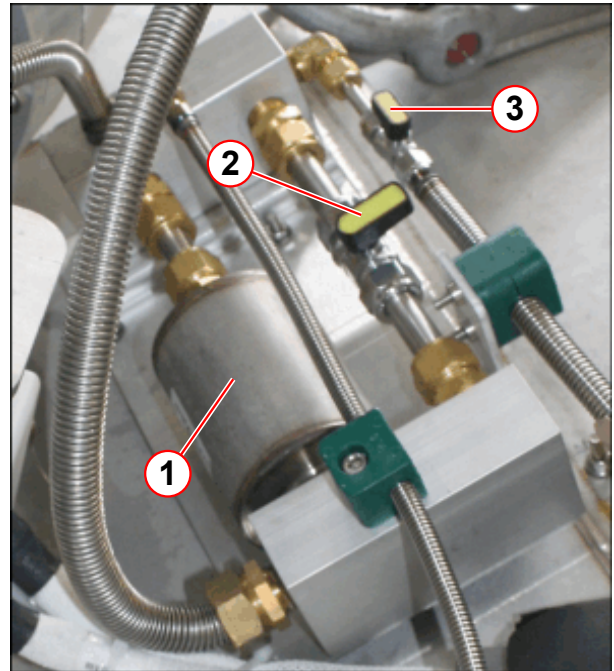
- Depressurize to within **0.2 psi (A)** of atmospheric pressure.

Fig. 53: 17 psi (A) venting valve



- (1) 17 psi (A) valve
- (2) Magnet venting bypass valve (shown closed)
- (3) Cold head bypass valve (shown closed)

Fig. 54: OR97 - 17 psi (A) venting valve



- (1) 17 psi (A) valve
- (2) Magnet venting bypass valve (shown closed)
- (3) Cold head bypass valve (shown open)

- **Fully open the magnet bypass valve to depressurize** the helium vessel.
- Wait until **zero psi** is displayed on the pressure gauge.
- The **pressure switch activates** and the cold head turns **off**.

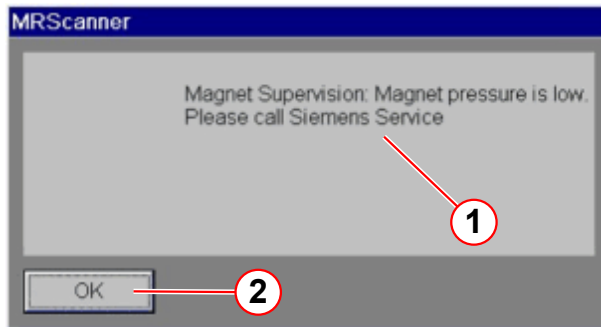


Perform the helium fill or cold head replacement at this point if the control switch is being verified during another procedure.

- **Close the magnet bypass valve to prevent air ingress**
- **Warm up the vent valve assembly.**

- **Take care** not to damage any **components** when using a **heat gun**.

Fig. 55: MSUP - Magnet pressure is low



- (1) Pop-up message
(2) Click OK to clear

On the **MR Scanner**:

- A pop-up **alarm message** is displayed.
 - » **Magnet pressure is low.**
 - » Click **OK** to clear the alarm.
- The **alarm box** will sound when **pressure falls to alarm limits**.
 - » The yellow **WARNING LED** lights on the alarm box.
 - » The alarm **buzzer is activated**
 - » Press **AUDIO ALARM OFF** to silence the alarm.

- **Repressurize** the helium vessel.¹

Activation of the pressure heater

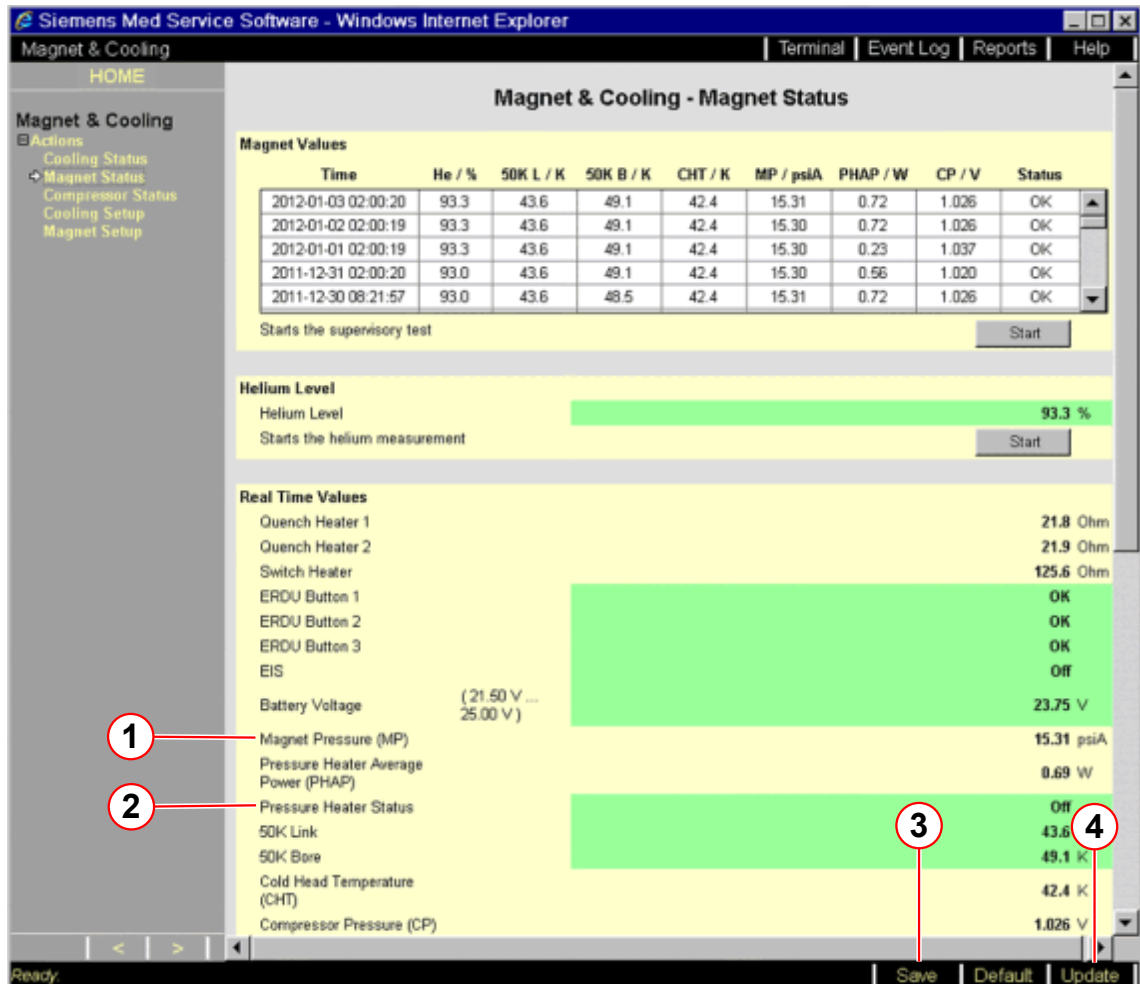
- At **MSUP Control** - **Sets the pressure heater mode** select the **On 100%** button.
(→ 1/Fig. 52 Page 57)
- **Check** that the bypass valve is locked securely in the **closed** position.
Click **UPDATE**.
- Check that the **compressor** activates **automatically** as the pressure rises above **0.2 psi (A)** atmospheric.
- At **Real-Time Values** - Monitor the **magnet pressure**.
 - Select **Update** to refresh data until pressure reaches **≥ 15.4 psi (A)**.
 - As the cold head recondenses the helium gas, the system pressure is returned to 15.3 psi (A).

1. If the verification is being carried out with another process (i.e. fitting the Magnet Service Syphon (except for Mobile systems), or replacing the Cold head), follow the **re-pressurization** part of this procedure when those processes are complete.



Data is only accurate from the last refresh of the page, or from when the page was opened last.

Fig. 56: Check magnet pressure



- (1) Magnet Pressure (MP)
- (2) Pressure Heater Status
- (3) Save
- (4) Update

- At **MSUP Control** - Sets the pressure heater mode select the **Automatic mode** button (→ 1/ Fig. 52 Page 57)
- Click **UPDATE**
- Cancel any **alarms**.

4.1 Quench tube check

- Perform a visual inspection of the quench tube.
 - » Refer to (→ Quench tube / M6-024.831)
- Check if the Installation protocol is signed by the Quench line manufacturer.



Ensure that the quench line installation is complete before performing the first ramping procedure.

The quench line acceptance protocol must be checked before the first ramp.

Check that the Quench Line Acceptance Protocol is signed off by the Quench Line Manufacturer.

The acceptance protocol can be found under:

Product Information Home > Planning > Planning Guides > MR Systems > Quench Line Design and Calculation Tool > Quench Line Calculation Tool

4.2 Identification of the controlled access area (0.5 mT area)

- Check that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone.
 - » Refer to: (→ Identification of the controlled access area (0.5 mT area) / M6-024.831)



WARNING

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Before ramping up the magnet, ensure that no ferrous tooling or equipment are present in the imaging suite.
- » Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone.
- » The sign must be posted in a way that it is visible even when the door to the controlled area is opened.

4.3 Magnet stop button test

- Perform the Magnet stop button test
 - » Refer to (→ ERDU / M6-024.831.03)

4.4 Venting leak check

4.4.1 General

For all recondensing systems, it is essential that the magnet system is leak tight to maintain a zero-loss cryogenic performance.

Individual leaks must be 1×10^{-4} mbar l/s or less.

Leak checking must be performed with an electronic leak detector capable of detecting leaks, at least as small as **1×10^{-4} mbar l/s**.

The magnet must be checked after any **service operations** have been performed, such as:

- During **installation**.
- After **helium filling**.
- After a **cold head exchange**.
- After any **exchange** or **disturbance** of a part on the **high-pressure** side of the magnet.



CAUTION

Risk of icing

Fine leaks may not show any change in pressure. Over time, they will contribute to air ingress and icing.

- » Leak check with a suitable leak check device.



For leak repair procedure - Refer to Troubleshooting Guide.

4.4.2 Venting system (Mk1 and Mk2)

The document describes the procedure for **OR105** magnet venting systems.

Fig. 57: Mk1 OR105 Venting system

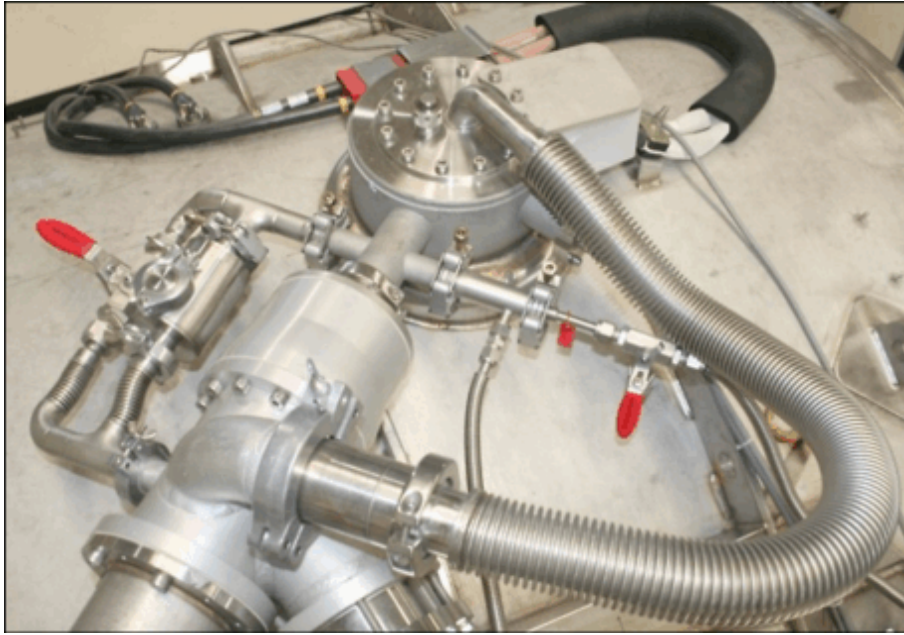


Fig. 58: Mk2 OR105 Venting System



4.4.3 Safety

! CAUTION

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following safety information may result in physical injury.

- » Fittings on the high-pressure side must not be disconnected without first depressurizing the helium vessel.

! WARNING

Magnet at field.

Personal injury or damage to equipment may occur.

- » Do not use magnetic tools for repairing or tightening seals when the magnet is at field.

4.4.4 Equipment required

- Heat gun.
- Paper towelling.
- Accepted leak detection equipment:

Fig. 59: Edwards Leak detector



- Use the Edwards hand-held leak detector (part no **10614850**).

This is the **recommended equipment** for leaks **greater than 1×10^{-4} mbar l/s**.

NOTE: The hand-held device is calibrated in **ml/sec**:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.



The functionality of the Edwards leak detector can be compromised by magnetic fields, if placed near the magnet bore.

- Check that the equipment is detecting helium.
 - Check the equipment using helium from a known gas supply (i.e. regulated gas bottle).

- Refer to the equipment manufacturer's operating instructions.
- The **calibration certificate** must be valid.

4.4.5 Pre-requisites for leak checking

- The helium vessel must at least be at **normal operating pressure (15.3 psi (A))** prior to leak checking.
 - Activate the pressure heater (if necessary) to raise pressure.

If pressure heater activation continues for more than **30 minutes** with **NO** increase in magnet pressure:-

 - Stop heater activation and check for any obvious gross leaks such as the syphon ring nut or burst disks.
- All joints are dry and at room temperature (use a heat gun).



After any service procedures are performed, the venting must be warmed to room temperature, and dried.
This must be completed **BEFORE** performing a leak check.

4.4.6 Using leak check equipment

- Switch on the leak detector and allow it to warm up (refer to manufacturer's instructions).
- Check the equipment is working correctly (detecting helium). This should be done both before and after use.
 - Set the leak detector to its maximum sensitivity.
 - Determine the background level in the magnet room.
- Note response time of the detector.



The background must be at least one order of magnitude lower than the smallest leak expected, (i.e. leaks of 10^{-4} mbar l/s, require background of order 10^{-5} mbar l/s).



Ensure that the atmospheric environment is not saturated with helium gas (i.e. Do not leak check soon after a helium fill, or depressurization).

Start at the **lowest** point:

- Hold the tip of the leak detector 5 mm above the surface of the component. **Keep the tip above any water..**
- Slowly move the detector around the seals or clamped joints.
- Hold the detector at a leak point until the reading has stabilized.
- Move the detector away and check that the indication disappears.

- Repeat the process to check that a leak is still indicated.

The following points must **always** be observed:-

- Joints on the '**HIGH**' pressure side of the venting system will generally be fitted with **metal seals**. Before dismantling a joint, ensure a replacement seal is available. (Refer to **Replacement of parts**).
- **DO NOT** use rubber seals as a substitute for metal seals.
- **DO NOT** re-use metal seals, always replace with a new seal.
- **DO NOT** use vacuum grease.



Do not use any form of vacuum grease on O-rings and seals. Vacuum grease dries over time and will cause a seal to leak.

4.4.7 Leak checks at installation and service

The venting and components on the **high-pressure** side of the magnet turret require leak testing to ensure that the re-condensing performance is achieved.

Installation

- Check the **system pressure** on arrival at the installation. If the system has **low**, or **zero** pressure then a **gross leak** could have occurred during shipment.
 - Check all seals, burst disks and fittings on the **high pressure side** of the vent valve **before** the cold head is turned on and the magnet run to field.
 - Check any **joints** (clamps and seals) that have been broken and re-made during installation.
 - If magnet pressure is still low, check for leaks through the **magnet vent valve**.

Service

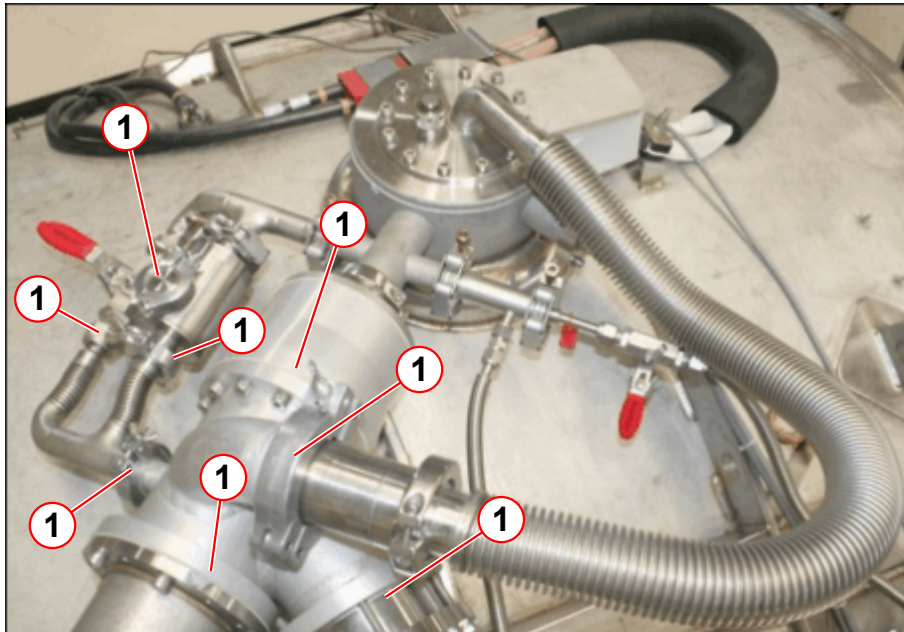
- The turret, venting valve assembly and quench valve must be **warmed up to room temperature** after any service procedures have been performed, **prior** to leak check.
- Leak check all seals that have been subjected to **low temperatures** and the subsequent build up of frost. For example, magnets that have been **helium filled, depressurized**, or the magnet **ramped**.
 - Check **joints** broken and re-made during servicing. **Replace** damaged seals as required.

4.4.8 Turret and venting components

Low pressure venting checks

- Check for **cold gas escaping** around **low pressure joints** during:
 - depressurization, or helium filling.
- Check that all **sealing nuts and flange screws** on the venting are tight.
 - **Replace gasket** seals if found damaged.

Fig. 60: Mk1 OR105 Low pressure check points



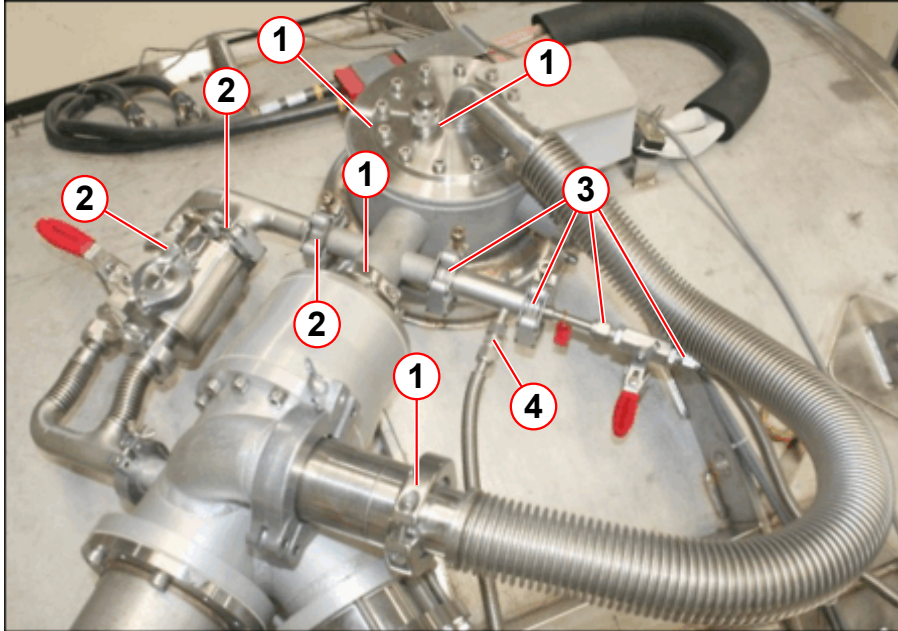
(1) Leak check points

High pressure leak checks

- Check that all clamps on the **high-pressure side** of the venting are tight.

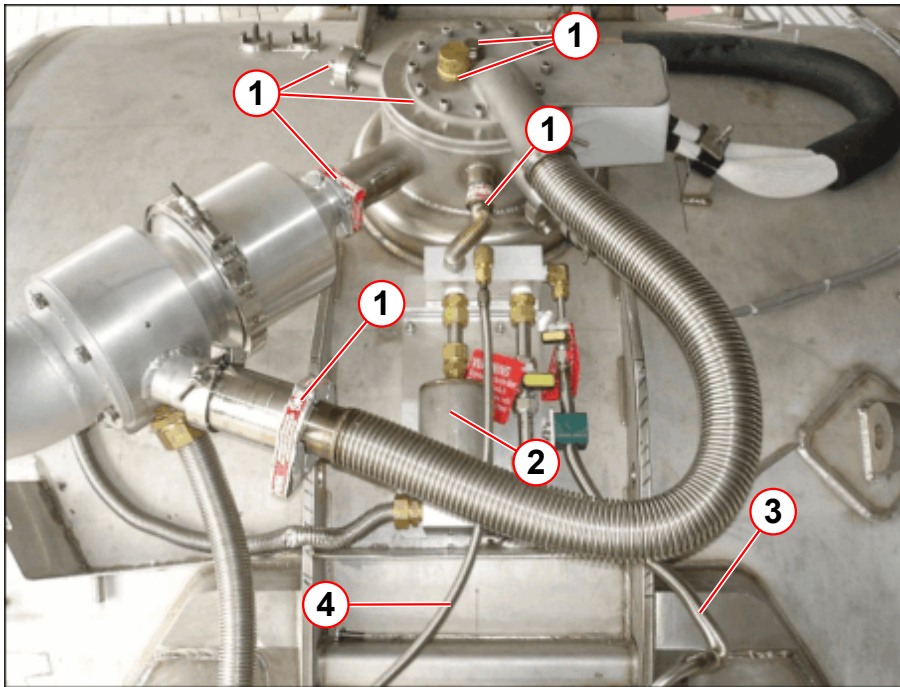
- **Do not** over-tighten clamps. This can cause a leak and make an existing leak worse. The figures below show the **leak check positions** on the **high-pressure** side of the magnet.

Fig. 61: OR105 Mk1 - Turret and venting leak checks (high pressure)



- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line

Fig. 62: OR105/122 Mk2 - Turret and venting leak checks (high pressure)



- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line



Replace a high load seal only with the same type. High load seals must only be used **ONCE**.

- After all **helium fills**, warm turret using a heat gun, and:-
 - Leak check the **syphon port**.
 - Ensure the **internal** port surface is clean and free of ice.
 - Fit a new, undamaged **copper sealing washer**. **DO NOT** over- tighten the syphon plug.



If a leak cannot be found, but there is a slow, steady helium loss, report it to your local Uptime Center for repair by a specialist.



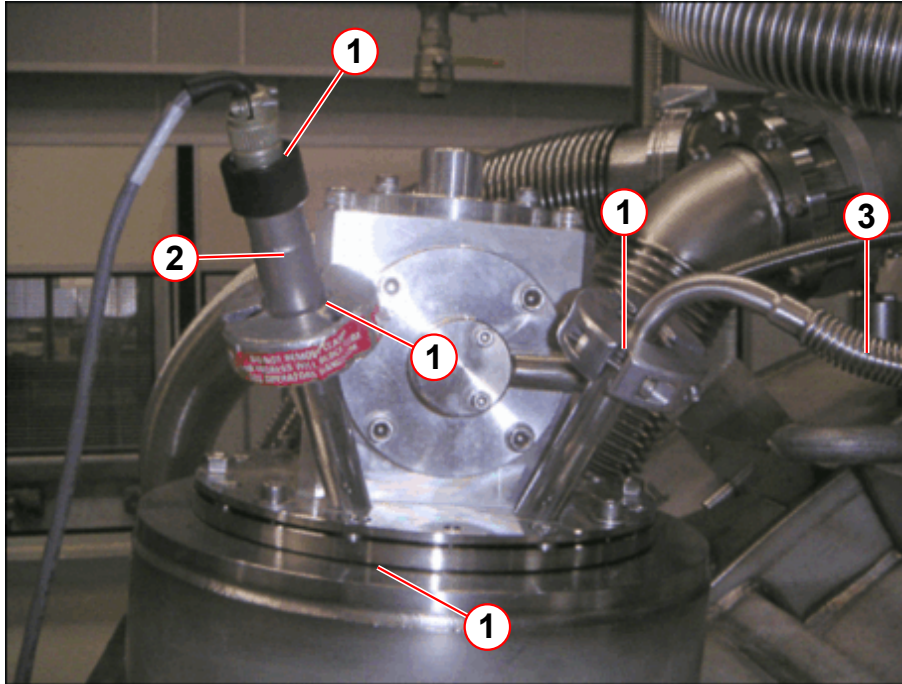
If a leak is found on the Turret entry seal, it must be reported to your local Uptime Service Center for repair by a specialist.

- **Check** the connections to the **16 psi (A) vent valve** assembly.
- **Check** the **Cold head** line.
- **Check** the **Pressure gauge** line.

4.4.9 Cold head

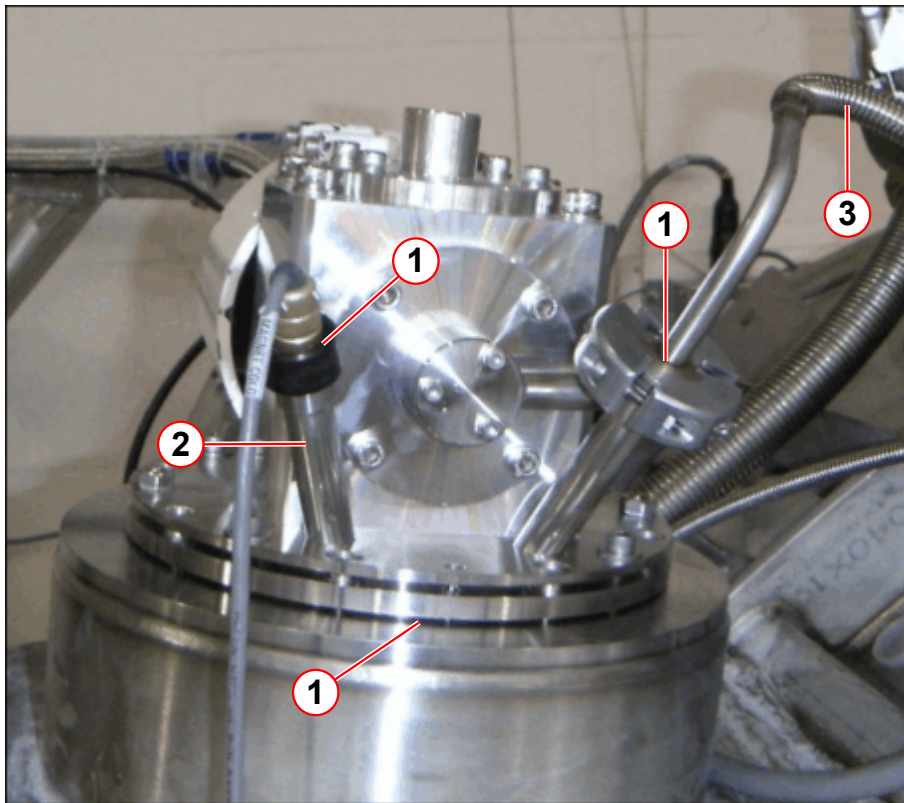
- Leak check around the cold head **O-ring seals**.
- Leak check the cold head bypass line seals at both ends.
- Leak check around the NW16 clamps and seals.
- Check that the **hand-valve seals** on the cold head vent line.

Fig. 63: Leak checking the cold head seals (early coldhead)



- (1) Cold head seals leak checkpoints
(2) Temperature sensor feed-through
(3) Cold head vent line

Fig. 64: Leak checking the cold head seals (later coldhead)



- (1) Cold head seals leak check points
- (2) Temperature sensor feedthrough
- (3) Cold head vent line



WARNING

Risk of cold burns. High helium gas pressure.

Risk of personal injury may occur.

- » Do not unscrew or remove the black temperature sensor sealing cap.
- » Remove the temperature sensor cable with a straight pull away from the sealing cap.

The cold head is fitted with **temperature sensors**:

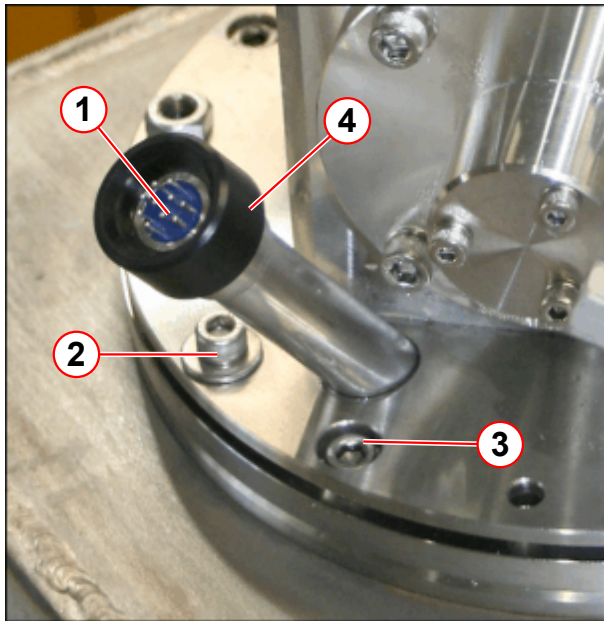
- The **feed-through** for the wires is sealed with an **O-ring**.
- Check the **feed-through** and **cold head seals**.
- **Check** the **torque** of the cold head **screws**.



If a leak is found with the magnet at field, it is permissible to re-torque the 4 off cold head flange screws to a higher setting of 5 Nm.

If the leak persists, send a report to your local Uptime Service Center for repair by a specialist.

Fig. 65: Cold head 10 pin seal



- (1) Leak check point
- (2) 4 off M6 x 25 mm cold head flange screws, torque to 4Nm
- (3) 8 off M6 x 16 mm skt hd screws O-ring clamp plate, torque to 8Nm
- (4) Black sealing cap

- **DO NOT** turn or try to remove the 'black' sealing cap
- Remove the temperature sensor cable (pos. 4), with a straight pull away from the black sealing cap.
- Leak check the 10 pin seal connector.
- Immediately refit the cable if no leak is found.

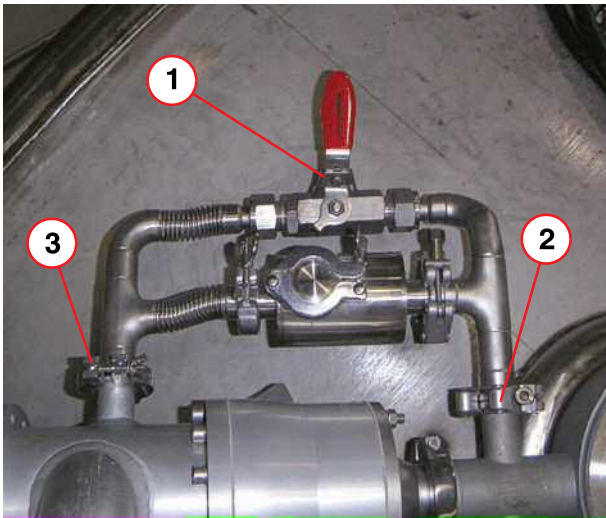
4.4.10 16 psi (A) vent valve assembly

- Leak check all joints on the 16 psi (A) valve **high-pressure** assembly.
- For leaks found at pipe couplings, first try tightening the clamp or securing screws. **DO NOT OVER-TIGHTEN!**
- If tightening does not solve the leak, the **complete** vent valve assembly must be replaced.
- If leaks are found at **connections** into the inlet aluminium block, the **complete** assembly **must be replaced**.

- Refer to Replacement of parts — Vent Assembly

Mk1 Vent valve assembly

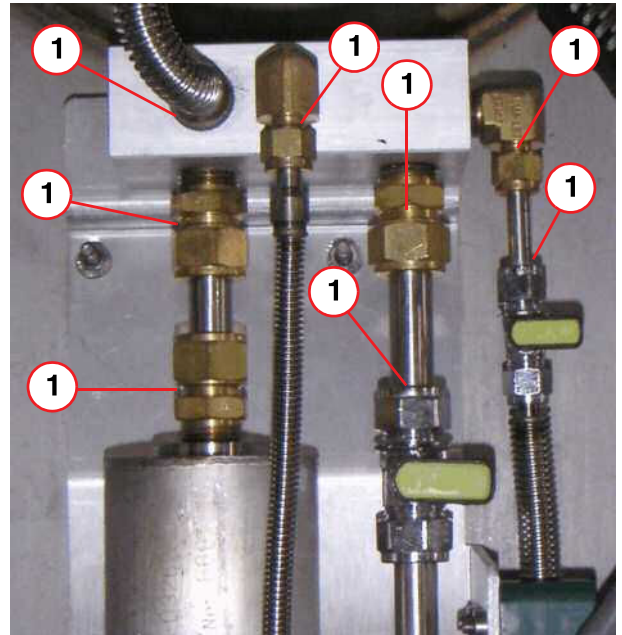
Fig. 66: OR105 Mk1 Vent valve assembly



- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

Mk2 Vent valve assembly

Fig. 67: Mk2 Venting 16 psi (A) Valve Assembly



- (1) Leak test points

4.4.11 Pressure control assembly

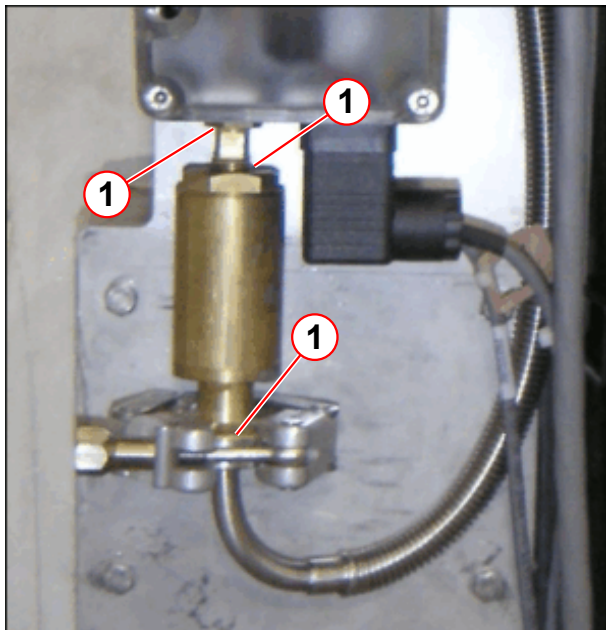
Check version of pressure control switch fitted:-

- **Mk1 only** — Leak check around the NW25 high load seal and clamp.
- For leaks found at pipe couplings, first try tightening the clamp or securing screws.
BEWARE OF OVER-TIGHTENING!

- If tightening does not solve the leak, the **metal seal** must be **replaced**. (Refer to **Replacement of parts**)

Mk1 Manocomb leak check points

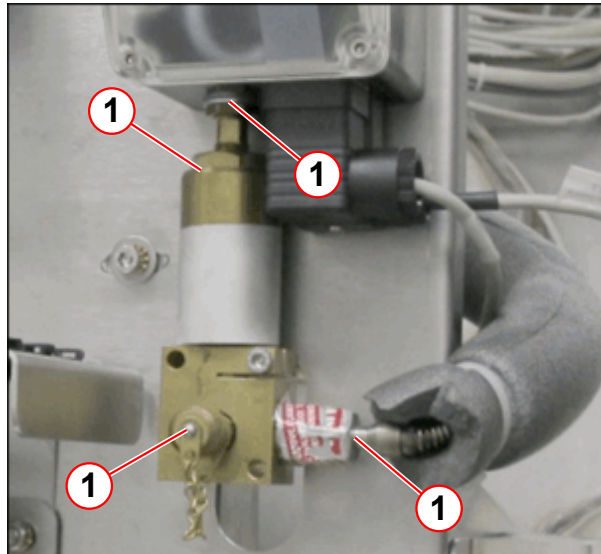
Fig. 68: Mk1 Pressure gauge - leak check



(1) Leak test points

Mk2 Manocomb leak check points

Fig. 69: Pressure switch assembly - leak checks

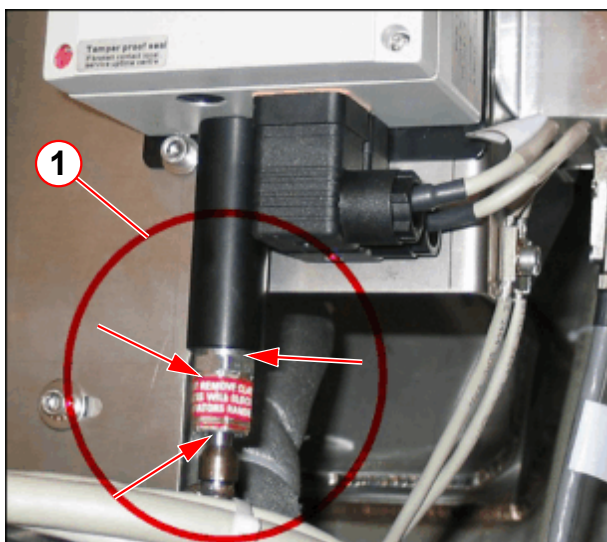


(1) Leak check points



The WIKA Pressure control switch is **NOT** directly interchangeable with the MK1 Manocomb switch.

Fig. 70: WIKA - Pressure switch assembly



(1) Leak check points

4.4.12 Auxiliary 1.8 bar metal bursting disk

MK1 Auxiliary burst disk

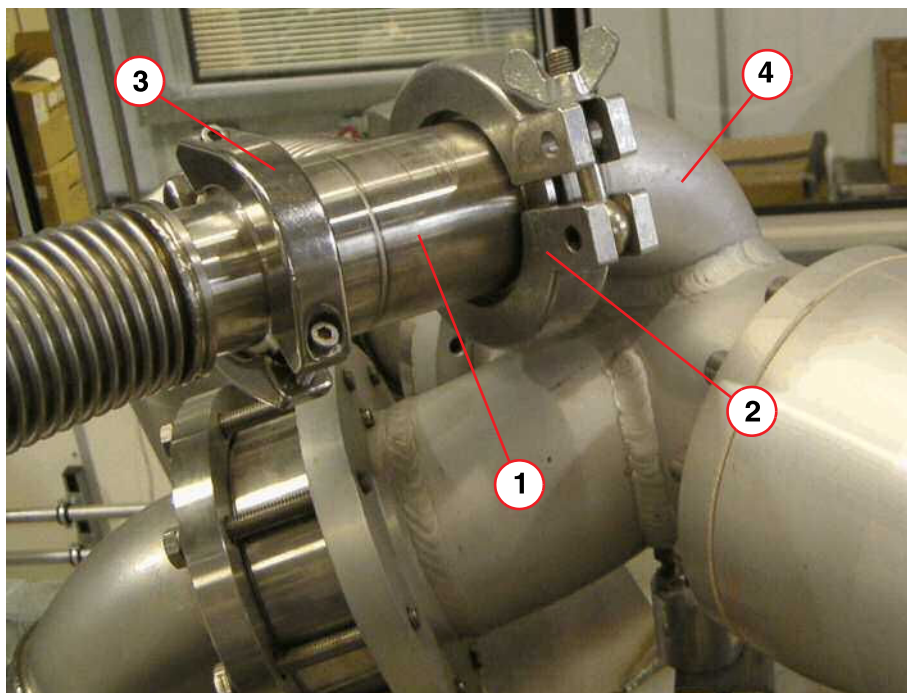
To check for any leak path through the **auxiliary metal bursting disk**:-

- Remove the **NW50 clamp** and low pressure seal connecting the auxiliary burst disk to the quench valve assembly.
- Visually inspect the **metal membrane** of the disk for damage.
- Visually inspect the low pressure **O ring** seal.



The membrane is very fragile - **DO NOT TOUCH THE SURFACE.**

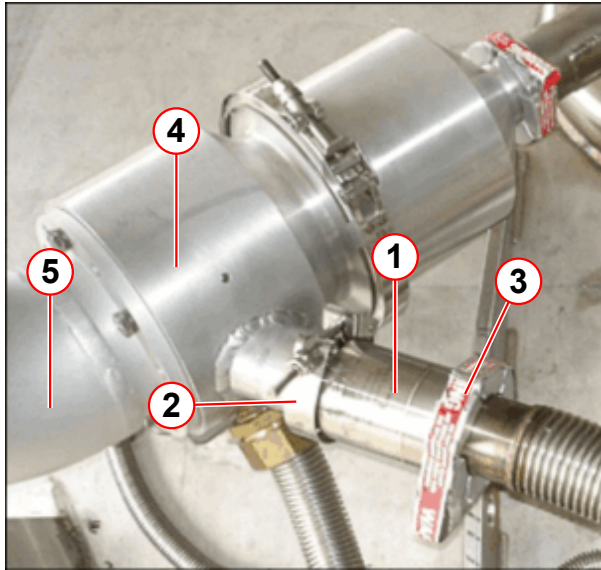
Fig. 71: OR105 Mk1 Auxiliary vent burst disc assy



- (1) Aux vent burst disc
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

MK2 Auxiliary Burst disk

Fig. 72: Mk2 Auxiliary burst disk assembly



- (1) Aux burst disk
- (2) V-clamp (low pressure)
- (3) NW40 clamp (high pressure)
- (4) Manifold
- (5) Quench elbow

To check for any **leak path** through the **metal auxiliary bursting disk**:

- Undo and remove the **hinged 'V' clamp** from the **low-pressure** side of the auxiliary burst disk.
- Visually inspect the **metal membrane** of the burst disk for damage.
- Visually inspect the **O-ring** on the **low-pressure** side.
- **Replace** the auxiliary burst disk if a leak is detected.

4.4.13 Quench valve

- Leak check the **quench valve**:
 - Check that the room background helium level is **less than** 10^{-5} mbar l/s.
 - If the background level is **above** 10^{-5} , **wait one hour** with the port **open**. This allows the helium gas to disperse.
 - When the background level is **less than** 10^{-5} , insert the probe of the leak checker **inside the quench valve port**
 - If a leak is detected, the **burst disk** and burst disk **O-ring** could be faulty. The quench valve will need to be **disassembled**. (Refer to **Replacement of Parts**).

4.4.14 Final work steps

- At the end of leak checking, ensure again that the leak detector is **working correctly** by re-checking its function with a known helium gas source.
- Check that all parts of the venting system have been re-fitted correctly.
- Check that the magnet is at its normal operating pressure.

5.1 OR105 Ramping



Only Siemens CSEs trained in ramping magnets are allowed to perform the magnet ramping procedure.



Before ramping check that the quench line is correctly installed from the magnet to the outlet.



WARNING

Magnet safety checks

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Ramping is not allowed if the visually inspection of the quench line was not performed successfully.
- » Before initially ramping up the magnet, the 'Testing the ERDU circuit' has to be performed.
- » Procedure described in the safety related test can be used if ERDU test load is available.



WARNING

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Before ramping up the magnet, ensure that no ferrous tooling or equipment are present in the imaging suite.
- » Only non-magnetic equipment may be taken into the magnet room.
- » Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone.
- » The sign has to be posted in a way that it is visible even when the door to the controlled area is opened.

**CAUTION**

Risk of asphyxiation and hypothermia, if helium gas is vented into the magnet room

Failure to observe the following safety warning may result in dizziness or loss of consciousness in personnel.

- » Quench valve must be connected to the Quench pipe.



Initial installation ramp helium level recommended to be > 60 %.



Minimum helium level for ramping must be > 40%.
Wait at least 2 h after helium filling.



Wait at least 1 h after completing a previous ramp before starting the next ramp, (including lead checks and aborted ramps).

Always

- Always restrict access to the magnet room when ramping the magnet. Use flashing lights and signs.
- Always inspect all equipment and tools for defects prior to use.
- Always inspect the Anderson ramp lead connections at the filter plate.
- Clean the area of the magnet room, dispose off empty cartons or other packing materials
- Always inspect magnet room for the presence of unidentified objects. Before ramping, the nature of those objects must be clarified.



Prepare a minimum of 4 pictures of the room **prior to first ramping**.

These pictures must be colored and have 2 to 4 megapixel resolution. They must show the complete room, all 4 corners and walls.

All cupboards in the room must be pictured with their doors open to show that they are empty.

Later those pictures are to be attached to the installation report.

5.1.1 Tools required

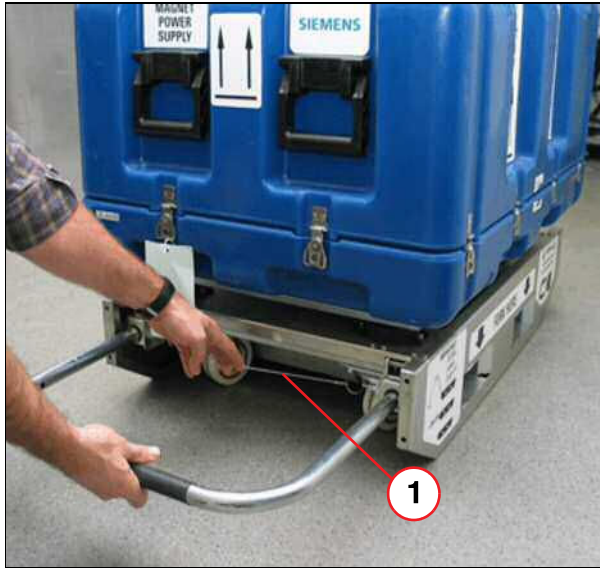
- Magnet Power Supply (3600 MPSU 3600); (order number: **8395613**). This includes:-
 - 3600 MPSU Power lead
 - CAN Open lead

- RS232 lead
- Heat gun.
- Host computer.

5.1.2 Preparation

- Move the 3600 MPSU into the equipment room close to the RF filter plate.

Fig. 73: MPSU



(1) Wire locking strap

- Lower the trolley handle into the horizontal position. This raises the wheels and allows the trolley to rest on the two skids.
- Pull the wire strap to unlock the handle, and slide the handle into the recess under the trolley.
- Remove the shipping case lid by unclipping the 12 spring clips.

Fig. 74: MPSU



(1) MPSU LID

- Open the hinged lid of the 3600 MPSU by pulling the 2 silver knobs and turning them through a 1/4 turn.
- Raise the lid.

5.1.3 Installation

Fig. 75: 3600 MPSU Power connections



- (1) X1 (MPSU power)
- (2) CB1 (Mains circuit breaker)

- Ensure the mains circuit breaker **CB1** is in the **OFF** position on the 3600 MPSU.

Fig. 76: Adapter plate (view from Equipment Room)



- Unscrew the securing screws and remove the **adapter plate** from the **RF filter plate**.
- Access to the magnet ramp leads are in the **magnet** room.



Do NOT connect the magnet ramp leads to the 3600 MPSU until **AFTER** the Transport test has been completed. See (→ Transport test / Page 89).

The test is only required the **FIRST TIME** time the MPSU is used on site.

Lead connections are detailed in the table below:-

Tab. 9 MPSU lead connections

Lead	From	To	Length
Magnet Ramp leads	MPSU X3 MPSU X4	Filter plate (grey) Filter plate (blue)	5m
CAN OPEN lead	MPSU X7	EPC X125	5m
RS232 lead	MPSU X6	MRC (MRAWP) Host, W10220/X12/MPSU Ca- ble W10220 to X128, disconnect cable at X12 on top of EPC.	5m
MPSU Power lead	MPSU X1	EPC X130	10m

Fig. 77: Avanto - Verio Ramp Configuration

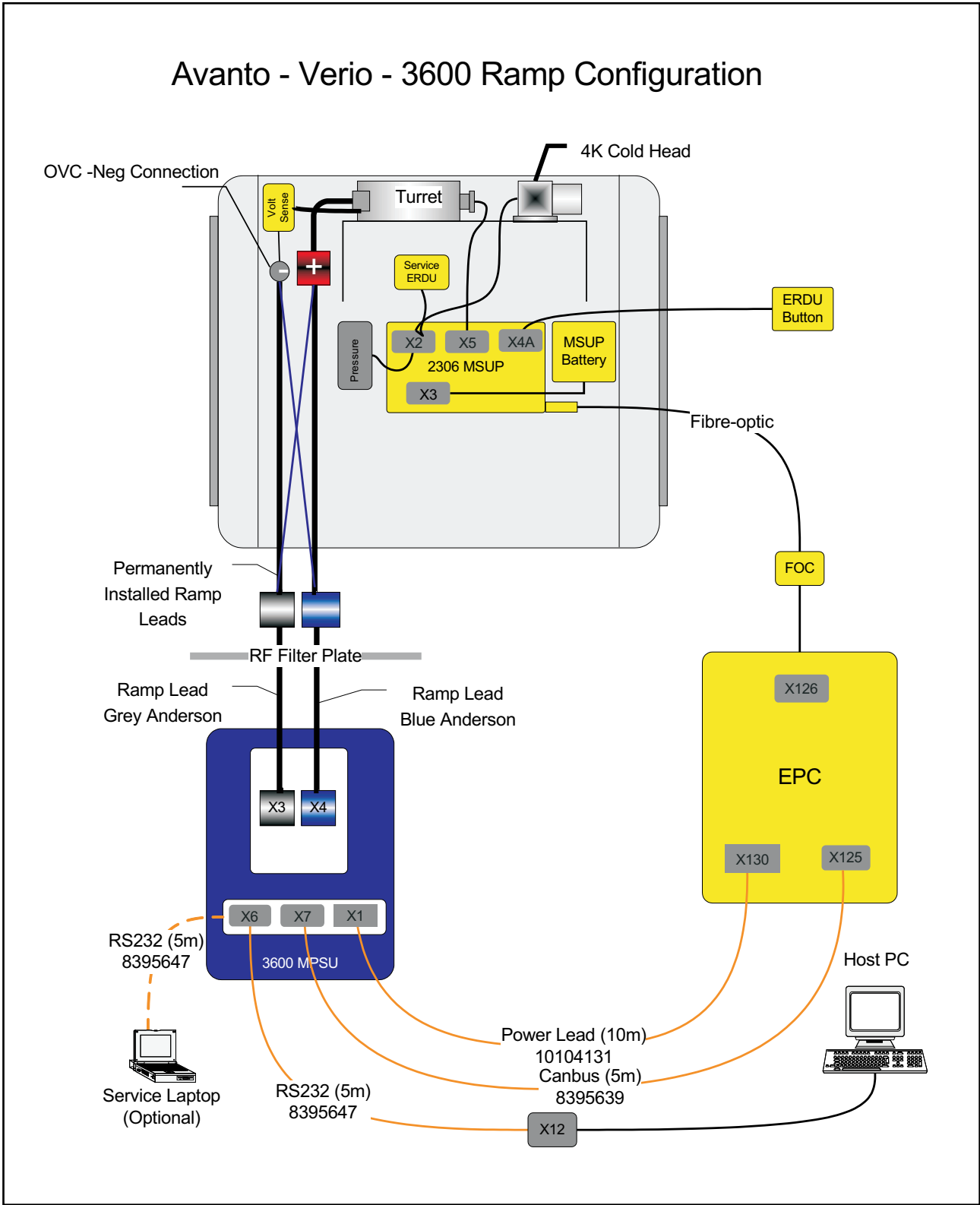
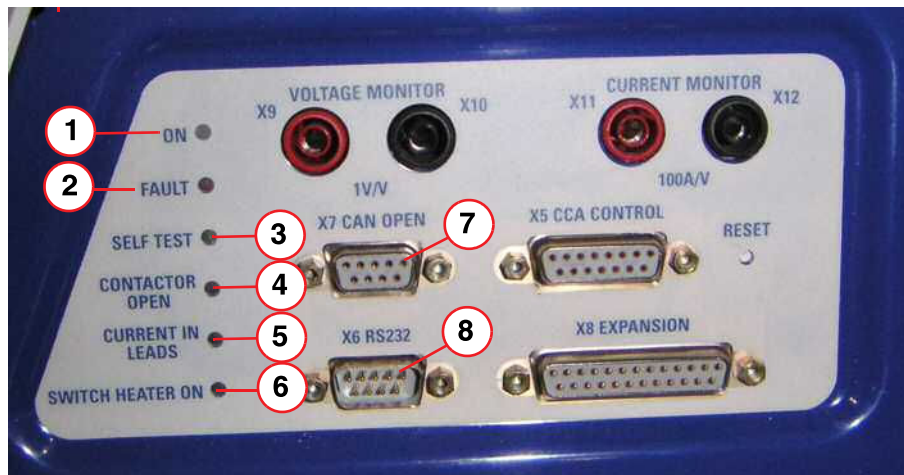


Fig. 78: 3600 MPSU Connector and indicator panel



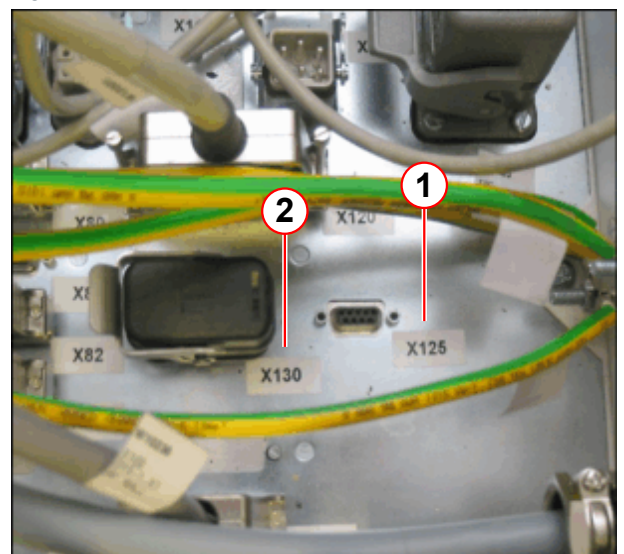
- (1) ON indicator
- (2) FAULT indicator
- (3) SELF TEST indicator
- (4) CONTACTOR OPEN indicator
- (5) CURRENT IN LEADS indicator
- (6) SWITCH HEATER ON indicator
- (7) X7 CAN OPEN connector
- (8) X6 RS232 connector

Fig. 79: 3600 MPSU leads



- (1) CAN OPEN lead
- (2) RS232 lead
- (3) MPSU power lead

Fig. 80: EPC connections

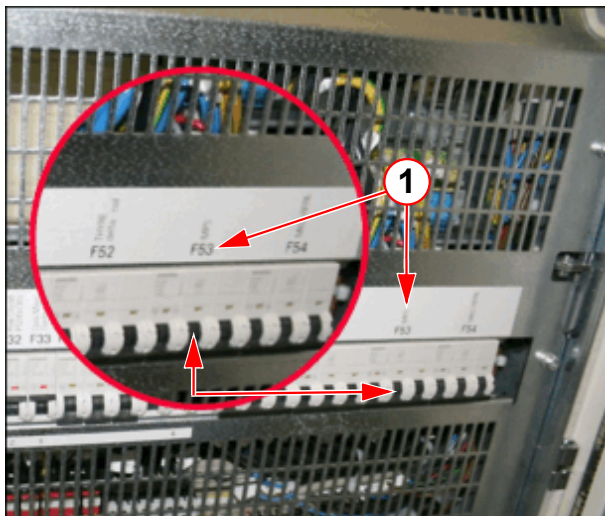


- (1) X125
- (2) X130

- Connect the **CAN OPEN**, and **RS232** leads. (refer to (→ Tab. 9 Page 83)).

- Connect 3600 MPSU **power lead** to X1 (→ 1/ Fig. 75 Page 82)

Fig. 81: MPSU Circuit breaker F53



(1) Circuit breaker F53

- Turn ON **F53** in the **EPC**.

5.1.4 Set up of the 3600 MPSU

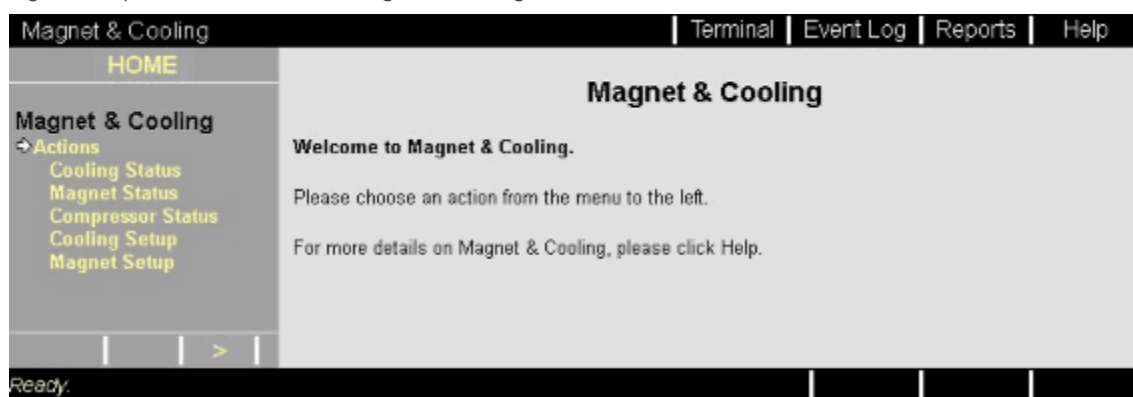


If the Host is not working, refer to Troubleshooting guide for ramping with the service laptop.

5.1.4.1 Configuring the Host

- Open the **Magnet & Cooling** Service platform.
- Select **Terminal** from the **toolbar**.

Fig. 82: Open the Terminal via Magnet&Cooling



- **Terminal** option: the program will **open**. No configuration is required.

Fig. 83: MPS3600 Serial Connection



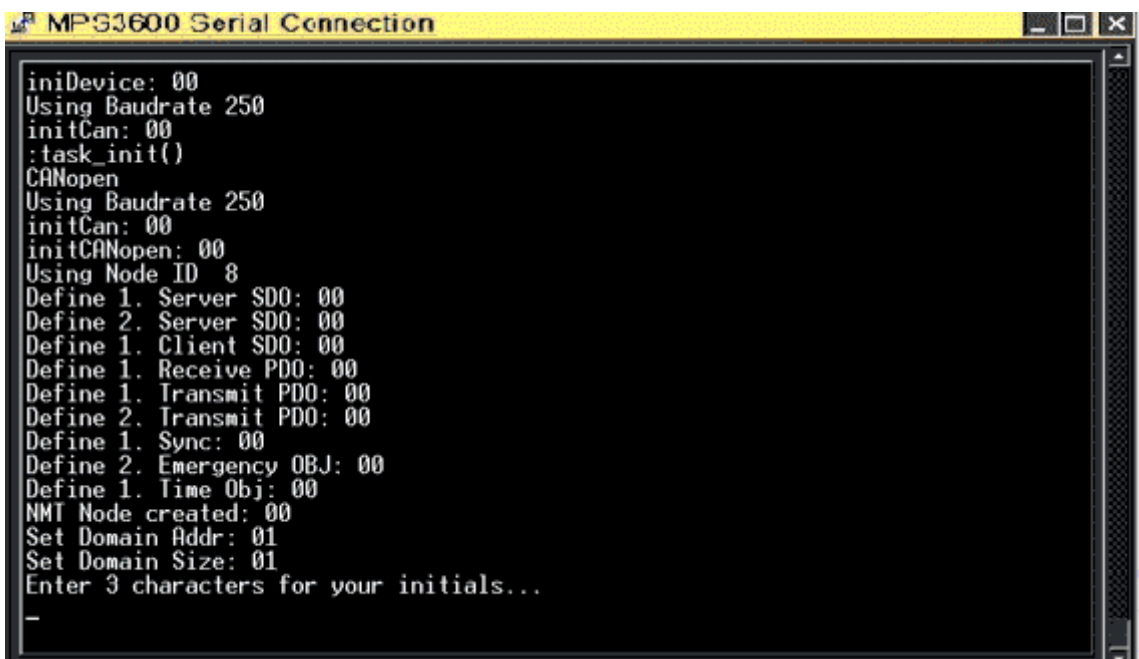
5.1.4.2 Initializing the 3600 MPSU



Ensure that there is a valid helium level reading before switching on the 3600 MPSU.

- Turn **ON** the 3600 MPSU using mains circuit breaker **CB1** on the 3600 MPSU.
(→ 2/Fig. 75 Page 82)
 - The green '**ON**' LED lights on the 3600 MPSU.(→ 1/Fig. 78 Page 85)
- The **boot-up** screen is displayed.

Fig. 84: Boot-up screen



If the screen does not appear, make sure that the correct Com port and settings have been selected. If they have, turn the 3600 MPSU off and then back on, using the mains circuit breaker CB1. (→ 1/Fig. 75 Page 82)



To update the software on the MPSU:-

Reboot the scanner with the MPSU switched on.

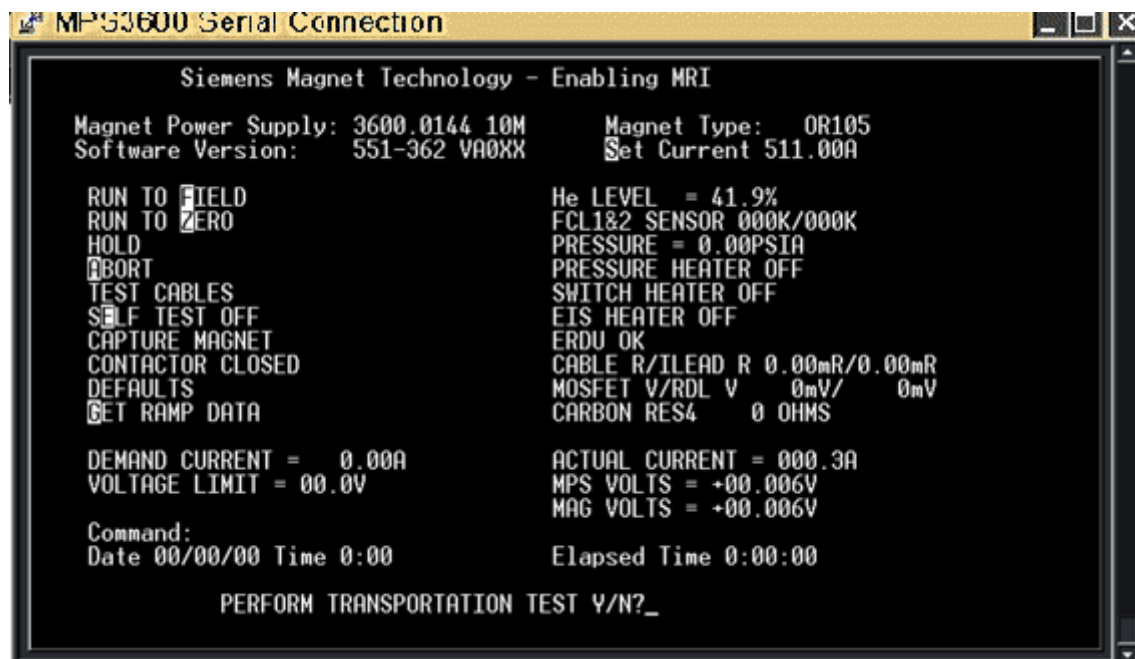
Repeat booting of the scanner if following warning message still persists: "MPS not operational"

- When prompted, enter your **initials**.

The 3600 MPSU now retrieves the **magnet type**, and **magnet serial number**, **last current set**, **He-level**, **date** and **time** from the MSUP.

- The following screen prompts to '**Perform Transportation test**':-

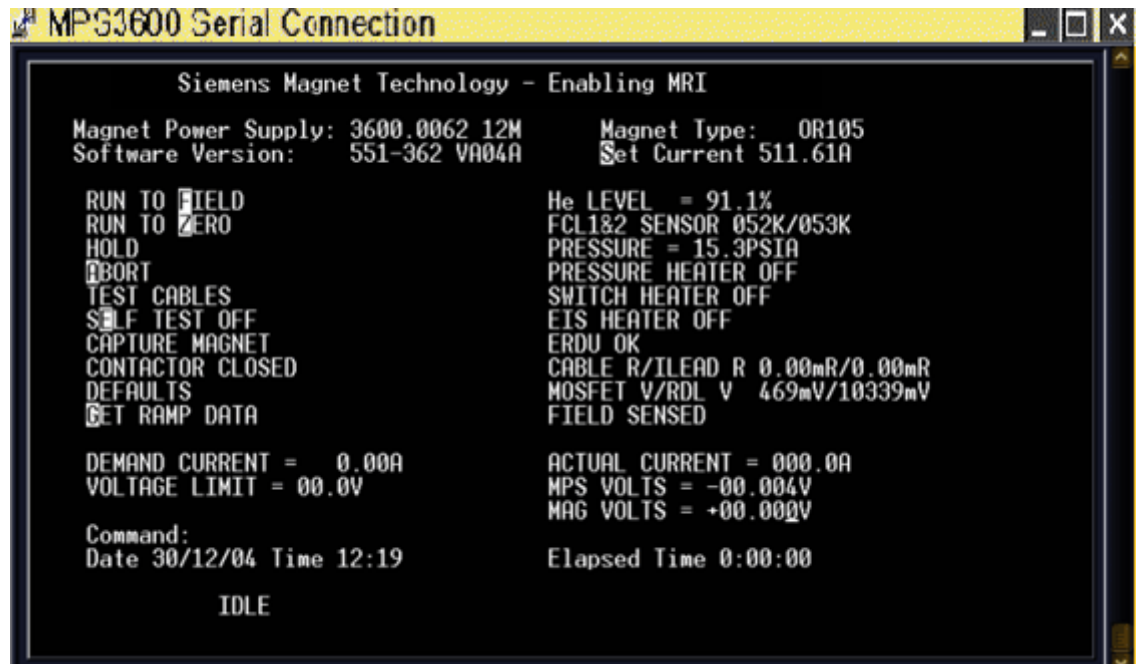
Fig. 85: Transport Test (example of Avanto screenshot)



- If the **Transport test** is required, continue with (→ Transport test / Page 89).
- If the test is **not** required:
 - » Enter **N<ENTER>**
 - » At the prompt 'ARE YOU SURE', enter **Y<ENTER>**
- The 3600 MPSU performs an **internal self-test**.
 - » The **self-test** LED indicator lights during the test. (→ 3/ Fig. 78 Page 85)

- When the self-test is complete, the **Operational screen** is displayed.

Fig. 86: Operational Screen (example of Avanto screenshot)



- Continue with (→ Tab. 9 Page 83)(→ Ramp lead connection / Page 92).

5.1.4.3 Transport test

During transport, it is possible that the 3600 MPSU may have suffered internal damage due to rough handling.



This test is only required the first time the 3600 MPSU is used on site.

5.1.4.4 Transport test procedure

Fig. 87: Transport test tool

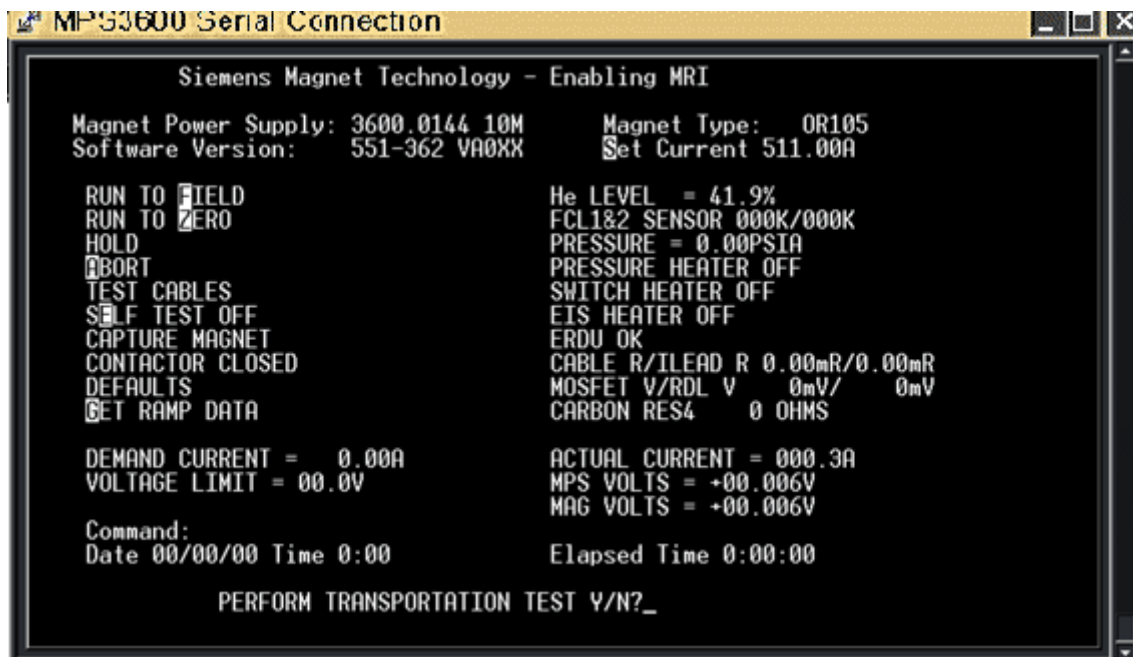


- (1) Transport test plug (fitted)
- (2) RS232 cable

The MPSU is supplied with the transport test plug connected.

- » Check that the transport test plug is connected to X3.

Fig. 88: Transport Test (example of Avanto screenshot)



- The screen prompts to 'Perform Transportation test'. Type Y <Enter>.

- The power supply will now perform a **Transit test**.

Failure of the Transport test displays briefly in the event of the test detecting an **internal fault** (→ Fig. 88 Page 90).

This is followed by a request to check the connections and repeat the test (→ Fig. 90 Page 91)

Fig. 89: Transport Test failed (example Avanto screenshot)

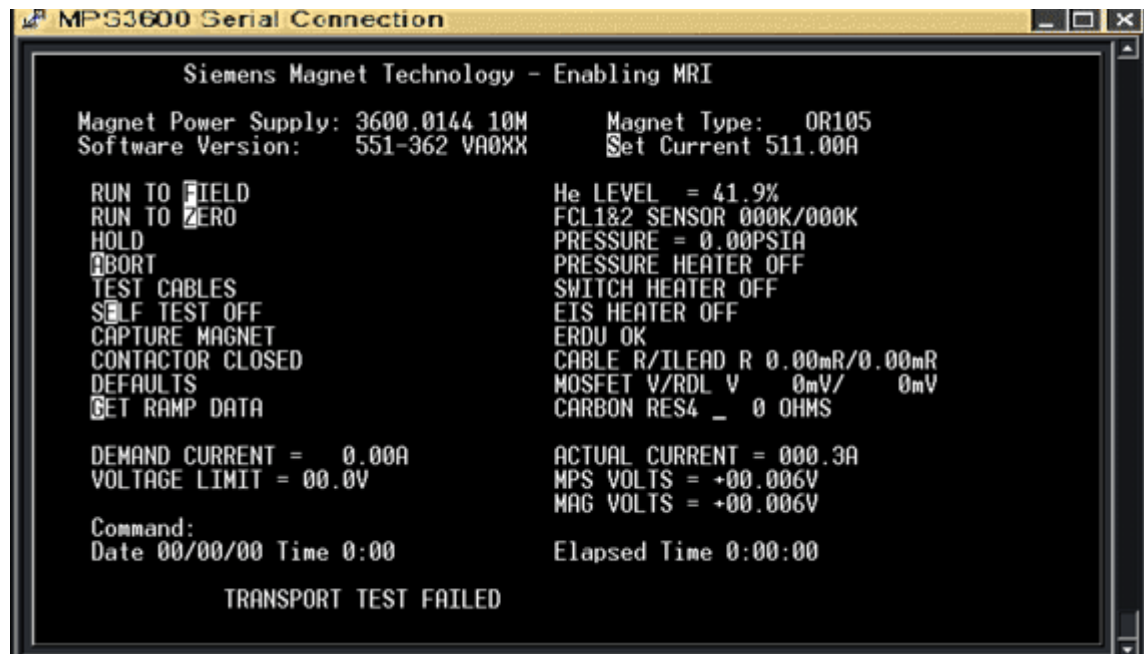
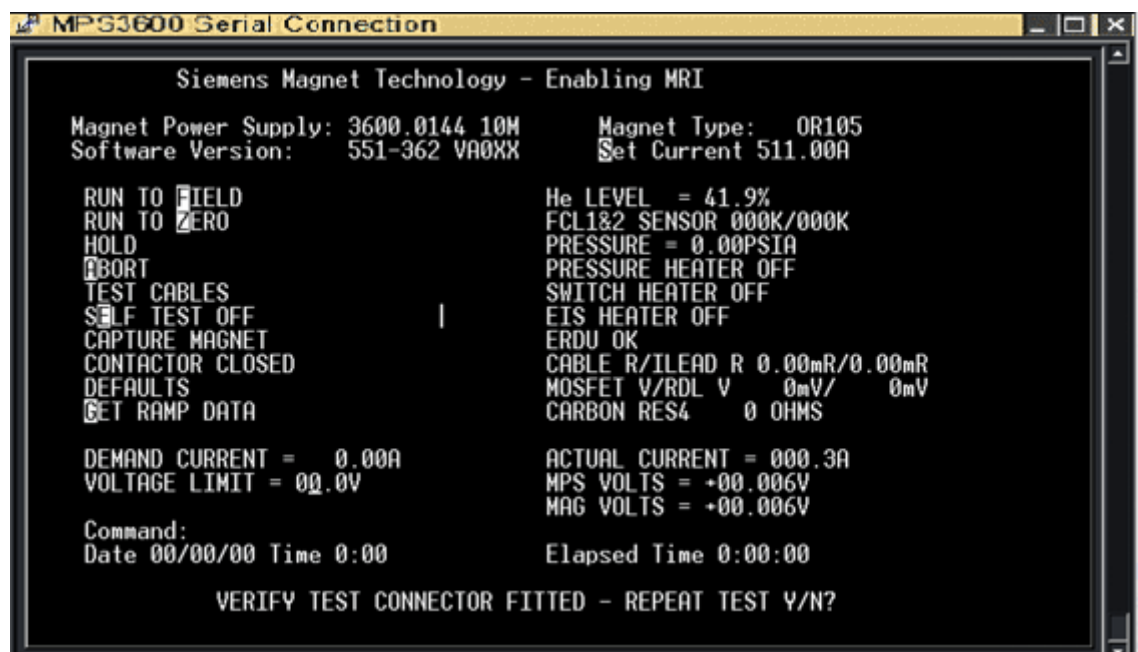


Fig. 90: Verify Test Connector (example Avanto screenshot)





After 3 successive Transport tests have **FAILED**, further operation of the 3600 MPSU is prevented.

Line power to the 3600 MPSU must be switched **OFF** then **ON** to enable further tests to be performed.



If all attempts fail, the 3600 MPSU must be returned for repair, and a replacement unit obtained.

If the test is **successful**, the power supply performs an **internal self-test**.

- The **self-test LED** indicator lights during the test. (→ 3/ Fig. 78 Page 85)
- When the self test is complete turn off the 3600 MPSU via CB1.

5.1.4.5 Ramp lead connection

After the Transport test has been successfully completed, (if applicable), the following connections must be made prior to ramping the magnet.

- **Remove the Transport test plug.**

Fig. 91: RF filter plate



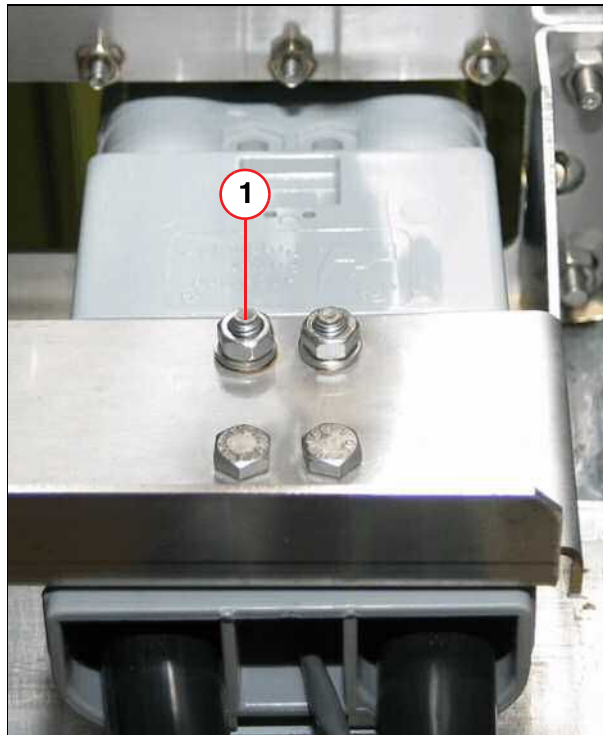
Equipment room

- Connect the **extension leads** to the **magnet leads** at the **RF filter plate**.



Ramp cables are heavy - to prevent stress on the connection, use tie wraps to support the weight if locking bolts are missing.

Fig. 92: Power lead connection



(1) Fastening bolts

Magnet room

- Prevent the **extension cables** from being accidentally **disconnected**.
- **Secure the leads** by tightening the fastening bolts.

Fig. 93: MPSU - ramp cables fitted



- Connect the **2 magnet extension ramp leads** to Anderson connectors **X3** and **X4** on the top of the 3600 MPSU.

5.1.5 Run To Field

5.1.5.1 Run To Field Prerequisites:

- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including lead checks).



Following a helium level fill, wait at least 2 hours before ramping.

- Helium level must be > 40%.
- Turn on the 3600 MPSU via CB1.
- The **boot-up** screen will be displayed.

Fig. 94: Boot-up screen

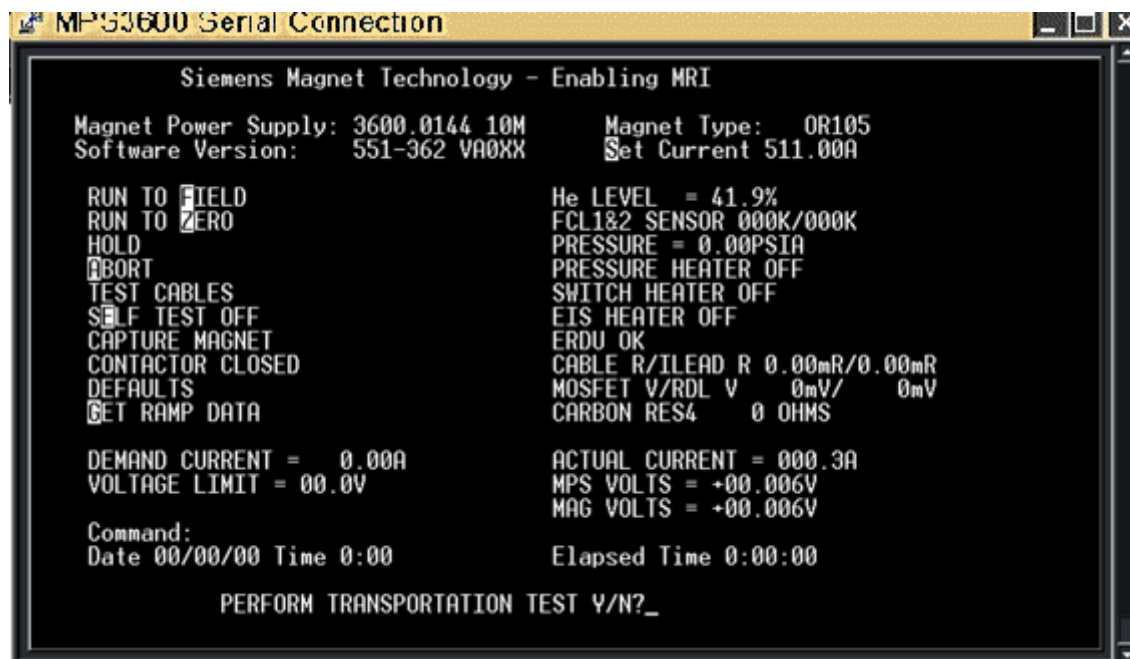
```
iniDevice: 00
Using Baudrate 250
initCan: 00
:task_init()
CANopen
Using Baudrate 250
initCan: 00
initCANopen: 00
Using Node ID 8
Define 1. Server SDO: 00
Define 2. Server SDO: 00
Define 1. Client SDO: 00
Define 1. Receive PDO: 00
Define 1. Transmit PDO: 00
Define 2. Transmit PDO: 00
Define 1. Sync: 00
Define 2. Emergency OBJ: 00
Define 1. Time Obj: 00
NMT Node created: 00
Set Domain Addr: 01
Set Domain Size: 01
Enter 3 characters for your initials...
_
```

- When prompted, enter your **initials**.

The 3600 MPSU will now retrieve the **magnet type**, **magnet serial number**, **last current set**, **He- level**, **date** and **time** from the MSUP.

- The following screen will prompt to '**Perform Transportation Test**':-

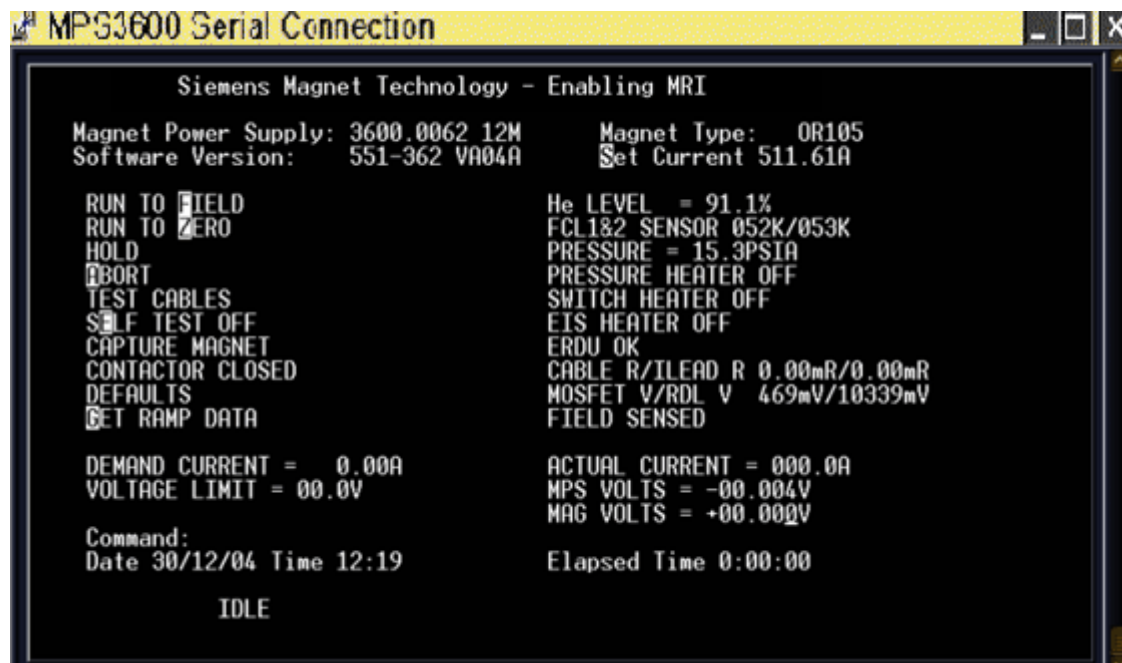
Fig. 95: Transport Test (example of Avanto screenshot)



- The **Transport Test** is not required at this point.
 - » Enter **N<ENTER>**
At the prompt 'ARE YOU SURE', enter **Y<ENTER>**
- The 3600 MPSU will now perform an **internal self-test**.
 - » The **self-test** LED indicator will light during the test. (→ 3/ Fig. 78 Page 85)

- When the self-test is complete, the **Operational screen** is displayed.

Fig. 96: Operational Screen (example of Avanto screenshot)



5.1.5.2 Setting the magnet current

- Enter the command: **S<space>514.05<ENTER>**
The numerical number may be varied as required.
- Enter the command for Run to field: **F <ENTER>**

Checks will be carried out and the 3600 MPSU will then perform an automatic ramp to field.

5.1.6 Run To Zero

5.1.6.1 Run to zero prerequisites:

- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including lead checks).



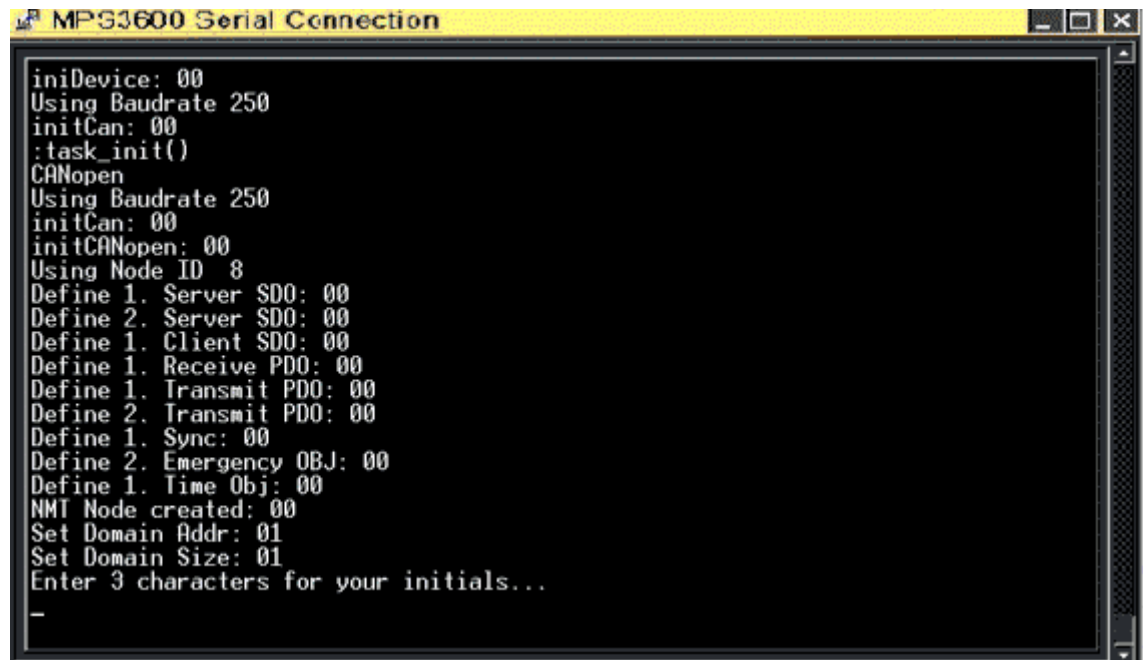
Following a helium level fill, wait at least 2 hours before ramping.

5.1.6.2 Run To Zero

- Turn on the 3600 MPSU via CB1.

- The **boot-up** screen will be displayed.

Fig. 97: Boot-up screen

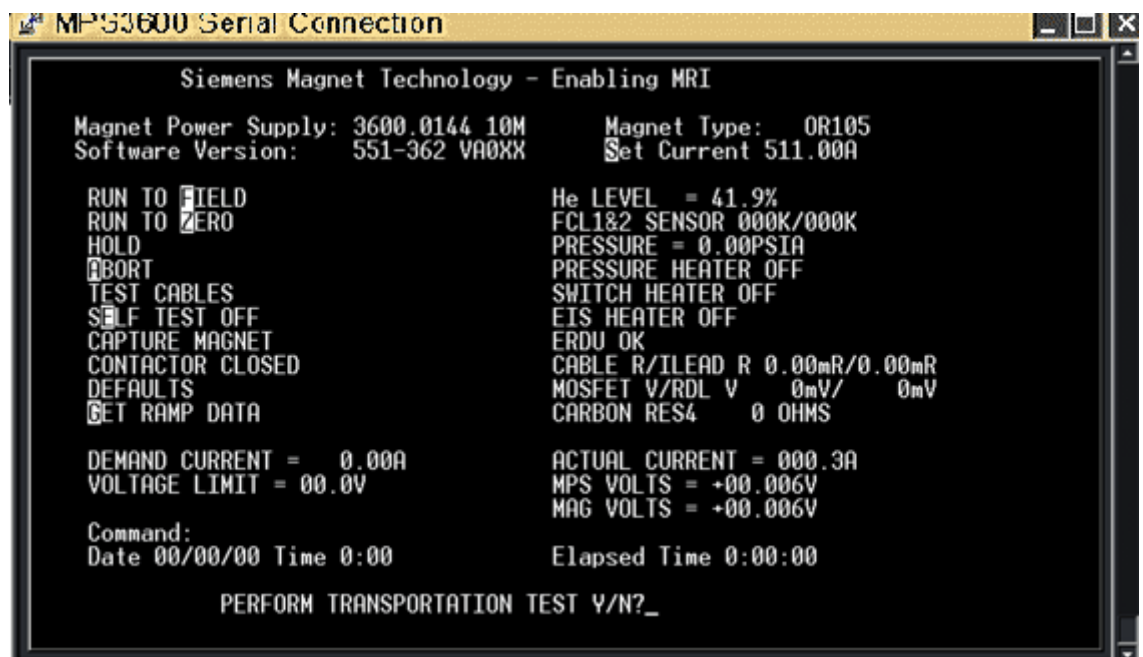


- When prompted, enter your **initials**.

The 3600 MPSU will now retrieve the **magnet type**, **magnet serial number**, **last current set**, **He- level**, **date** and **time** from the MSUP.

- The following screen will prompt to '**Perform Transportation Test**':-

Fig. 98: Transport Test (example of Avanto screenshot)



- The **Transport Test** is not required at this point.
 - » Enter **N<ENTER>**
 - At the prompt 'ARE YOU SURE', enter **Y<ENTER>**
- The 3600 MPSU will now perform an **internal self-test**.
 - » The **self-test** LED indicator will light during the test. (→ 3/ Fig. 78 Page 85)
- When the self-test is complete, the **Operational screen** is displayed.

Fig. 99: Operational Screen (example of Avanto screenshot)

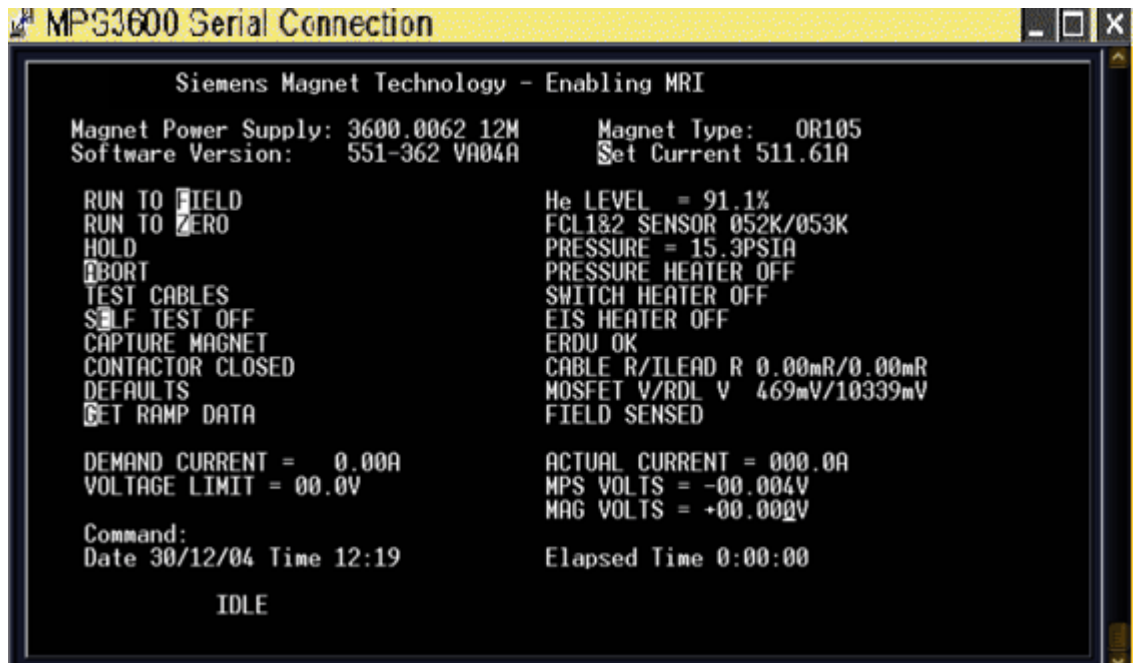
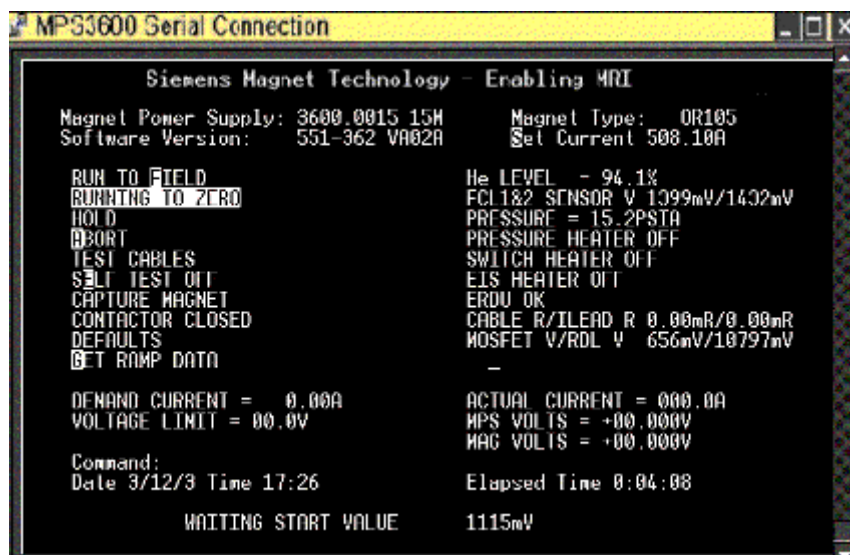


Fig. 100: Ramp to zero



- Enter the command **Z<ENTER>** for ramping down the magnet.

Checks will be carried out and the 3600 MPSU will then perform an automatic ramp to zero.

5.1.7 Ramping the magnet to the required field strength

The magnet is ramped to the upper limit of the frequency limit. Bumping the magnet involves making a small change to the magnetic field strength either up or down. The procedure is the same as ramping the magnet.

5.1.7.1 Magnet current setting

Refer to the **Magnet Acceptance Document/MSUP** to obtain the operating **current value**.



Avanto range of frequency:
Maximum frequency: 63,700 000 MHz
Minimum frequency: 63,500 000 MHz

Use the mathematical formula below to calculate the current to determine the **center frequency**.

Between ramps, the magnet system's frequency slowly reduces over time. To increase the period between ramps, aim to achieve for the **upper sector** of the frequency window.

$$\text{Magnetom Avanto: } I_{\text{final}} = (63,680 \text{ MHz} \times I_{\text{actual}}) / \text{Freq actual}$$

5.1.7.2 Bumping prerequisites:

- 3600 MPSU must be connected.
- Wait at least 1 h after completing any previous ramp, or aborted ramp (including lead checks).



After a helium level fill, wait at least 2 h before ramping.

- Helium level must be > **40%**.

The last ramp current is stored in the magnet supervisory (MSUP). The ramp current is displayed on-screen when the 3600 MPSU is switched on.

- Enter the new current value: **S<space>510.05<ENTER>**
 - The demand current (numerical number) may be changed as required.
- Enter the command **F<ENTER>** for 'bumping' the magnet.
 - Checks are carried out and the 3600 MPSU then performs an automatic bump sequence.

5.1.8 Mobile ramping



Follow the Ramping procedure (→ Run To Field / Page 94)

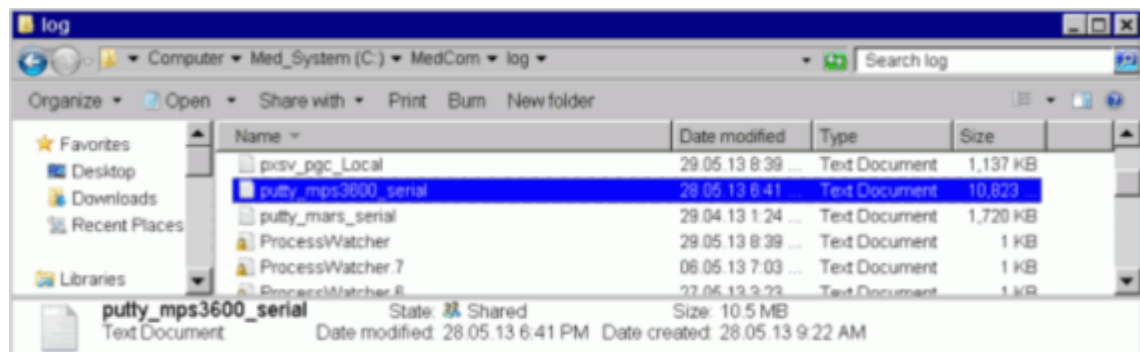
5.1.9 Work after ramping the magnet

5.1.9.1 Downloading the MPSU log file



The MPSU log file is saved under MedCom/log.

Fig. 101: MPS_logfile



- Navigate to the location C:\MedCom\log\putty_MPS3600_serial.log, zip this file, and send the log file to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>).

5.1.9.2 Switching off the 3600 MPSU



NEVER attempt to remove the magnet ramp leads when the “CURENT IN LEADS” LED is flashing.

- Set the mains circuit breaker **CB1** to the **OFF** position. (→ 2/ Fig. 75 Page 82)
- Set the circuit breaker **F53** in the **EPC** on the MPSU to the **OFF** position.
- **Disconnect** the **extension leads** from the **magnet leads** at the **RF filter plate**.

- Re-fit the adapter plate using the securing screws.

Fig. 102: RF filter plate

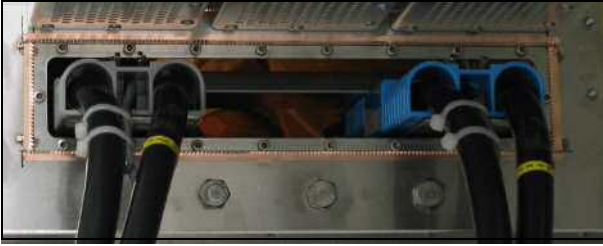


Fig. 103: Adapter plate (view from Equipment Room)



CAUTION

Overheating of FCL's/Ramp leads

Failure to observe the following warning may result in equipment damage.

» SHORTING LINKS MUST NOT BE FITTED!!

5.1.10 Final work steps

5.1.10.1 Save the magnet current to the HOST

- Access the service menu in the **SeSo** at **MRC**.
- Select menu item **"Magnet & Cooling"**.

- Select "Magnet Setup" in the "Action" menu

Fig. 104: Magnet Setup - Dot Upgrade

Siemens Med Service Software - Windows Internet Explorer

Magnet & Cooling | Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

- Actions
 - Cooling Status
 - Magnet Status
 - Compressor Status
 - Cooling Setup
 - Magnet Setup**

Magnet & Cooling - Magnet Setup

Magnet

Helium Level Probe 1	Min. (0%)	85.1	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe 2	Min. (0%)	85.1	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Warning	Min.	55	%
Pressure Heater Resistance		20.5	Ohm
Shield Temperature Warning 50K Link	Max.	74.0	K
Shield Temperature Warning 50K Bore	Max.	84.0	K
CCR Calibration	Room Temperature	0.0	Ohm
	Nitrogen Temperature	0.0	Ohm
	Helium Temperature	0.0	Ohm
Time Setting	Daily at	02:00	O'Clock
MSUP	Serial Number	1054	
	Revision Number	0	
Magnet	Serial Number	21176154	
	Revision Number	0	
Magnet Current		466.64 A	

MSUP Data

Downloads MSUP data from MSUP to host Download (1)

Uploads MSUP data from host to MSUP Upload

MSUP Logfiles

Downloads all MSUP logfiles to the host Download (2)

☐ Save values as default

Ready. | Save | Default | Update

(1) Download of MSUP data

(2) Download of MSUP files

- Click "Download" to save the magnet-current value on the HOST.

5.1.10.2 Warm Turret & Venting

Fig. 105: Warm the turret and venting with a heat gun



Ensure the **turret and venting** remains **leak tight** and the valves continue to function correctly.

- Warm up the **turret and venting** with a heat gun.

5.1.10.3 Pack the 3600 MPSU

Fig. 106: MPSU - ramp cables fitted



- Disconnect the two **3600 extension ramp leads** from the Anderson connectors **X3** and **X4**.
- Re-fit the **Transport test plug** into **X3**.
- Disconnect the remaining leads and pack inside the transport case.
- Close the lid and secure by pulling the 2 silver knobs and turning them through a 1/4 turn.

Fig. 107: MPSU



(1) Wire locking strap

- Replace the shipping case lid and secure using the 12 spring clips.
- Pull the wire strap to unlock the handle, and slide the handle out of the recess under the trolley.
- Release the wire strap to lock the handle in the extracted position.
- Raise the handle to lower the wheels and lift the trolley off the 2 skids.
- The 3600 MPSU and trolley can now be safely moved.

5.1.11 Mains failure during ramping



CAUTION

If the supply to the 3600 MPSU fails while the magnet is ramping through the ramp leads up or down, the magnet current will continue to flow.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » **Never remove the ramp leads from the magnet while current is in the magnet ramp leads.**

Current is in the ramp cable whenever:

- » The yellow “**CURRENT IN LEADS**” LED is flashing.
- » Magnet was ramping up or down and there was a mains failure.
- » Magnet was ramping up or down and the 3600 MPSU detected a fault such as over temperature, which would shut down the 3600 MPSU.
- » Normal ramp up or down of the magnet.



If the cause of the shutdown was failure of the mains, nothing can be done until both the 3600 MPSU and the Magnet Supervisory (MSUP) are working again.

When the 3600 MPSU and the MSUP are operational, ramp the magnet to zero.

Do not delay. Ramp down as soon as mains power returns.

5.1.12 Ramp failure due to FCL over-temperature

The FCLs must not exceed pre-defined temperature limits, otherwise irreparable damage will occur to the turret internals. The 3600 MPSU continuously monitors this temperature and will automatically ramp the magnet to zero if this temperature is exceeded. Providing there is a continuous flow of cold gas through the FCLs during ramping, this condition should never occur.

Possible reasons for the gas flow to be interrupted may be as follows:-

- **16 psi (A) valve** is jammed closed.
- **Blockage** of the neck tube.
 - » Ice must be considered but is unlikely during ramping.

Failure due to FCL over-temperature must be reported to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)

5.2 What to do after a Quench?

5.2.1 Post-Quench Procedure

5.2.1.1 General

If the magnet is quenched deliberately (via the ERDU) or accidentally, a large volume of helium gas is generated within the helium vessel. The **quench valve opens**, and stays open until the flow subsides. The valve then re-seats, re-sealing the venting and thus preventing any air (caused by cryo-pumping) from being sucked back into the helium vessel.

The MSUP automatically maintains the system pressure above atmospheric. **If not, contact the Magnet HSC immediately.** (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)



The Magnet HSC must be notified if a quench is detected.

Fill out the **Quench Report database**, (→ Fig. 135 Page 129) and send to Magnet HSC. (→ Quench Report, <https://erlh1uqa.ww005.siemens.net/mrquality/mr/format/Index1.htm>)



CAUTION

If the graphite or auxiliary burst disk breaks.

Failure to observe the following will result in ICE FORMING within the magnet.

- » Replacement of burst disk(s) must be performed by a qualified CSE.
- » Replace the graphite or auxiliary vent bursting disk IMMEDIATELY.
- » DO NOT RAMP THE MAGNET.
- » Inform Magnet HSC. (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)



For all types of quench, allow up to 1 h for the quench pressure to subside before opening the magnet.

Safety



WARNING

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogenes.

- » Ensure that the room is ventilated.
- » O₂ monitors must be used in the imaging suite.



CAUTION

RISK OF ICING!!

Air ingress will cause INTERNAL ICING OF THE TURRET.

- » Replace broken or damaged burst disk/O-rings IMMEDIATELY.
- » Check that there is a free flow of gas through the magnet bypass valve. If there is no gas flow, there may be an ice blockage.
- » Report the icing to Magnet HSC IMMEDIATELY.
- » Do not fill or ramp the magnet until advised to do so by Magnet HSC.
- » DO NOT OPEN OR REMOVE THE TURRET TOP PLATE.

Required tools

- Heat gun
- Standard tool kit
- Helium leak detector (part no **10614850**)
- Laptop and RS232 cable
- O₂ monitor
- Eye protection
- Gloves

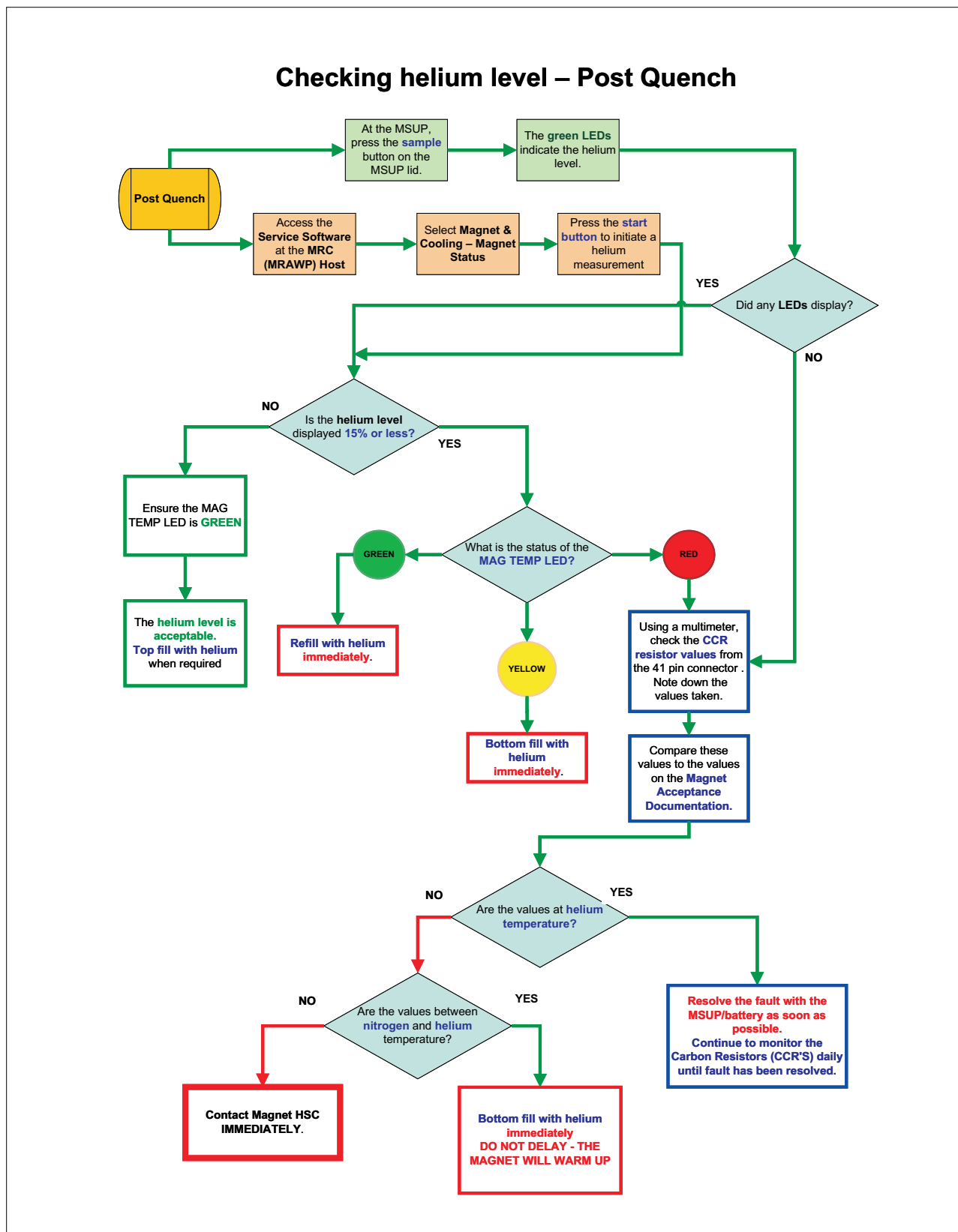
Post quench steps

Follow the work steps in the basic guide below for the different types of quench.

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent quench or ERDU button pressed
Step 1	- Download the 3600 MPSU ramp log files. - Isolate the MPSU from the magnet, and the mains power.	Stop filling.	Check that no ERDU buttons are pressed. Re-set ERDU button.
Step 2	Check the liquid helium level. at the Host, MRC - Magnet & Cooling. (→ Fig. 108 Page 110) Check the ceramic carbon resistors at the MRC. (→ Check the Ceramic Carbon Resistor (CCR) values / Page 112)		
Step 3	Order about 1000 liters liquid helium ¹		
Step 4	Warm up and check the quench valve and aux vent burst disk are intact. Refer to Checking the quench valve and auxiliary burst disks. (→ Checking the quench valve and auxiliary burst disks / Page 115)		
Step 5	After 2 h: Download the Mag Supervisory log files. (→ Log file capture / Page 111) Helium fill. Refer to Start up.		
Step 6	Leak check the quench valve and venting. Refer to Turret leak check.		
Step 7	Fill out the Quench report. Refer to Quench report (→ Fig. 135 Page 129).		
Step 8	Ramp the magnet. Refer to Start up		
Step 9	- Perform the shim check » If the shim check exceeds the specifications, a re-shim is necessary.		
Step 10	Contact Magnet HSC. (→ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com) » Email Mag 3600 Power Supply and Mag Supervisory logs to: Mag-net HSC with magnet serial no. and all log data. Refer to Startup		

-
1. The quantity of the order depends on the helium volume already on site.

Fig. 108: Check helium level - Post Quench



5.2.1.2 Unattended quench

- If the **Magnet STOP (ERDU)** has been activated:
 - » the yellow **WARNING LED** lights on the alarm box.
 - » the alarm **buzzer** is activated.
 - » Ensure the ERDU button is **still not activated**.
- If the **helium level** is low:
 - » the yellow **WARNING LED** lights on the alarm box.
 - » the alarm **buzzer** is activated.
- If the **magnet pressure** is very low, the
 - » **cold head** switches off.
 - » the yellow **WARNING LED** lights on the alarm box.
 - » the alarm **buzzer** is activated.

Press the **AUDIO ALARM OFF** button on the alarm box to silence the audible alarm.

5.2.1.3 Log file capture

Magnet Supervisory

- The **quench data is lost**, if:
 - a) The system is **re-booted** within 2 h of a quench
 - b) The system is **turned off** within 2 h of a quench
 - c) The data is downloaded within 2 h of a quench
- Download the MSUP log files (refer to TSG - **Downloading MSUP 2306 data to the MRC**).

If the magnet quenches during ramping:-

- Download the **Magnet Power Supply** log file (see Installation - Startup - Downloading MPS Log files).



The MPSU log files **MUST** be downloaded **BEFORE** the next ramp is performed, otherwise previous data is overwritten!

If the magnet quenches during helium fill:

- **Stop** the transfer.

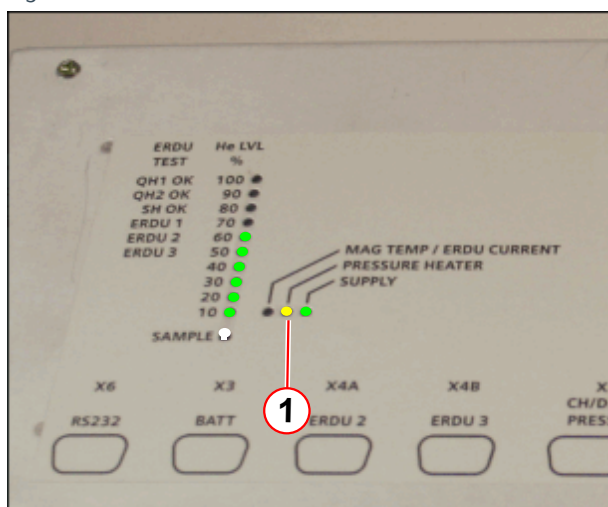


Do not remove or disconnect the syphon. The magnet is at high pressure.



For all types of quench, allow up to 1 h for the quench pressure to subside before opening the magnet..

Fig. 109: MSUP - Pressure heater LED



(1) Pressure heater LED - yellow

Check the **pressure heater status**

- Check the MSUP **PRESSURE HEATER LED** is lit, **or**
- Access the **Service Software** at the **MRC (MRAWP) Host**.
(→ Fig. 114 Page 116)

If the pressure control system is **NOT** working, the pressure inside the cryostat falls to atmospheric levels. Ensure the compressor **switches off**.

5.2.1.4 Check the Ceramic Carbon Resistor (CCR) values



Compare the temperatures with the values on the Magnet Acceptance Document.

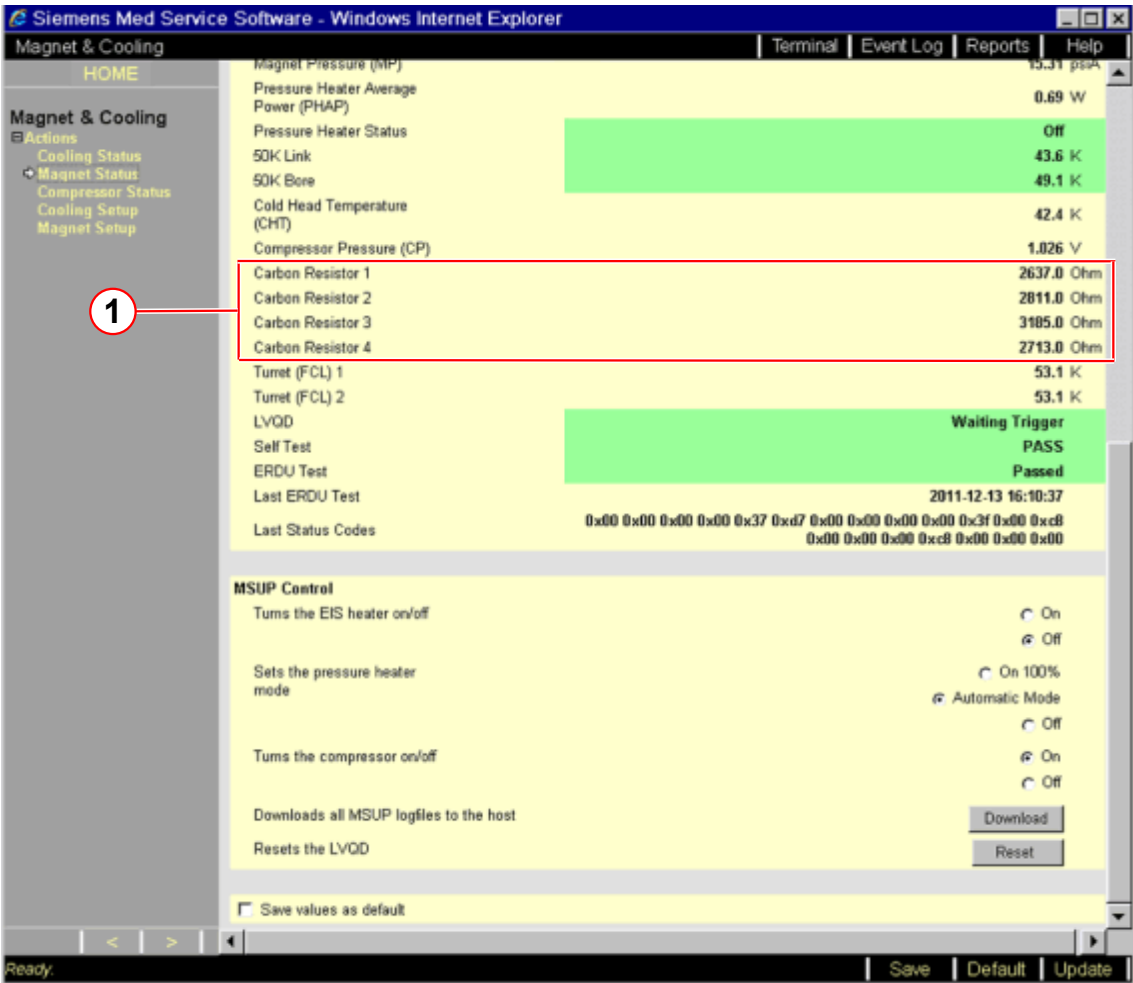
Fig. 110: OR105 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation data MR Magnet Technology Quality Regulations	
Cryostat number	21326022	SMT reference	21330107 12.11
System description	1.494T Actively Shielded Superconducting Magnet System OR105		
Measured magnet centre (Offset)	+1 (P) mm		
If magnetic offset is outside the spec (± 5mm), the magnet will be marked (± 5mm).			
 This system requires <u>0</u> Z2 rings and <u>2.92 kg of iron</u> to shim. This is a theoretical calculation that may change due to site conditions.			
Operating parameters			
Description	Specification value	Measured value	
Operating current @ 1.494T	> 500 Amps	510.01 Amps	
Field/Current Ratio	N/A	12.48 kHz/0.1 Amp	
Magnet type:	OR105-01		
TDC calibration values		Carbon resistor values	
Room temperature (23 °C)	1009 Ohms	Calibration values	
Liquid nitrogen temperature (77K)	1309 Ohms		
Liquid helium temperature (4.2K)	2994 Ohms		
Position	Room temperature	Helium temperature	
TDC d - e	1034 Ohms	2837 Ohms	
Top a - e	980 Ohms	3014 Ohms	
Middle b - e	1110 Ohms	2681 Ohms	
Bottom c - e	1006 Ohms	2930 Ohms	
Helium level probe resistance (when using a 250mA supply)			
	Probe 1	Probe 2	
@ 0 resistance	85.36 Ohms	85.36 Ohms	
0% calibration	774	774	
100% resistance	0.22 Ohms	0.22 Ohms	
100% calibration	2	2	
Diode table no.	-1		
Pressure heater resistance	20.5 Ohms		
Final temperature sensor recon values			
Shield bore	Shield Link	Coldhead	Recon margin
47K	47K	42K	658mW
Checked by	Signed		
V Turner			
	Date 23/12/2011		
	Page 1 of 1 25-AD EPS 000 01		

- Access the **Service Software** at the **MRC (MRAWP) Host**.

- Select the menu item **Magnet & Cooling - Magnet Status**.

Fig. 111: Checking the Ceramic Carbon Resistors



(1) Resistor values

Tab. 10 Expected CCR resistor values

41 way plug, measured in Ohms	Expected values room temperature	Expected values 77 K temperature	Expected values 4.2 K temperature
All ceramic carbon resistors (CCRs)	880 - 1050	1150 - 1350	2800 - 3380

i

The magnet helium level can be recovered if the temperature of the CCR is still colder than nitrogen temperature.

The CCR (CARBON RES 1) reading must be in the range of between 1200 and 3400 ohms.

i

If temperatures are between N2 and helium temperature (77K), a BOTTOM FILL must be performed IMMEDIATELY.

5.2.1.5 Checking the quench valve and auxiliary burst disks

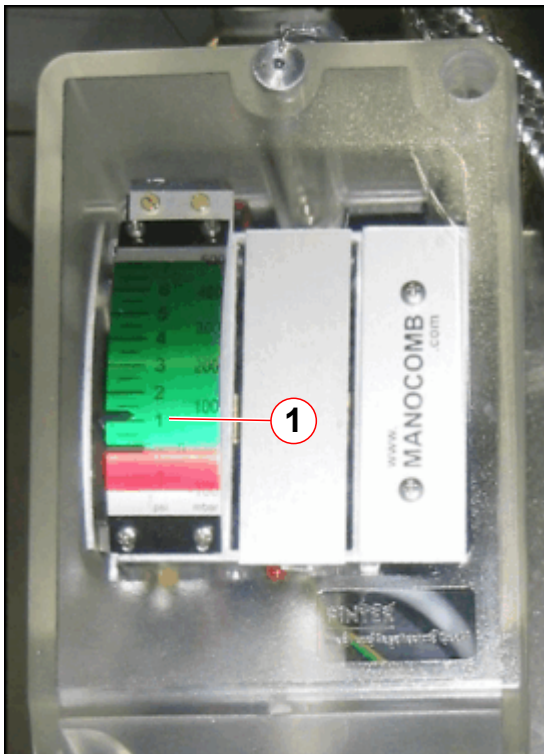
After helium filling, the quench valve and venting must be leak checked. Ensure that the helium vessel has a **positive** pressure.

- at the **pressure gauge**.
- at the **MRC (MRAWP) Host**.

Monitor the helium vessel pressure

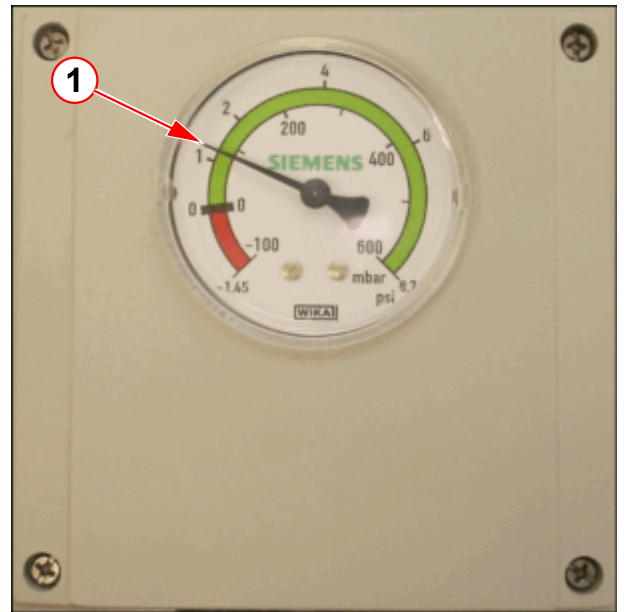
- If the **pressure gauge** is within the **RED zone**, the magnet is **subatmospheric**.
- **DO NOT** open the magnet **bypass valve**, or disturb any high-pressure fitting.

Fig. 112: Pressure gauge (expected pressure)



(1) Expected pressure 1.3 psi (G)

Fig. 113: WIKA pressure switch - expected pressure



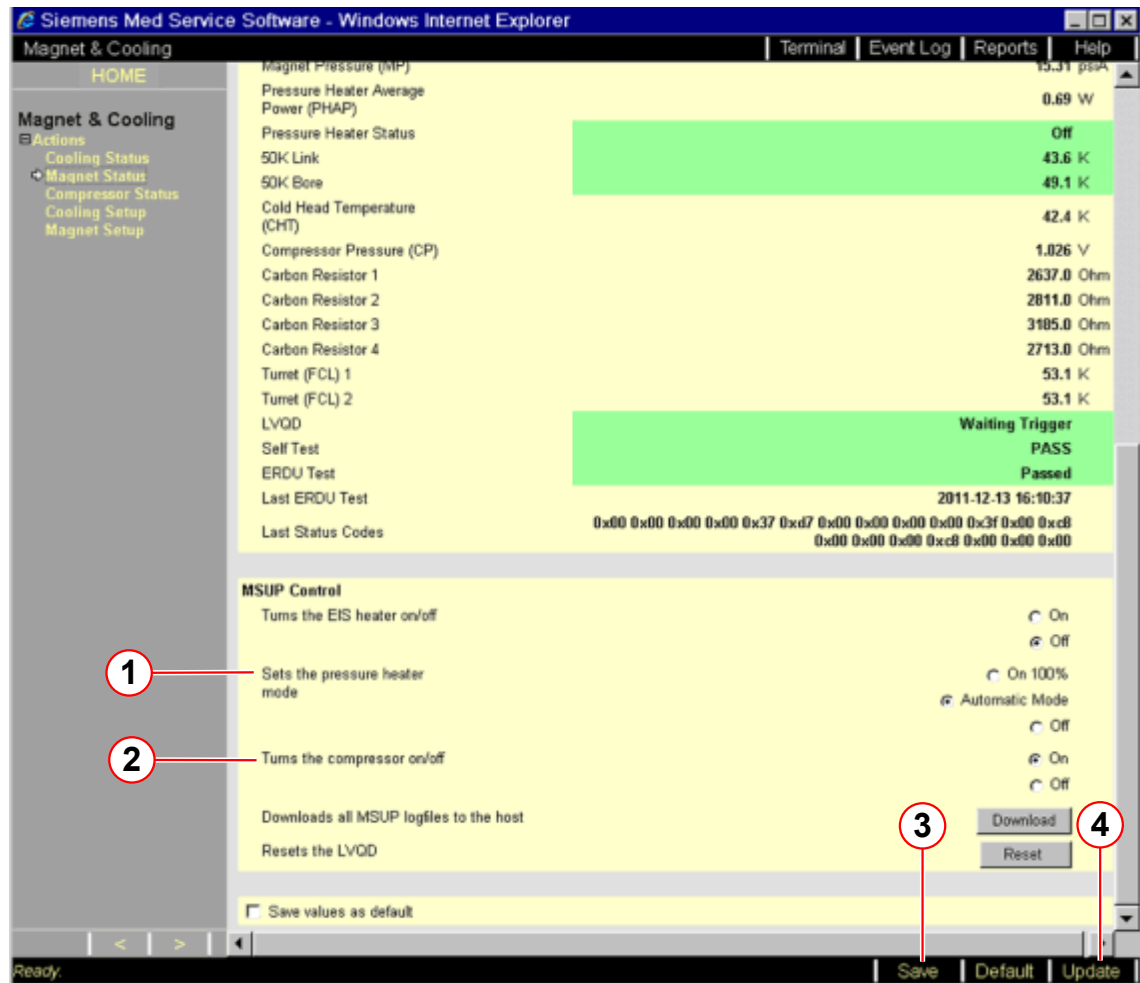
(1) Expected pressure 1.3 psi (G)

Deactivate the pressure heater

- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- At **MSUP Control** - Sets the pressure heater mode select the **OFF** button.

- Select **SAVE**.

Fig. 114: Service screen Mag&Cooling - Magnet Status - MSUP Control



- (1) Sets the pressure heater mode
- (2) Sets the compressor on/off
- (3) Save
- (4) Update



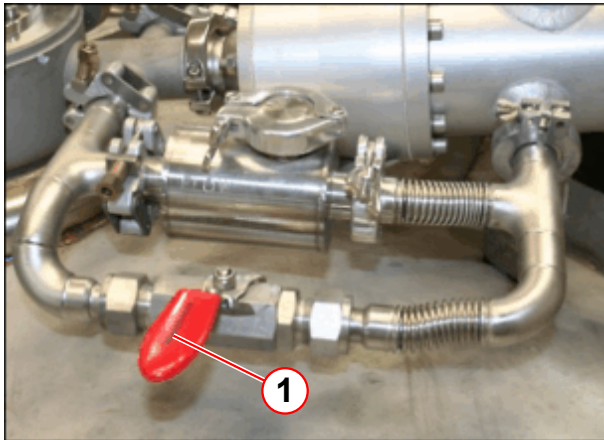
The pressure heater reactivates within 60 min.

5.2.1.6 Check the quench valve and aux burst disks are intact

- Check the **aux burst disk** first, as it is the most accessible.
- Under normal quenching circumstances, the **OVC burst disk** remains intact. **It must be inspected for damage to remove the risk of cold gas reaching the OVC POD.**

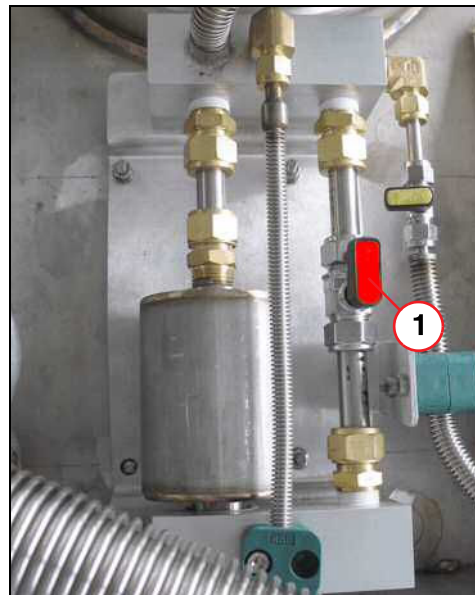
Depressurize the helium vessel

Fig. 115: Mk1 Vent valve assembly



(1) Bypass valve

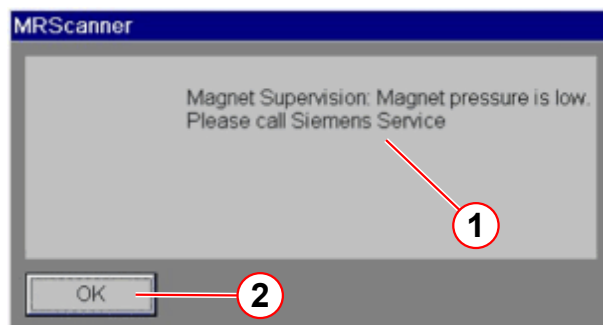
Fig. 116: Mk2 Venting bypass valve



(1) Bypass valve (show open)

- **Fully open** the **magnet bypass valve** to **depressurize** the helium vessel.
- Wait until **zero psi** is displayed on the pressure gauge.
- The **pressure switch** activates.
 - » the **cold head** turns **off**.
- **Close** the magnet bypass valve to prevent **air ingress**.
- **Warm up** the vent valve assembly.
 - » **Take care** not to damage any components when using a **heat gun**.

Fig. 117: MSUP - Magnet pressure is low



(1) Pop-up message
(2) Click OK to clear

On the MR Scanner:

- A pop-up **alarm message** is displayed.
 - » **Magnet pressure is low.**
 - » Click **OK** to clear the alarm.
- The **alarm box** sounds when **pressure falls to alarm limits**.
 - » The yellow **WARNING LED** lights on the **alarm box**.
 - » The alarm **buzzer** is **activated**
 - » Press **AUDIO ALARM OFF** to silence the alarm.

- When the magnet has reached **zero pressure**, warm up the **quench valve and venting** to room temperature, using a **heat gun**.

5.2.1.7 Auxiliary vent burst disk



CAUTION

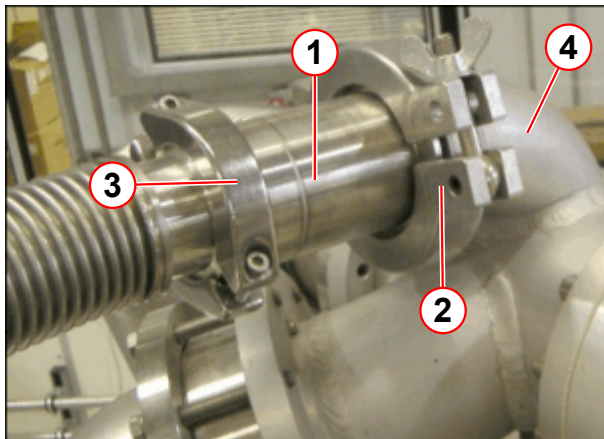
If the graphite or auxiliary burst disk breaks.

Failure to observe the following will result in **ICE FORMING** within the magnet.

- » Replacement of burst disk(s) must be performed by a qualified CSE.
- » Replace the graphite or auxiliary vent bursting disk **IMMEDIATELY**.
- » **DO NOT RAMP THE MAGNET.**
- » Inform Magnet HSC. (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)

Mk1 Auxiliary burst disk

Fig. 118: OR105 Mk1 Auxiliary burst disk assembly

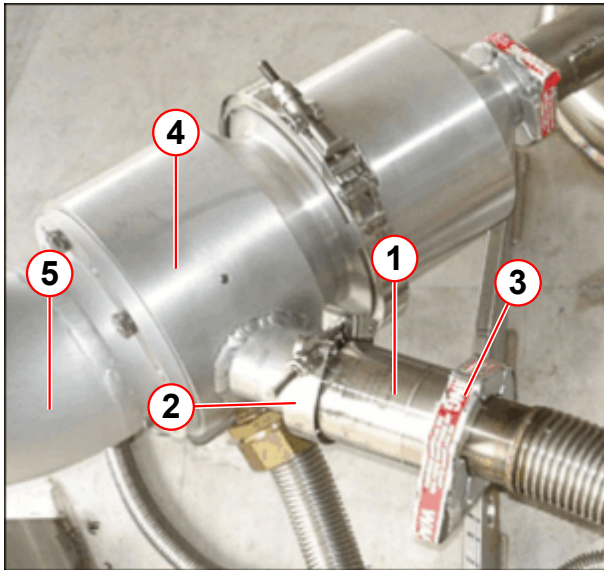


- Remove the **NW50 clamp** from the manifold on the **low-pressure** side of the aux burst disk.

- (1) Aux burst disk assy
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

Mk2 Auxiliary burst disk

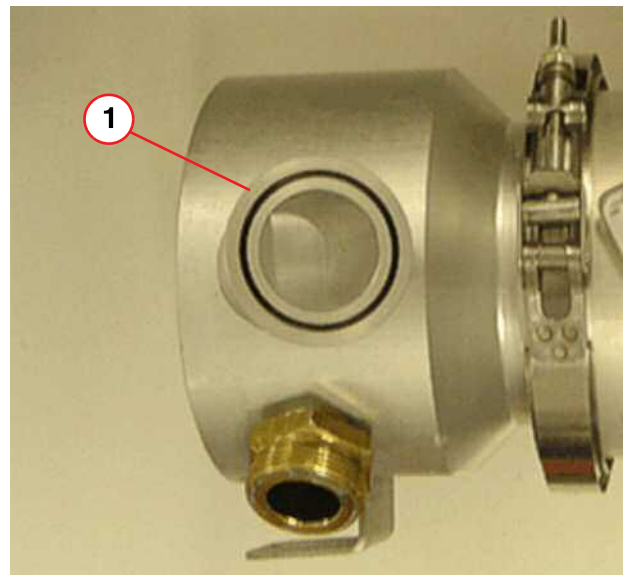
Fig. 119: Mk2 Auxiliary burst disk assembly



- (1) Aux burst disk
- (2) V-clamp (low pressure)
- (3) NW40 clamp (high pressure)
- (4) Manifold
- (5) Quench elbow

Mk2 vent interface

Fig. 120: Mk2 Auxiliary vent interface



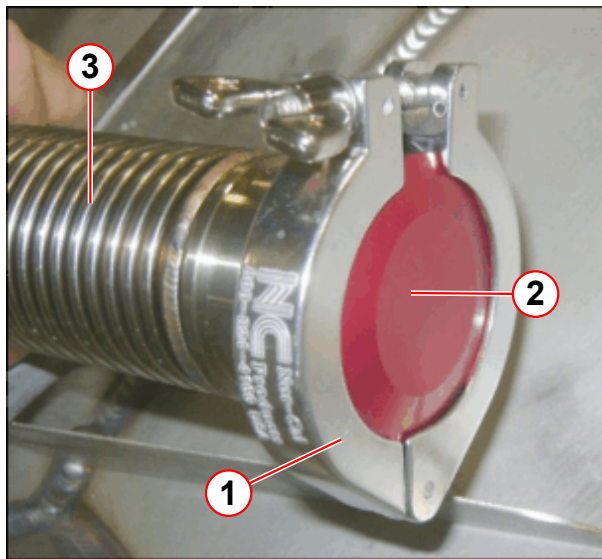
- (1) O-ring seal for aux vent

- Undo and remove the **hinged V-clamp** from the **low-pressure** side of the aux burst disk
- Visually inspect the **internal membrane** of the burst disk for damage.
- Visually inspect the **O-ring** on the low-pressure side.



The membrane is very fragile - DO NOT TOUCH THE SURFACE.

Fig. 121: Aux vent - fitting the NW40 blanking plate



- (1) NW40 high-load clamp
- (2) NW40 red anodized blanking plate
- (3) Aux vent line

If damaged in any way:

- Undo and remove the **NW40 high-pressure clamp**
- **Remove** the broken burst disk and metal seal.
- Fit the **red NW40 blanking plate** in place of the burst disk.
- **Refit** the NW40 high-pressure clamp.
 - » Contact Magnet HSC **immediately**
 - » order a replacement aux bursting disk
- Place the blanked **aux vent line** to one side and continue with checking the **quench valve**.



WARNING

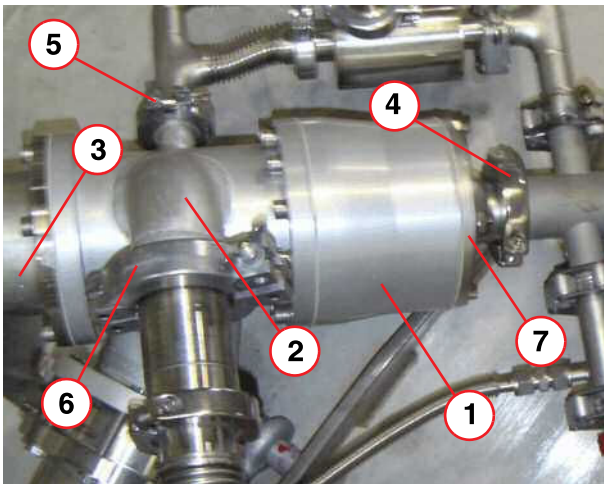
Safety device removed!!!!

The magnet must not be ramped with the auxiliary burst disk removed or broken.

- » **Replace the auxiliary vent burst disk before ramping.**

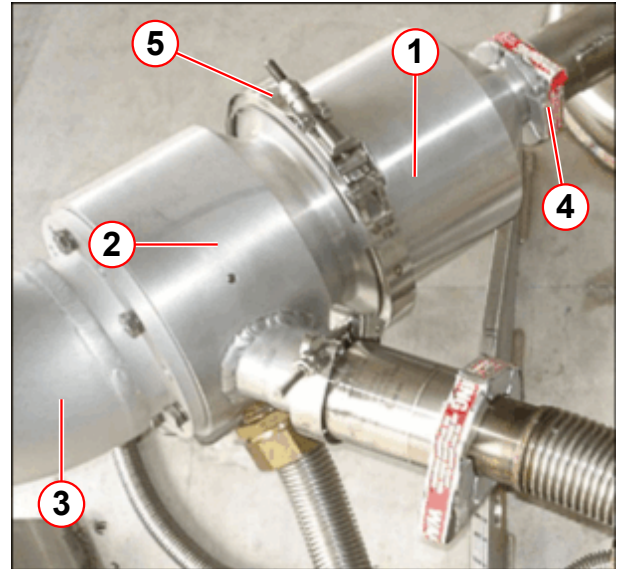
5.2.1.8 Quench valve burst disk and O-ring seal

Fig. 122: OR105 Mk1 Manifold/Quench Valve



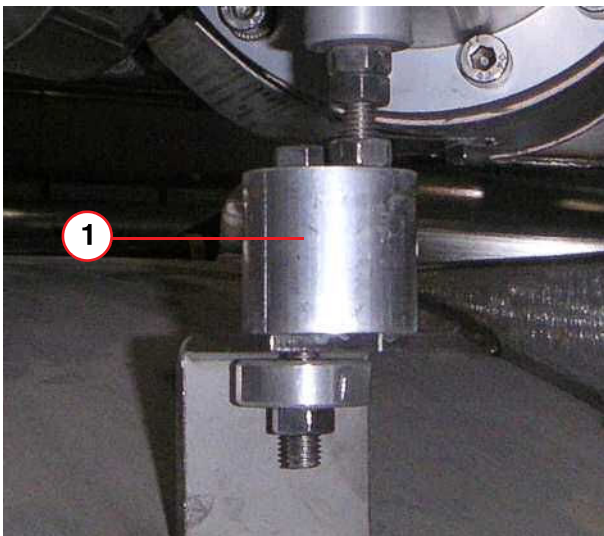
- (1) Quench valve
- (2) Mainfold
- (3) Quench elbow
- (4) NW40 high pressure clamp
- (5) NW25 low pressure clamp
- (6) NW50 low pressure clamp
- (7) Turret flange

Fig. 123: Mk2 Manifold/Quench Valve



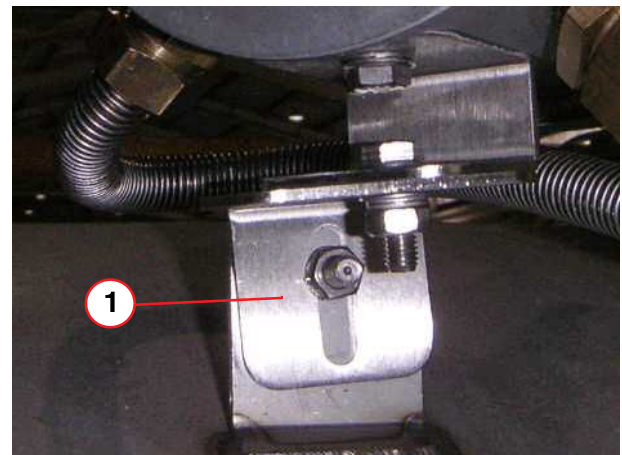
- (1) Quench valve
- (2) Manifold
- (3) Quench elbow
- (4) NW40 clamp (high pressure)
- (5) V-clamp

Fig. 124: Mk1 Support Bracket



- (1) Support Bracket

Fig. 125: Mk2 Support bracket



- (1) Support bracket

- **Remove the quench elbow**, or straight coupling if fitted. (→ 3/ Fig. 122 Page 121) (→ 3/ Fig. 125 Page 121)
- **Slacken the support bracket** underneath the **quench valve manifold**, (→ Fig. 125 Page 121), (→ Fig. 124 Page 121)

■ **Mk1 -**

- **Remove** the NW25 (low pressure) **clamp** to release the **16 psi (A) valve assembly**.
(→ 5/Fig. 122 Page 121)
- **Remove** the NW50 (low pressure) **clamp** to release the **aux vent line**.
(→ 6/Fig. 122 Page 121)
- **Remove** the (12 off) M8 cap **screws** securing the **quench valve manifold** to the quench valve. Place the manifold to one side, allowing access to the quench valve.
- **Mark the orientation** of the quench valve, then **remove** the 4 off M8 nuts and Nordlock washers that secure the **quench valve** to the **turret flange**.
(→ 7/Fig. 122 Page 121)



Remove the 4 off M8 nuts and nordlock washers in a diametric sequence. Undo the nuts slowly and evenly - the valve is spring loaded.

■ **Mk2**

- **Remove** the **V-clamp** securing the **quench valve manifold** to the **quench valve**. Place the manifold to one side, allowing access to the quench valve.
(→ 5/Fig. 123 Page 121)
- **Mark the orientation** of the quench valve, then remove the **NW40 (high pressure) clamp and seal**, securing the **quench valve** to the **turret**. (→ 4/Fig. 123 Page 121)

- **Re-open** the magnet **bypass valve** (to ensure zero pressure).
- **Lift the quench valve** from the turret flange.

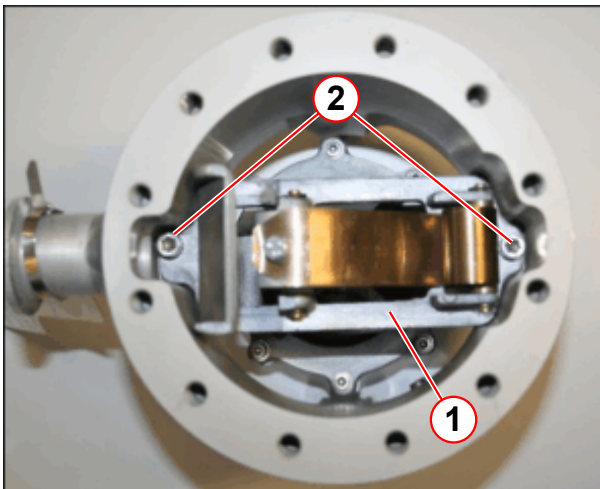


To minimize air ingress, immediately cover the open turret port temporarily, using the NW40 blank.

Use well compacted paper towelling if the blank is not available.

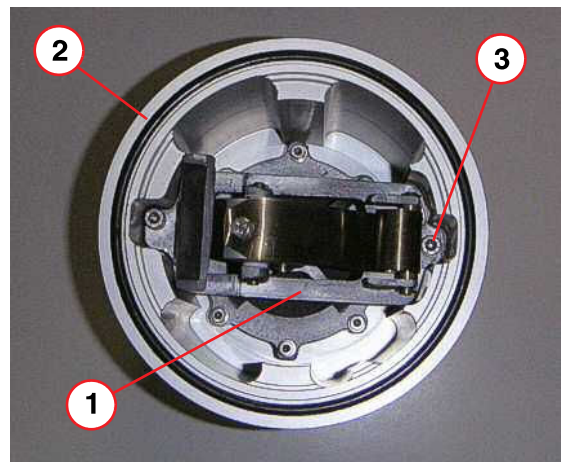
Quench Valve Inspection

Fig. 126: Mk1 Quench valve assembly (low pressure side)



- (1) Burst disk assembly
- (2) M5 cap head screws (2 off)

Fig. 127: Mk2 Quench valve



- (1) Burst disk assembly
- (2) O-ring
- (3) M5 Cap head screws (2 off)

- **Evenly** unscrew and remove the (2 off) M5 cap head screws and lift out the **burst disk assembly** from the quench valve body.

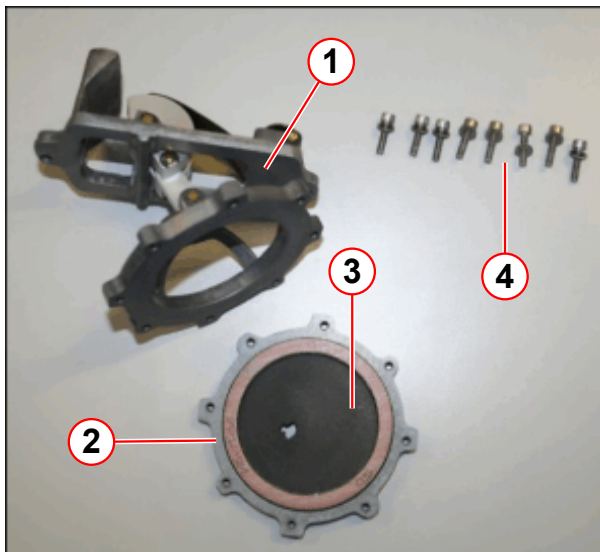


There is no requirement to remove the spring plate on the Mk1 quench valve.



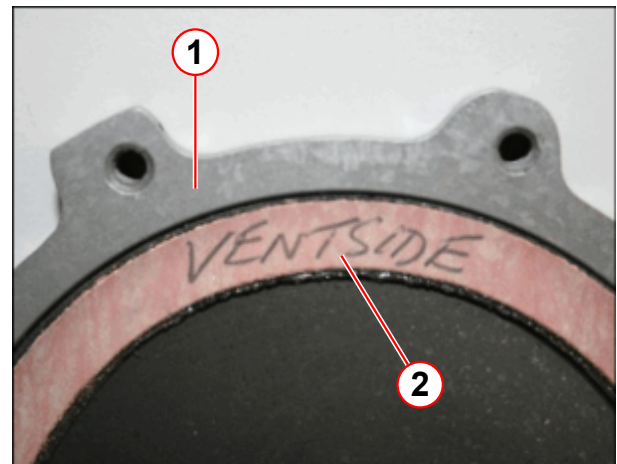
The spring exerts significant force on these screws.

Fig. 128: Burst disk assembly



- (1) Spring plate assembly
- (2) Disk housing
- (3) Broken disk
- (4) M4 screws (8 off)

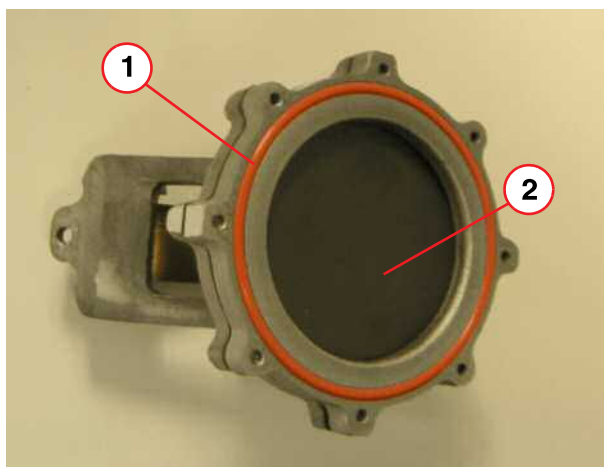
Fig. 129: Burst disk fitment



- (1) Burst disk housing
- (2) Burst disk (Vent side uppermost)

Burst disk Inspection

Fig. 130: Quench Valve Bursting Disk Assembly



- (1) O-ring
- (2) Burst disk

- **Inspect the burst disk.**
 - » If the **burst disk** has ruptured, or is cracked, it must be replaced **immediately**.
 - » (Refer to **Replacement the Quench Valve Burst disk (REP)**).
- Inspect the **burst disk 'O-ring**. Replace if damaged.
- **Clean** the O-ring groove and O-ring before re-fitting.
- **Ensure** the O-ring is **seated evenly** all around the groove.
- Re-fit the **burst disk assembly** into the **quench valve body**.
- **Check** the quench valve manifold **O-ring** is in good condition. (→ 2/ Fig. 127 Page 123) **Replace if damaged**.
- **De-ice** the turret flange.
- **Remove** the paper towelling (or NW40 blank if used) from the turret port. Ensure the surfaces are clean and dry.

- **Immediately** re-fit the **quench valve assembly** to the **turret flange**, using **NEW** seals.
 - Ensure correct **orientation**.
- **CLOSE** the magnet **bypass valve**.
- **Leak check** the burst disk.
- **If no leaks are detected**, re-fit the **quench manifold** to the **quench valve**.
 - **Mk1**- using the 12 off M8 cap screws (→ 5/ Fig. 122 Page 121)
 - **Mk2** - using the V-clamp (→ 4/ Fig. 123 Page 121)
- **Tighten** the support **bracket** underneath the quench valve **manifold**. (→ Fig. 125 Page 121)
- **Mk1** - Re-fit
 - the **16 psi (A) valve** assembly
 - **aux vent** burst disk
 - the **OVC burst disk**
- **Mk2** - Re-fit
 - **aux vent** burst disk



If the auxiliary vent burst disk was broken, do not attempt to refit it until a replacement is on site.

Refer to Replacement of Parts procedure.

5.2.1.9 OVC Burst disk



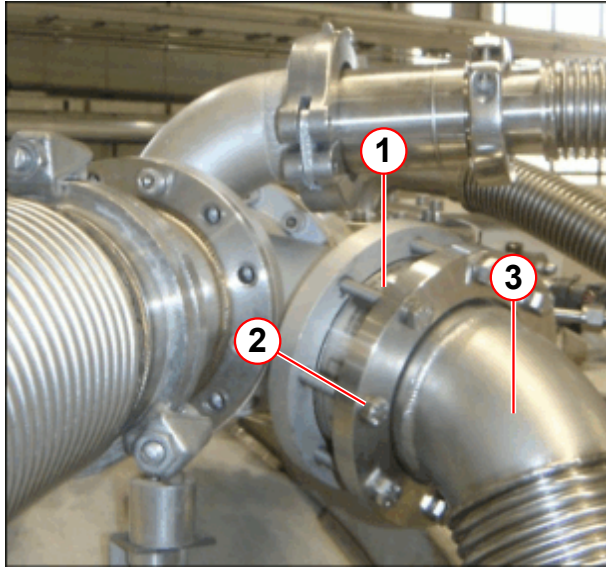
The OVC (DOP) port burst disk and vent line has been removed from the Mk2 magnet design. See Table for system serial number change.

Tab. 11 Venting configurations

Venting	OR105 Avanto serial #	OR122 Esprea serial #
Mk1	Before System MR25575	Before System MR30125
Mk2	From System MR25575	From System MR30125
Mk 2 (No DOP)	From System MR57232	From System MR31427

Mk1 OVC burst disk assembly

Fig. 131: OR105 Mk1 OVC burst disk assembly



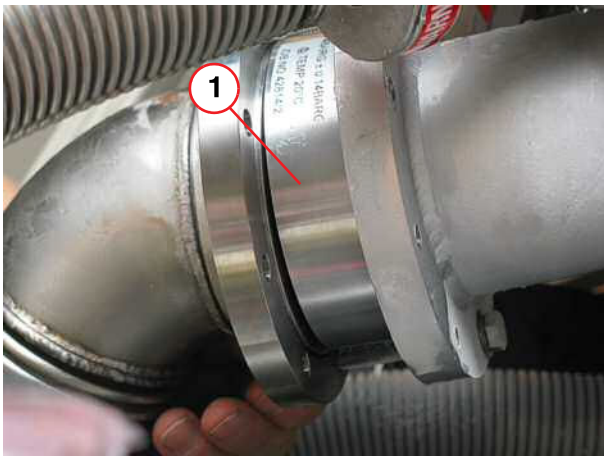
- (1) OVC burst disk
- (2) Screws (8 off)
- (3) OVC vent tube

- **Remove the (8 off) screws** holding the OVC burst disk in place.



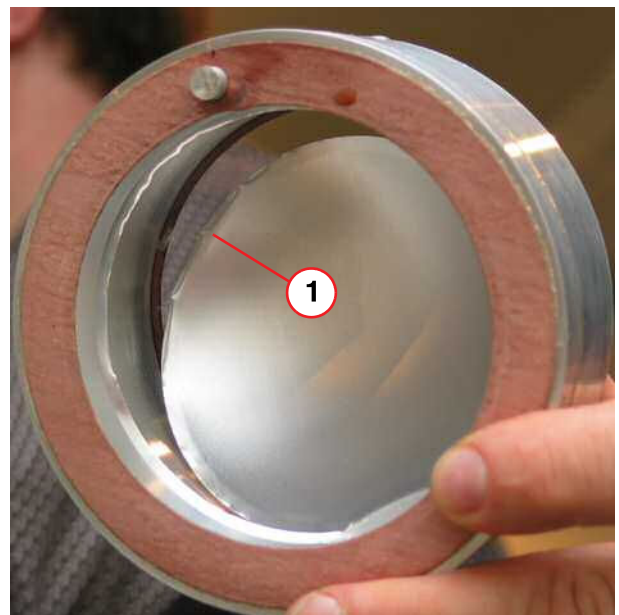
The membrane is very fragile - DO NOT TOUCH THE SURFACE.

Fig. 132: OVC burst disk



- (1) OVC burst disk

Fig. 133: OVC burst disk



- (1) Broken OVC burst disk



The burst disk has a metal peg in one face to ensure correct orientation of the disk.

- Carefully lift out the burst disk.
- Visually inspect the **internal metal** membrane for damage.
- Replace if damaged in any way - refer to **Replacement of Parts**.
- **Re-fit** the **OVC burst disk**.
- There is **no requirement** to check the Mk2 OVC burst disk, post quench.

5.2.1.10 Reactivate the pressure heater

- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- At **MSUP Control - Sets the pressure heater mode** select the **On 100%** button.
- Select **SAVE**.
- **At Real-Time Values** - Monitor the **magnet pressure**.
 - Select **Update** to refresh data until pressure reaches ≥ 15.3 psi (A).

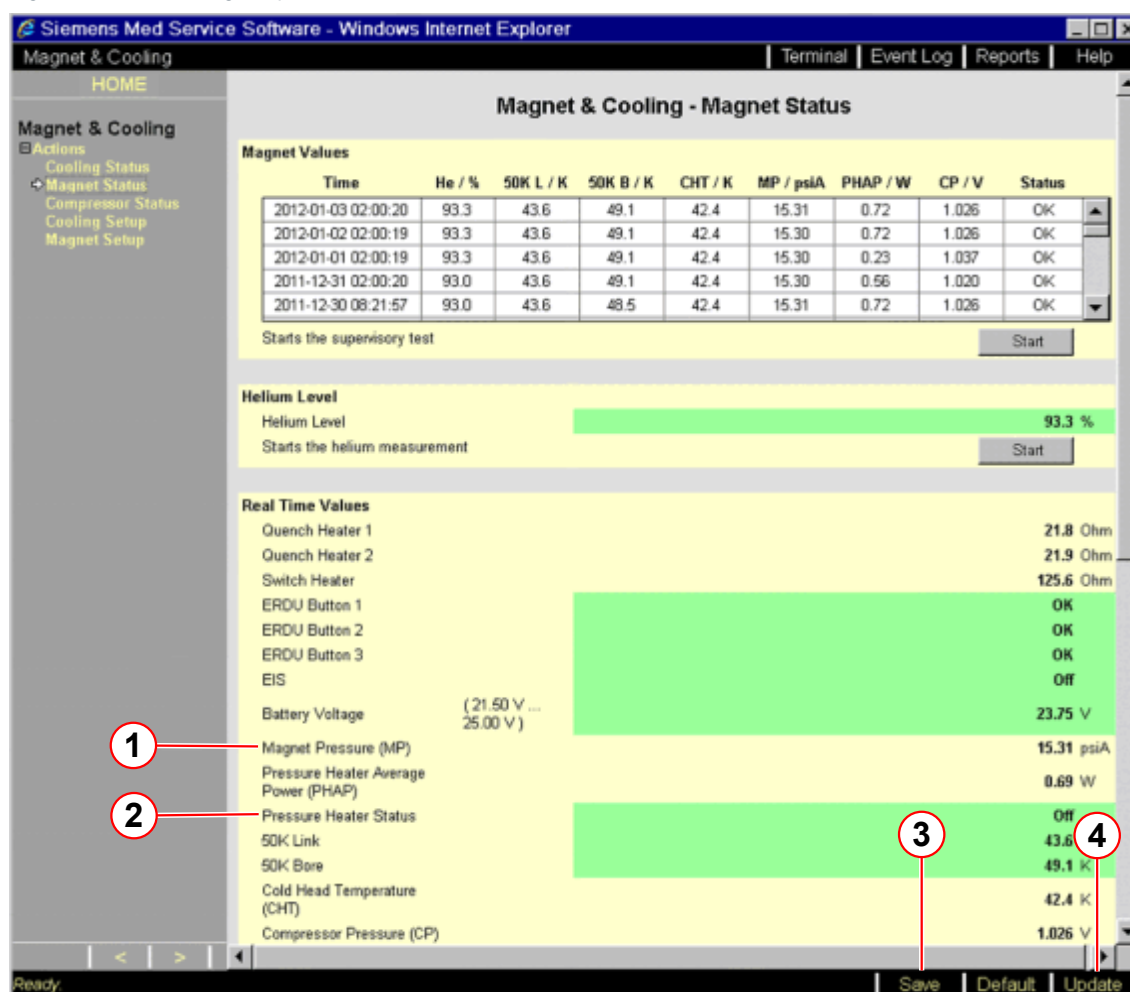


Data is only accurate from the last refresh of the page, or from when the page was opened last.



During pressurization, the cold head switches on automatically.

Fig. 134: Check magnet pressure



- (1) Magnet Pressure (MP)
- (2) Pressure Heater Status
- (3) Save
- (4) Update

- At **MSUP Control** - Sets the pressure heater mode select the **Automatic mode** button.
- Select **SAVE**

5.2.1.11 Final Steps

- **Helium fill** (refer to **Startup**).
- Perform a **system leak check** - (refer to **Turret Leak Check**).
- **Ramp the magnet** to field.

- Fill out the **Quench Report report** . (→ Quench Report, <https://erlh1uqa.ww005.siemens.net/mrquality/mr/format/Index1.htm>) (→ Fig. 135 Page 129)

Fig. 135: Quench report screenshot

SIEMENS Find Contact Home | Siemens Intranet

Quality Evaluations Quench Report RDIAG Coverage Reports Online Support

MR

System Data

System Type:

☒ Stationary ☐ Mobile

Magnet Ser.Nr.: System Ser.Nr.:

Quench Data

Date (dd-mm-yy hh:mm) of Quench:

6 - 2 - 12 01 : 00

System Status:

Circumstances:

Additional comments:

Helium Level:

% Before Quench* Number of days since last magnet ramp

% After Quench* Sent Magnet Supervisory to Healthcare Magnet WS CS HSC ? ☒ No ☐ Yes

Sender's Data

e-mail address:

* Required

if Persistent > We need a comment on possible cause and what the system was doing at the time of the quench
 if Ramping > We need the current (mandatory) and coil if known
 if ERDU > Was the ERDU pressed or some other reason (water, earthquake)
 if Refilling > Was an Auto, high efficiency or service Syphon being used
 if De-icing > What was the gas pressure, has the person doing the de-ice been trained

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Serviceability

6.1 Prerequisites for Shimming

6.1.1 Only for Systems with a Tim Dockable Patient Table



If the system has only one dockable patient table, no further action is necessary.

If the system has more than one dockable patient table, you have to check which patient table is registered.

Only one dockable patient table is registered in the factory.

A correctly registered patient table is necessary for shimming.

1. Check which dockable patient table is already registered.

- Open a command tool and enter the command:

```
C:\Users\meduser>MrVerifyPTAB
```

Example:

Fig. 136: MrVerifyPTAB

```
86); Numaris 4 VD13A ; 25-Sep-2011 00:51; .
```

```
-----
current system settings:
```

```
mobile.....:1
frequency.....:123200000Hz
field strength.....:2.9T|
```

```
current PTAB settings read from HW:
```

```
material number.....:10433522
serial number.....:1002
```

```
PTAB devices registered on host:
```

```
device 0:
material number.....:10433522
serial number.....:1002
-----
```

```
finished successfully.
```

2. Use the registered dockable patient table for shimming.



Check the correct adjustment of the dockable patient table.

Refer to the Installation Instructions (→ Mobile Patient Table (Tim Dockable Table) / M6-024.892.01) if an adjustment is necessary.

If the system has a 2nd or more dockable patient tables, these additional patient tables need to be registered in the system software.

For detailed information, please refer to document (→ Additional Tim Dockable Table / MR07-000.814.05)

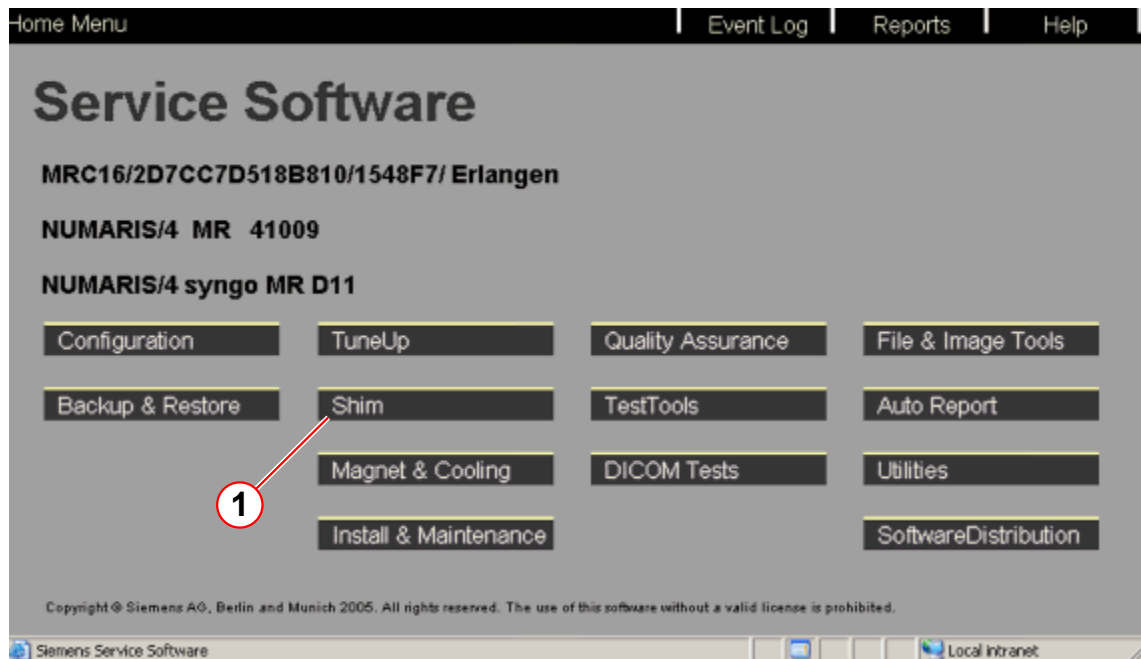
6.2 Passive Shim



Information regarding passive Shim is not published as hardcopy documentation. For more information, refer to the Online Help

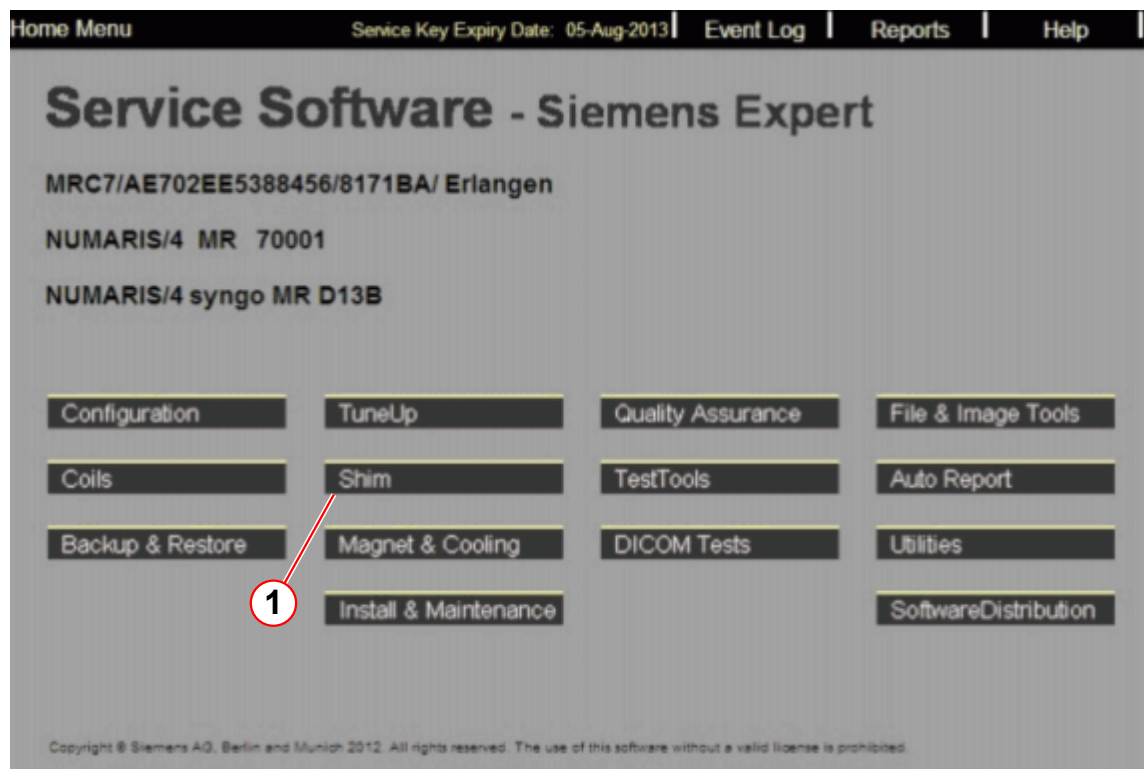
- Start the shim workflow from the SeSo platform.

Fig. 137: Service Software



(1) Shim workflow

Fig. 138: SeSo



(1) Shim Workflow

Version 02

- Chapter 01
 - Update of IButton configuration.

Version 03

- Chapter 05, Ramping the Magnet:
 - Update: The OVC (DOP) port disk and vent line have been removed from the Mk2 magnet design. (Charm MR_00453494)
- Chapter 04 prerequisites for ramping the magnet
 - Update with warning: Do not unscrew or remove the black sealing cap of the temperature sensor. (Charm MR_00450528)

Version 04

- All
 - Minor editorial changes.
- Chapter 03, Filling the Magnet:
 - Quench Line Acceptance Protocol: New path to the acceptance protocol and deleted the sign off by the project manager.
- Chapter 04, Prerequisites for Ramping the Magnet
 - Quench Line Acceptance Protocol: New path to the acceptance protocol and deleted the sign off by the project manager.
- Chapter 05, Ramping the Magnet:
 - Transport test, updated Charm MR_00455599)
 - Warning for visual inspection of quench tube added. (→ OR105 Ramping / Page 79)

There are no Hazard IDs in this document.

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